

# A Secular Mass-Ratio Construction for the Electroweak Pole Spectrum: String, Kaluza–Klein, and $G_2$ Interpretation Targets

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# Abstract

The Rivero–de Vries electroweak observation is formulated as a pole-spectrum mass-ratio problem organized by a two-branch secular equation. The algebraic input inserts  $J = s(s+1)$ , historically a quadratic-Casimir eigenvalue, into  $x^2 + Jx - J = 0$ . The positive-root quotient  $1 - x_+(3/4)/x_+(2) = 0.22310132\dots$  is proposed for comparison with the W/Z complex-pole mass-ratio observable  $s_{\text{pole}}^2$ , with the low-energy pole spectrum taken as the working placement hypothesis. The negative branch belongs to the same diagonalization and is treated as a candidate scalar/order-parameter datum whose physical meaning requires a gauge-invariant scalar functional. Endpoint, interval Kaluza–Klein, localized  $G_2$ , and electroweak-superconnection settings supply candidate arenas for a two-channel kernel. The unresolved derivation is the ordered assignment  $(J_W, J_Z) = (3/4, 2)$ , the source of the DeVries kernel entries, and the pole-matched scalar interpretation of the partner branch.

## I. INTRODUCTION

The electroweak sector contains a dimensionless pole-spectrum ratio,

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}, \quad (1)$$

which is fixed experimentally by the pole positions of the charged and neutral weak vector bosons. In the Standard Model at tree level the same ratio is written as an angular datum in coupling space, while the Higgs vacuum expectation value fixes the common radial scale of the vector masses. This separation between projective and radial data gives the working language of the present paper.

The Rivero–de Vries observation inserts  $J = s(s+1)$ , a quadratic-Casimir eigenvalue, into a dimensionless two-channel ansatz. In the normalized notation used below, the equation is

$$x^2 + Jx - J = 0, \quad J = s(s+1). \quad (2)$$

The positive branch gives

$$s_{\text{dV}}^2 = 1 - \frac{x_+(3/4)}{x_+(2)} = 0.22310132\dots \quad (3)$$

The original note emphasized the proximity of this number to the mass-shell weak angle and motivated the operator through Casimir invariants and high-spin behavior [1]. In this

paper Eq. (3) is used as a descriptive spectral clue. The source-theory derivation of the two-channel ansatz is open. The central question asks which physical object carries the square root in Eq. (2), and why that object is read at the low-energy W/Z pole ratio.

This placement question is the conceptual core. A standard grand-unified lineage uses representation theory to obtain a clean weak-angle value at a high scale. In  $SU(5)$ , and in  $Spin(10)$  with the usual fermion content, the famous group-theoretic value is  $\sin^2 \theta_W = 3/8$ ; contact with the observed low-energy angle is then made through running and thresholds [24]. The Rivero–de Vries construction places the scale burden on the pole-mass ratio itself. The required source theory must explain why a spectral datum attaches to the pole spectrum. A unification-scale boundary condition would define a different reading of the same algebra.

The string-theoretic setting uses the older dual-model and Regge ambition as part of the physical context. Open strings, endpoint data, Kaluza–Klein spectra, and boundary conditions are treated as possible mechanisms for organizing low-energy particle structure. String theory also incorporates Kaluza–Klein ideas, grand-unified representations, and supersymmetric structure in transmuted form [3]. The physical question is whether those tools can produce an electroweak secular datum at the observed pole spectrum. The numerical clue remains a clue until that mechanism is written.

The construction has four layers. The first layer is algebraic: a two-channel secular equation has roots whose ratio is fixed by the Casimir variable. The second layer is electroweak: the W/Z mass ratio is a projective datum along the broken-to-unbroken ray of the gauge-Higgs system. The third layer is spectral: the square root must be assigned to a pole, running, compactification, or boundary-condition object. The fourth layer is dynamical: strings, branes, Kaluza–Klein boundary conditions, or singular  $G_2$  geometries may generate the secular equation or its branch structure.

The logical structure used below separates assumptions from theorem targets:

- A1 :  $Q(J)$  is the algebraic two-channel object,
- A2 :  $s_{\text{pole}}^2$  is read with complex pole masses,
- A3 :  $(J_W, J_Z) = (3/4, 2)$  is the ordered electroweak assignment,
- Target I : place the quotient in the pole scheme,
- Target II : derive A3 from gauge-Higgs or source data,
- Target III : derive the route kernel entries,
- Target IV : identify the  $x_-$  branch.

The algebraic calculation is explicit. The pole-mass comparison is a placement hypothesis whose scheme conventions are fixed in Sec. III. In this paper the ordered assignment  $A3$  is an explicit working assumption for the conditional pole-ratio proposal. Its derivation is retained as Target II: a source package must derive the charged order-parameter sample  $J_H = 3/4$ , the photon-orthogonal neutral current sample  $J_{\text{adj}} = 2$ , and the pole-scheme remainder before the relation becomes a source-derived electroweak result. The negative branch is kept as a candidate scalar/order-parameter datum. The electromagnetic coupling  $\alpha$  lies outside the claim of this paper. A claim for  $\alpha$  would require a source-fixed scalar normalization, a derived  $U(1)_{\text{em}}$  generator and charge lattice, and a specified pole/running matching scale.

Target II has an electroweak-superconnection formulation. In that formulation the odd Higgs/order-parameter component is projected to  $J_H = 3/4$ , while the even transverse current component is projected to  $J_{\text{adj}} = 2$ :

$$P_W : \Phi_{\text{odd}} \mapsto J_H = \frac{3}{4}, \quad P_Z : F_{\text{even}}^{\gamma\perp} \mapsto J_{\text{adj}} = 2.$$

A successful version uses one gauge-Higgs complex, one inner product, and one pole-matching map. Coquereaux's  $SU(2|1)$  superconnection source supplies the electroweak grading vocabulary [39]; the ordered pole assignment remains the theorem target.

The paper is organized as follows. Section II gives the secular operator and its roots. Section III defines the pole observable and the placement problem. Section IV formulates the projective electroweak interpretation. Section V records the negative branch and its possible Higgs-scale role. Sections VI–XIII develop the string, endpoint, Kaluza–Klein, and  $G_2$  mechanism tests in a common two-channel language. Sections XIV and XV delimit the flavor and global-gauge-group issues. Section XVI gives the conclusion and outlook.

## II. TWO-CHANNEL SECULAR MASS-RATIO OPERATOR

Let  $J = s(s+1)$  be the historical quadratic-invariant input. The normalized two-channel operator used in this paper is

$$\frac{Q(J)}{\mu^2} = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}. \quad (4)$$

Its characteristic equation is

$$\det \left( xI - \frac{Q(J)}{\mu^2} \right) = x^2 + Jx - J = 0. \quad (5)$$

The two roots are

$$x_{\pm}(J) = \frac{-J \pm \sqrt{J^2 + 4J}}{2}. \quad (6)$$

They obey

$$x_+(J) + x_-(J) = -J, \quad x_+(J)x_-(J) = -J. \quad (7)$$

For  $J > 0$ , the positive branch is positive and the negative branch is negative.

Historically, the electroweak number is obtained by sampling the positive branch at the two Casimir values

$$J_H = \frac{3}{4}, \quad J_{\text{adj}} = 2. \quad (8)$$

This gives

$$s_{\text{dV}}^2 = 1 - \frac{x_+(3/4)}{x_+(2)}. \quad (9)$$

The expression is dimensionless; the overall scale  $\mu$  cancels. The cancellation is essential for an electroweak interpretation because Eq. (9) can then be compared with a projective W/Z mass ratio. In the electroweak reading developed here,  $J_H = 3/4$  is the Higgs/order-parameter doublet invariant and  $J_{\text{adj}} = 2$  is the adjoint/current invariant. The ordered use of these two samples in the W/Z quotient remains the central derivation problem.

The relation to the original Rivero operator is obtained by restoring the Poincare Casimir eigenvalues  $C_1 = m^2$  and  $C_2 = -m^2 s(s+1)$ . In the original construction, a mass-squared operator is formed from combinations of Casimir invariants, with a high-spin condition designed to preserve the asymptotic mass behavior relevant for Regge trajectories [1]. Equation (4) is the reduced dimensionless representative of that branch problem. Its use here requires a later source-theory derivation; Casimir language identifies the input datum and the historical clue.

### A. What the algebra fixes

The operator in Eq. (4) fixes three algebraic facts. First, the trace is  $-J$ . Second, the determinant is  $-J$ . Third, the two eigenvalues are tied by the same invariant:

$$\text{tr} \frac{Q(J)}{\mu^2} = -J, \quad \det \frac{Q(J)}{\mu^2} = -J. \quad (10)$$

Thus a physical derivation has to explain a simultaneous trace and determinant assignment. A model that gives only a square-root branch, or only a  $J$ -dependent diagonal term, leaves the DeVries construction underdetermined. The required two-channel matrix has the schematic form

$$\mathcal{M}_J = \begin{pmatrix} 0 & \kappa_J \\ \kappa_J & -\tau_J \end{pmatrix}, \quad \kappa_J^2 = J, \quad \tau_J = J. \quad (11)$$

This is the compact target for the endpoint, Kaluza–Klein, and  $G_2$  sections.

Trace and determinant statements in this paper are matrix-specific. The trace of  $Q(J)$ , the trace of any first-seed kernel  $K_{\text{seed}}^{(1)}$ , and the trace of any  $\sigma_3$ -projected negative-sector kernel  $K_-^{\sigma_3}$  belong to distinct vector spaces until a common source operator, field basis, projection map, and normalization identify them. A coefficient transfer between these matrices requires an explicit source-theory theorem.

The square-root radical is then the discriminant of this two-channel problem:

$$\Delta_J = J^2 + 4J. \quad (12)$$

The radical has to attach to a physical object. The radical-placement section lists the main options. This paper follows the pole-spectrum placement because the positive and negative branches are interpreted as a single low-energy spectral clue.

## B. Minimal assumptions and theorem target

A source derivation must state the minimal assumptions that select Eq. (4). The current minimal list is:

1. dependence on available Poincare or internal invariants;
2. dimensional consistency after restoring the mass scale  $\mu$ ;
3. a two-branch secular structure;
4. preservation of the high-spin or Regge asymptotic datum in the originating problem;
5. a trace/determinant condition equivalent to Eq. (10).

These assumptions define the uniqueness target. The theorem to prove would read: within a stated class of two-channel operators built from the allowed invariants, the high-spin

condition and trace/determinant constraints select Eq. (4) up to similarity transformations and normalization.

This theorem would also clarify the status of the negative branch. Since  $x_-(J)$  is fixed by the same trace and determinant as  $x_+(J)$ , a derivation of Eq. (4) automatically derives both branches. The electroweak interpretation can then ask which fields or boundary data realize the two eigenvectors.

### III. POLE OBSERVABLE AND ELECTROWEAK SCHEME SEPARATION

The observable used for comparison is the pole-mass ratio

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}. \quad (13)$$

This ratio is distinct from a running  $\overline{\text{MS}}$  weak angle and from effective weak mixing angles extracted from neutral-current asymmetries. This paper therefore treats Eq. (13) as the primary target and later relates other schemes to it.

The following placement convention will be used. The Rivero–de Vries radical is read as a spectral object attached to the W/Z pole data. Other readings could attach the same radical to a high-scale boundary value, to a running  $\overline{\text{MS}}$  coupling, or to an internal compactification eigenvalue. Each reading changes the physical content of the proposal. The pole reading gives the word “spectrum” its literal electroweak meaning.

A vector-boson pole is defined by the complex pole of the propagator,

$$s_V = M_{V,\text{pole}}^2 - iM_{V,\text{pole}}\Gamma_{V,\text{pole}}. \quad (14)$$

Throughout this paper,  $M_{V,\text{pole}}$  and  $\Gamma_{V,\text{pole}}$  refer to the parameterization in Eq. (14). Martin and Robertson also record the alternative reporting convention

$$s_{\text{pole}} = (M - i\Gamma/2)^2. \quad (15)$$

Any input reported in that convention must be translated to Eq. (14) before Eq. (13) is formed [10]. For an unmixed charged vector,  $s_V$  is the zero of the inverse transverse propagator after renormalization,

$$\Delta_{T,V}^{-1}(s_V) = 0. \quad (16)$$

In the neutral sector,  $s_Z$  is the massive eigenvalue pole of the transverse  $(\gamma, Z)$  inverse-propagator matrix. Equivalently, after the photon null direction is separated,

$$\det \Delta_{T,\gamma Z}^{-1}(s_Z) = 0 \quad \text{on the massive neutral subspace.} \quad (17)$$

When experimental inputs are quoted in a variable-width Breit–Wigner line-shape convention, the conversion to pole parameters must be performed before using Eq. (13). With a common W/Z variable-width convention, the following equations define a conditional convention map with  $\gamma_V$  formed from the quoted variable-width parameters:

$$\gamma_V = \frac{\Gamma_{V,\text{BW}}}{M_{V,\text{BW}}}. \quad (18)$$

The pole parameters in Eq. (14) are then obtained as

$$M_{V,\text{pole}} = \frac{M_{V,\text{BW}}}{\sqrt{1 + \gamma_V^2}}, \quad \Gamma_{V,\text{pole}} = \frac{\Gamma_{V,\text{BW}}}{\sqrt{1 + \gamma_V^2}}. \quad (19)$$

Martin and Robertson summarize the corresponding vector-boson mass relation as

$$M_{V,\text{BW}}^2 = M_{V,\text{pole}}^2 + \Gamma_{V,\text{pole}}^2, \quad V = W, Z. \quad (20)$$

Equations (19)–(20) apply after the quoted W and Z inputs have been verified to share the same variable-width Breit–Wigner convention. Fixed-width line-shape inputs, already-pole inputs, and inputs using Eq. (15) require their stated convention map. This conversion is part of the source check for the pole comparison [10].

The resulting convention chain is

$$(M_{V,\text{BW}}, \Gamma_{V,\text{BW}}) \longrightarrow (M_{V,\text{pole}}, \Gamma_{V,\text{pole}}) \longrightarrow s_V \longrightarrow s_{\text{pole}}^2. \quad (21)$$

### A. Scheme map

The comparison uses the following scheme map before any weak-angle quantity is interpreted:

| Object                                | Definition level                                   | Role in this paper                                               |
|---------------------------------------|----------------------------------------------------|------------------------------------------------------------------|
| $s_{\text{pole}}^2$                   | complex-pole mass ratio                            | primary low-energy spectral target                               |
| Breit–Wigner masses                   | experimental line-shape convention                 | input convention requiring Eq. (20)                              |
| $\hat{s}^2(Q)_{\overline{\text{MS}}}$ | running coupling ratio in a renormalization scheme | translation and loop-scale comparison                            |
| $\sin^2 \theta_{\text{eff}}^\ell$     | neutral-current effective angle                    | phenomenological comparison after electroweak corrections        |
| GUT boundary value                    | high-scale representation datum                    | comparison class for structural weak-angle values                |
| compactification eigenvalue           | internal spectral datum                            | possible origin for $\mathcal{O}_J$ after a dynamical derivation |

The pole reading selects the first row. The remaining rows are translation targets. The source-checked electroweak comparison begins from the quoted experimental convention, maps to complex pole quantities, forms Eq. (13), and then compares with Eq. (9).

The inherited numerical illustration uses

$$M_W = M_Z \sqrt{1 - s_{\text{dV}}^2}. \quad (22)$$

Appendix B records the inherited arithmetic for provenance.

## B. PDG W/Z input set

The Particle Data Group live tables and 2025 listings quote the W and Z masses as mass-dependent-width Breit–Wigner parameters. The W listing gives  $M_{W,\text{BW}} = 80.3692 \pm 0.0133 \text{ GeV}$  and  $\Gamma_{W,\text{BW}} = 2.14 \pm 0.05 \text{ GeV}$ . The Z listing gives  $M_{Z,\text{BW}} = 91.1879 \pm 0.0020 \text{ GeV}$  and  $\Gamma_{Z,\text{BW}} = 2.4955 \pm 0.0023 \text{ GeV}$  [12, 13]. Applying Eq. (19) gives the descriptive pole inputs

| Boson | $M_{\text{BW}}$ (GeV) | $\Gamma_{\text{BW}}$ (GeV) | $M_{\text{pole}}$ (GeV) | $\Gamma_{\text{pole}}$ (GeV) |
|-------|-----------------------|----------------------------|-------------------------|------------------------------|
| W     | $80.3692 \pm 0.0133$  | $2.14 \pm 0.05$            | $80.3407 \pm 0.0134$    | $2.1392 \pm 0.0499$          |
| Z     | $91.1879 \pm 0.0020$  | $2.4955 \pm 0.0023$        | $91.1538 \pm 0.0020$    | $2.4946 \pm 0.0023$          |

The uncertainty propagation in the last two columns treats the quoted mass and width errors as uncorrelated. Correlations in a future electroweak-average combination would change the uncertainty assignment, while the convention map itself is fixed by Eq. (19).

With the same uncorrelated-input convention, the descriptive pole-ratio value is

$$(s_{\text{pole}}^2)_{\text{PDG input}} = 1 - \frac{(80.3407 \text{ GeV})^2}{(91.1538 \text{ GeV})^2} = 0.2231768 \pm 0.0002608. \quad (23)$$

This is the W/Z input line used for descriptive comparison in this paper. The CDF-II measurement and the CMS 2026 measurement  $M_W = 80.3602 \pm 0.0099 \text{ GeV}$  are important W-mass inputs [14, 15]; they are not used as standalone replacements for Eq. (23) without a common Z input, width convention, covariance prescription, and averaging rule.

The Standard Model parameter literature distinguishes several vacuum and renormalization conventions. In particular, pure  $\overline{\text{MS}}$  treatments may define the Higgs vacuum expectation value as the minimum of the loop-corrected effective potential in Landau gauge, eliminating tadpole diagrams at the cost of gauge-choice dependence in intermediate parameters [10]. Such scheme details are relevant for comparing a mass-ratio statement with Lagrangian parameters, while Eq. (13) remains the pole-spectrum target.

Loop effects are therefore a conceptual discriminator. Pole, effective, and running definitions are related by scheme-dependent self-energies and threshold terms. They ask what object the radical belongs to and at what scale it is to be read. A high-scale reading would make the DeVries value analogous to a GUT boundary datum. The pole-data reading makes it a low-energy spectral datum and transfers the burden to string/Kaluza–Klein dynamics.

### C. Limited claim

The formal claim of this section is limited:

$$s_{\text{dV}}^2 \quad \text{is compared first with} \quad 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}. \quad (24)$$

Every comparison to  $\overline{\text{MS}}$  inputs, effective angles, or line-shape masses requires an explicit map. Thus a disagreement among schemes diagnoses the placement of  $\sqrt{J^2 + 4J}$  and the scale at which the radical is read.

#### IV. ELECTROWEAK EMBEDDING AS A PROJECTIVE DATUM

The gauge-Higgs sector gives the tree-level vector masses from the Higgs kinetic term

$$(D_\mu H)^\dagger (D^\mu H), \quad D_\mu = \partial_\mu - \frac{ig}{2} \sigma^a W_\mu^a - \frac{ig'}{2} B_\mu, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h \end{pmatrix}. \quad (25)$$

The charged fields

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp iW_\mu^2) \quad (26)$$

receive a mass term  $g^2 v^2 W_\mu^+ W^{-\mu}/4$ . The neutral fields have mass matrix

$$\frac{v^2}{8} \begin{pmatrix} W_\mu^3 & B_\mu \end{pmatrix} \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix} \begin{pmatrix} W^{3\mu} \\ B^\mu \end{pmatrix}. \quad (27)$$

It is diagonalized by

$$Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu, \quad A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu, \quad \tan \theta_W = \frac{g'}{g}. \quad (28)$$

This gives

$$M_W^2 = \frac{g^2 v^2}{4}, \quad M_Z^2 = \frac{(g^2 + g'^2) v^2}{4}, \quad M_\gamma^2 = 0. \quad (29)$$

The ratio is

$$1 - \frac{M_W^2}{M_Z^2} = \frac{g'^2}{g^2 + g'^2}. \quad (30)$$

Thus the vacuum scale  $v$  fixes the radial size of the massive vector pair, while the quotient fixes a projective direction in the two-coupling plane [5, 9].

##### A. Custodial structure and the electroweak ray

The Standard Model doublet can be packaged as the bidoublet

$$\Phi_H = \begin{pmatrix} H^{0*} & H^+ \\ -H^{+*} & H^0 \end{pmatrix}. \quad (31)$$

In the scalar sector this makes the approximate global  $SU(2)_L \times SU(2)_R$  structure visible; the vacuum proportional to the identity preserves the diagonal custodial  $SU(2)$  subgroup [9]. Hypercharge gauging and Yukawa splittings break the symmetry, while the tree-level doublet mass matrix preserves the relation

$$\rho = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W} = 1. \quad (32)$$

This is the electroweak reason the DeVries comparison is read as a projective statement about the W/Z pair. The broken-to-unbroken deformation scales the vector masses with  $v$  while preserving the coupling-space direction.

The DeVries proposal is read in the same projective language. In a source route the pole statement has the form

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(3/4)}{x_+(2)} + \Delta_{\text{O1}}. \quad (33)$$

Here  $\Delta_{\text{O1}}$  collects field normalization, threshold, source-scale, and complex-pole matching data. The tree-level analogy is the pre-pole comparison

$$\Delta_{\text{O1}}^{\text{tree}} = \frac{g^2}{g^2 + g'^2} - \frac{x_+(3/4)}{x_+(2)}, \quad (34)$$

using Eq. (29). A source theorem must derive the value of  $\Delta_{\text{O1}}$  in the complex-pole scheme. Sec. III and Appendix C 4 state that matching obligation. The common scale is then supplied by the electroweak vacuum or, in a deeper model, by the compactification or brane scale that generates the effective gauge-Higgs sector.

The unbroken limit is the ray

$$v \rightarrow 0 \quad \text{with} \quad \frac{g'^2}{g^2 + g'^2} \text{ fixed}. \quad (35)$$

Along this ray,  $M_W$  and  $M_Z$  vanish together and the photon remains massless. This is the electroweak deformation relevant to the construction. The projective datum remains meaningful as the limiting direction even when the radial mass scale is taken to zero.

Equivalently, introduce a parameter  $t$  by

$$H_t = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ tv + h \end{pmatrix}, \quad g(t) = g, \quad g'(t) = g', \quad 0 \leq t \leq 1. \quad (36)$$

Then

$$M_W^2(t) = t^2 \frac{g^2 v^2}{4}, \quad M_Z^2(t) = t^2 \frac{(g^2 + g'^2) v^2}{4}, \quad M_\gamma^2(t) = 0. \quad (37)$$

This is a tree-level ray statement before the pole-matching map of Sec. III. Coupling-space paths such as  $g \rightarrow 0$ ,  $g' \rightarrow 0$ ,  $g'/g \rightarrow 0$ , or  $g'/g \rightarrow \infty$ , and custodial-breaking shifts such as  $\delta M_W^2 \neq \delta M_Z^2 \cos^2 \theta_W$ , define separate deformation problems. A source route for the DeVries determinant must preserve the single radial parameter, the photon null direction, and the projective W/Z direction. Routes with different behavior enter as failure modes or as separate models.

The ordered assignment used in assumption A3 is

$$(J_W, J_Z) = \left( \frac{3}{4}, 2 \right). \quad (38)$$

The electroweak gauge fields are gauge-algebra fields before symmetry breaking:  $W^1, W^2, W^3$  live in the adjoint of  $SU(2)_L$ , while  $B$  is the gauge field of the abelian  $U(1)_Y$  factor. The value  $J = 2$  has a direct  $SU(2)_L$  adjoint reading through  $T(T+1)$  with  $T = 1$ . The value  $J = 3/4$  belongs to the Higgs/order-parameter side, where the symmetry-breaking field is an  $SU(2)_L$  doublet with  $T = 1/2$  and  $T(T+1) = 3/4$ .

These representation labels define the available samples:

$$J_Z = T_{\text{adj}}(T_{\text{adj}} + 1) = 2, \quad J_H = T_H(T_H + 1) = \frac{3}{4}. \quad (39)$$

The source-completion target is the ordered sampling rule that turns the pair  $(J_H, J_{\text{adj}})$  into the W/Z quotient in Eq. (33). A derivation must start from the gauge-Higgs Lagrangian, a current-algebra construction, or a compactification boundary problem and produce Eq. (38) from the same determinant that supplies the negative branch.

## B. The assignment problem as a theorem target

The theorem target can be stated mechanism-independently. Let  $\mathcal{E}$  be the electroweak gauge-Higgs system, including its projective W/Z pole data and scalar order-parameter channel. A DeVries derivation is a map

$$\mathcal{D} : \mathcal{E} \longrightarrow \mathcal{M}_J = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix} \quad (40)$$

with the following outputs:

$$\mathcal{D}(\text{order-parameter channel}) \mapsto J_H = 3/4, \quad (41)$$

$$\mathcal{D}(\text{adjoint/current channel}) \mapsto J_{\text{adj}} = 2, \quad (42)$$

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{match}}. \quad (43)$$

The exact pole claim requires a derived vanishing of  $\Delta_{\text{match}}$  or a computed matching remainder with stated scheme and source-theory origin. The theorem must also account for the negative branch in the same basis, so that the vector ratio and scalar clue arise from one operator.

The assignment target is a same-source projection package. A source variable  $u_{\text{EW}}$  must produce

$$u_{\text{EW}} \longmapsto (\mathcal{H}_J, \langle \cdot, \cdot \rangle_{u_{\text{EW}}}, \Lambda_J, P_W, P_Z, P_\gamma, \mathcal{R}_{\text{pole}}, \mathcal{F}_{\text{sc}}), \quad (44)$$

where  $P_W$  is the charged pole sample,  $P_Z$  is the neutral massive pole sample,  $P_\gamma$  is the photon null sample,  $\Lambda_J$  fixes the dimensionless source normalization,  $\mathcal{R}_{\text{pole}}$  is the source-to-complex-pole map, and  $\mathcal{F}_{\text{sc}}$  is the scalar functional associated with the negative branch. The ordered electroweak assignment then has the target form

$$P_W(h_J) = J_H = \frac{3}{4}, \quad P_Z(a_J^{\gamma^\perp}) = J_{\text{adj}} = 2, \quad P_\gamma(a_\gamma) = 0, \quad (45)$$

The neutral entry requires a hypercharge normalization condition. Let  $\mathcal{N}_T^\partial = \text{span}\{W_T^3, B_T\}$  be the transverse neutral trace space with the source product  $\langle \cdot, \cdot \rangle_{u_{\text{EW}}}$ . The standard neutral directions on this trace space are

$$a_\gamma^{(0)} = \frac{g'W_T^3 + gB_T}{\sqrt{g^2 + g'^2}}, \quad z^{(0)} = \frac{gW_T^3 - g'B_T}{\sqrt{g^2 + g'^2}}. \quad (46)$$

After projection to the current subspace, the photon trace is normalized in the source product:

$$\tilde{a}_\gamma = P_\gamma a_\gamma^{(0)}, \quad a_\gamma = \frac{\tilde{a}_\gamma}{\sqrt{\langle \tilde{a}_\gamma, \tilde{a}_\gamma \rangle_{u_{\text{EW}}}}}. \quad (47)$$

The massive neutral sample is

$$\Pi_{\gamma^\perp}^{(T)} = 1 - |a_\gamma\rangle\langle a_\gamma|_{u_{\text{EW}}}, \quad \tilde{a}_J^{\gamma^\perp} = P_{a,J} \Pi_{\gamma^\perp}^{(T)} z^{(0)}, \quad a_J^{\gamma^\perp} = \frac{\tilde{a}_J^{\gamma^\perp}}{\sqrt{\langle \tilde{a}_J^{\gamma^\perp}, \tilde{a}_J^{\gamma^\perp} \rangle_{u_{\text{EW}}}}}, \quad (48)$$

This definition carries admissibility conditions. The source product is required to be positive on the projected photon and  $Z$  traces,

$$\langle \tilde{a}_\gamma, \tilde{a}_\gamma \rangle_{u_{\text{EW}}} > 0, \quad \langle \tilde{a}_J^{\gamma^\perp}, \tilde{a}_J^{\gamma^\perp} \rangle_{u_{\text{EW}}} > 0. \quad (49)$$

The source theory must also state the ordered projection

$$P_{a,J} \Pi_{\gamma^\perp}^{(T)}$$

or prove a commutation relation for  $P_{a,J}$  and  $\Pi_{\gamma^\perp}^{(T)}$  on  $\mathcal{N}_T^\partial$ . These data belong to the same theorem package as the hypercharge remainder below. The theorem data are

$$P_Z(a_J^{\gamma^\perp}) = 2 + \Delta_Y(u_{\text{EW}}), \quad P_\gamma(a_\gamma) = 0. \quad (50)$$

Here  $\Delta_Y$  collects the hypercharge normalization,  $W^3/B$  mixing convention, source product, and photon-reference subtraction. The theorem target is a derived  $\Delta_Y$ ; the special closure outcome is  $\Delta_Y = 0$  in the selected convention. The corresponding pole package is

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{O1}}(u_{\text{EW}}). \quad (51)$$

Here  $\Delta_{\text{O1}}$  includes field normalization, threshold, scheme, and source-to-pole remainders. An exact pole statement is a theorem about  $\Delta_{\text{O1}}$ .

| Datum                       | Source-supported content                                                   | Target II theorem datum                                 |
|-----------------------------|----------------------------------------------------------------------------|---------------------------------------------------------|
| $h_J$                       | Higgs doublet/order-parameter charged channel with $C_2(\mathbf{2}) = 3/4$ | projector $P_W$ from the same $\mathcal{H}_J$           |
| $a_J^{\gamma^\perp}$        | adjoint/current channel with $C_2(\text{adj}) = 2$                         | neutral massive projector $P_Z$ after photon removal    |
| $a_\gamma$                  | unbroken electromagnetic direction                                         | null projector $P_\gamma$ and reference zero            |
| $\Lambda_J$                 | shared dimensionless source scale                                          | one normalization for $J_H$ and $J_{\text{adj}}$        |
| $\mathcal{R}_{\text{pole}}$ | pole-scheme conversion map                                                 | complex-pole quotient with $\Delta_{\text{O1}}$         |
| $u_-$                       | negative-branch scalar channel                                             | same-source scalar functional $\mathcal{F}_{\text{sc}}$ |

Coquereaux's algebraic  $SU(2|1)$  superconnection gives source vocabulary in which Higgs fields enter generalized connection data [39]. Here that source supports the grading language for  $h_J$  and  $a_J$ . The ordered quotient in Eq. (51) requires the projectors, normalization, and pole remainder above.

Appendix C 14 records the interval-superconnection realization of this package. In that realization the CHM transverse Dirichlet-to-Neumann kernel, Coquereaux-type finite gauge-Higgs connection, photon subtraction, and source scale enter one reduced kernel  $\hat{K}_J$ . Its closure condition is the simultaneous derivation of  $P_W(h_J) = 3/4$ ,  $P_Z(a_J^{\gamma^\perp}) = 2$ ,  $P_\gamma(a_\gamma) = 0$ , and  $\Delta_{O1/O10}$  in the complex-pole scheme.

A natural current to test is the Higgs isospin current

$$\mathcal{J}_\mu^a = iH^\dagger \frac{\sigma^a}{2} D_\mu H - i(D_\mu H)^\dagger \frac{\sigma^a}{2} H, \quad (52)$$

which transforms as an adjoint object and is built from the doublet order parameter. This current is a candidate bridge between the doublet source data and the adjoint current data. Turning this bridge into Eq. (43) is the remaining electroweak theorem target. In a general gauge, write

$$H = \begin{pmatrix} G^+ \\ (v + h + iG^0)/\sqrt{2} \end{pmatrix}.$$

The Higgs kinetic term contains the Goldstone-vector mixing terms

$$\mathcal{L}_{\partial GV} = iM_W [(\partial_\mu G^-)W^{+\mu} - (\partial_\mu G^+)W^{-\mu}] + M_Z(\partial_\mu G^0)Z^\mu. \quad (53)$$

This source-side expansion, reviewed in Ref. [9], is the equation-level test for the ordered current insertion: the charged insertion must satisfy

$$P_W(h_J) = \frac{3}{4} + \Delta_W(u_{EW}), \quad \Delta_W(u_{EW}) \stackrel{\text{closure}}{=} 0, \quad (54)$$

and the neutral insertion samples the photon-orthogonal current direction of Eqs. (48)–(50).

## V. NEGATIVE BRANCH AND THE SCALAR SIDE OF THE PROBLEM

The negative branch in Eq. (6) is fixed by the same secular equation as the positive branch. The motivating datum is therefore the paired spectrum. Hans de Vries identified the positive-branch proximity to the mass-shell weak angle. Under the temporary normalization

$M_Z^2 = \mu^2 x_+(2)$ , the negative magnitudes define provisional scalar-function targets. That arithmetic observation supplies the main source-selection question: can one source theory attach  $x_-(J)$  to a gauge-invariant scalar functional generated by the same reduced operator that gives  $x_+(J)$ ?

The root identities imply

$$x_-(J) = -J - x_+(J) = -\frac{J}{x_+(J)}. \quad (55)$$

Any physical use of the positive branch automatically carries a partner branch.

The Higgs potential can be written as

$$V(H) = m^2 H^\dagger H + \lambda (H^\dagger H)^2, \quad m^2 < 0 \quad (56)$$

in the broken phase. The sign of the quadratic term makes the scalar interpretation a concrete target. If the normalization is fixed by  $M_Z^2 = \mu^2 x_+(2)$ , the two descriptive branch magnitudes are

$$M_{-,1} = M_Z \sqrt{\frac{|x_-(2)|}{x_+(2)}}, \quad M_{-,1/2} = M_Z \sqrt{\frac{|x_-(3/4)|}{x_+(2)}}. \quad (57)$$

Under this normalization, the two branch magnitudes are descriptive arithmetic. They become physical scalar identifications after the same determinant, normalization, selected  $J_*$ , and scheme are derived from gauge-Higgs or source-theory data.

A valid identification must be gauge-invariant and scheme-controlled. Possible targets include  $m^2$ ,  $\lambda v^2$ , the Higgs pole mass, a Wilson-line curvature, or an eigenvalue in a compactification problem. The branch values associated with  $J = 3/4$  and  $J = 2$  generate definite dimensionless numbers. The map from those numbers to scalar observables remains open.

### A. Logical role of the negative root

The negative root changes the character of the clue. The full quadratic gives a paired object: a pole-ratio candidate and a scalar-side candidate arise from one secular equation. This pairing supplies the main conceptual reason to retain the negative branch. A dynamical theory that produces Eq. (2) has to say what both branches mean in the same basis.

The two-root structure can be stated as a spectral chain,

$$\mathcal{M}_J \longrightarrow \begin{cases} x_+(J) & \text{projective vector datum,} \\ x_-(J) & \text{candidate scalar/order-parameter target.} \end{cases} \quad (58)$$

The first line is the pole-ratio reading developed in Sec. III. The second line is a target for the Higgs, boundary, brane, or compactification sector. The chain is meaningful because both entries come from the same diagonalization. Any proposed derivation that keeps the positive branch acquires the negative-branch obligation automatically.

This obligation also sharpens the electroweak assignment. The ordered pair  $(J_H, J_{\text{adj}}) = (3/4, 2)$  already suggests two electroweak structures: the doublet order parameter and the adjoint current. The negative branch asks whether the doublet/order-parameter side can be read as a scalar instability, a vacuum scale, or a deformation mode in the same operator that gives the vector pole quotient. The scalar reading therefore ties together three open calculations: the ordered  $J$ -assignment, the determinant derivation, and the branch-to-Higgs map.

## B. Gauge-invariant scalar language

The scalar sector has several layers. The Higgs doublet  $H$  itself is gauge-covariant. The radial composite

$$\varphi^2 = 2H^\dagger H \quad (59)$$

is the local invariant order-parameter language used here after renormalization and scheme choice. In the broken phase, a gauge choice may represent the vacuum by a real radial expectation value, yet the physical statement should be expressible through invariant quantities, pole positions, or matched parameters. The negative branch therefore requires a defined scalar functional before it can carry a physical claim.

An abstract target is

$$\mathcal{F}_{\text{sc}} = \mathcal{F}_{\text{sc}}(H^\dagger H, (D_\mu H)^\dagger D^\mu H, V(H), \text{boundary or compactification data}), \quad (60)$$

with a branch assignment

$$\mathcal{F}_{\text{sc}}(J_\star; \Lambda, s_{\text{ren}}, u) = C_{\text{sc}}(\Lambda, s_{\text{ren}}, u) |x_-(J_\star)|. \quad (61)$$

Here  $J_\star$ ,  $C_{\text{sc}}$ ,  $\Lambda$ , the renormalization scheme  $s_{\text{ren}}$ , and the source datum  $u$  are part of the theorem. The expression may describe a pole mass, a curvature of the scalar potential, a vacuum-ray coordinate, a brane separation, a Wilson-line curvature, or a compactification deformation. These readings remain candidate targets until one route derives a specific object.

The same-source package needed for a physical reading is

$$u \longmapsto (\mathcal{H}_J, \langle \cdot, \cdot \rangle_u, P_J, \Lambda_J, \mathcal{R}_{\text{pole}}, \mathcal{F}_{\text{sc}}), \quad (62)$$

with  $\mathcal{H}_J$  the reduced two-channel space,  $P_J$  the projection,  $\Lambda_J$  the source scale, and  $\mathcal{R}_{\text{pole}}$  the pole map used for the vector branch. A dimensionless scalar target then has the form

$$\widehat{\mathcal{F}}_{\text{sc}}(u; J_\star) = \Lambda_J^{-d_{\text{sc}}} \mathcal{F}_{\text{sc}}(J_\star; \Lambda_J, s_{\text{ren}}, u) = C_{\text{sc}}(s_{\text{ren}}, u) |x_-(J_\star)|, \quad (63)$$

where  $d_{\text{sc}}$  is the mass dimension of the chosen observable. The source derivation must state whether  $\widehat{\mathcal{F}}_{\text{sc}}$  is a vacuum-ray coordinate, a potential curvature, a pole observable, a boundary modulus, or a compactification eigenvalue.

The target can be summarized by the commutative source diagram

$$\begin{array}{ccc} \mathcal{L}_{\text{src}}(u) & \longrightarrow & (\mathcal{H}_J, \langle \cdot, \cdot \rangle_u, P_J) \longrightarrow Q(J) \\ & & \downarrow \\ & & \left\{ \begin{array}{l} x_+(J) \text{ vector pole quotient,} \\ x_-(J) \text{ candidate scalar functional,} \\ x_-(J) \text{ auxiliary or gauge-fixed branch.} \end{array} \right. \end{array} \quad (64)$$

The third line is the explicit failure outcome for a scalar interpretation.

A sharper downstream target appears if the scalar functional is the vacuum-ray normalization. In that case the Standard Model identity  $M_W^2 = g^2 v^2 / 4$  can turn the pair  $x_+(3/4)$  and  $x_-(2)$  into a candidate construction coupling. Appendix C 11 records this as an electromagnetic endpoint theorem target: the scalar branch, the charged-vector normalization, the  $D = 9$   $U(1)_{\text{em}}$  kinetic term, and the matching pair  $(Q_\alpha, \Delta_{\alpha, \text{match}})$  must be derived before  $\alpha$  acquires construction status.

The scalar pole target is the most restrictive because it demands the same pole discipline as the vector comparison. The potential target is more sensitive to matching conventions, because  $m^2$ ,  $\lambda$ , and  $v$  depend on the renormalization scheme and on how the effective

potential is defined. The compactification target is broader:  $x_-(J)$  may first appear as an eigenvalue or deformation parameter in a higher-dimensional problem, with the four-dimensional Higgs interpretation emerging after reduction.

### C. Vacuum ray and branch pairing

The electroweak ray separates scale from direction. Along the broken-to-unbroken ray, the radial coordinate scales the vector masses, while the ratio

$$\frac{M_W^2}{M_Z^2} = \frac{g^2}{g^2 + g'^2} \quad (65)$$

is projective at tree level. The DeVries quotient addresses this projective datum. The negative branch is a scalar-side target because it shares the same two-channel operator while carrying the opposite eigenvalue.

In a route with fields  $h_J$  and  $a_J$ , the branch vectors have the schematic form

$$u_{\pm}(J) = \alpha_{\pm}(J)h_J + \beta_{\pm}(J)a_J. \quad (66)$$

The scalar-side reading becomes a statement about the  $h_J$  component of  $u_-(J)$ . The vector-side reading becomes a statement about the projective pole ratio extracted from  $u_+(J)$ . A source derivation must therefore give a branch basis, an inner product, and a normalization. These data determine whether the negative eigenvector genuinely selects an order-parameter amplitude or whether it belongs to an auxiliary sector of the chosen source theory.

The electroweak ray also supplies a consistency check. Taking the radial vacuum coordinate toward the symmetric point should collapse the vector masses together with the Higgs order parameter. The DeVries ratio can remain as a projective direction in coupling or spectral space. A scalar interpretation of  $x_-$  should respect this ray structure by scaling with the radial data or by controlling the deformation that sets the radial scale.

### D. Candidate scalar targets

The Standard Model scalar sector gives several inequivalent objects that can carry a scalar-side interpretation [8, 9]. They have different scheme and gauge properties:

| Target                                                     | Meaning                                         | DeVries requirement                                     |
|------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------|
| $m^2(s_{\text{ren}}, \Lambda)$                             | quadratic term in the Higgs potential           | define the scheme and invariant extraction              |
| $\lambda(s_{\text{ren}}, \Lambda)v^2$                      | curvature scale along the radial mode           | connect branch value to the electroweak ray             |
| $M_{h,\text{pole}}^2$                                      | scalar pole position                            | derive the scalar pole from the same determinant as W/Z |
| $\partial_\alpha^2 V_{\text{eff}}(\alpha) _{\alpha_\star}$ | Wilson-line or Hosotani curvature               | identify the holonomy and source normalization          |
| radion/modulus eigenvalue                                  | compactification scalar or boundary deformation | identify the compact operator and its normalization     |
| brane separation mode                                      | endpoint or brane order parameter               | show coupling to the adjoint/current channel            |

The pole interpretation is the sharpest target. It asks the same object that governs the vector pole quotient to generate a scalar pole or scalar instability. The potential interpretation is more flexible, because  $m^2$ ,  $\lambda$ , and  $v$  depend on conventions and matching. The compactification interpretation is broader still; it may identify the negative branch with a modulus or boundary deformation whose four-dimensional scalar meaning emerges after reduction.

### E. Route-specific scalar maps

The endpoint route maps the negative branch to brane scalar data. In a D-brane system, transverse positions of branes appear as scalar fields on the worldvolume, while strings stretched between branes acquire masses controlled by separation data [2, 3]. A DeVries endpoint derivation would identify  $u_-(J)$  with a brane-Higgsing direction, a stretched-string scalar, or a boundary order-parameter field. The map has to use the same endpoint algebra that supplies the current trace and off-diagonal product.

The interval route maps the negative branch to boundary or modulus data. A five-dimensional gauge field supplies four-dimensional vector modes and scalar components; the compactification radius, brane terms, and boundary conditions determine the light spectrum

[7, 16]. A DeVries interval derivation would identify  $u_-(J)$  with a boundary scalar amplitude, an  $A_5$ -like scalar, a radion-like deformation, or a brane-localized order parameter. The map has to preserve the same boundary determinant that gives the W/Z branch quotient. The Hosotani version of this target uses the holonomy variable

$$\alpha_H \sim g_5 \int_I dy A_5, \quad \mathcal{F}_{\text{sc}}^{\text{Hol}} = \left. \frac{\partial^2 V_{\text{eff}}(\alpha_H; \Lambda, s_{\text{ren}})}{\partial \alpha_H^2} \right|_{\alpha_H = \alpha_\star}, \quad (67)$$

and asks whether the same interval kernel that yields the transverse vector pole also yields  $\widehat{\mathcal{F}}_{\text{sc}}^{\text{Hol}} = C_{\text{sc}} |x_-(J_\star)|$  [29, 30, 34].

The  $G_2$  route maps the negative branch to singularity deformation data. Local  $G_2$  constructions place gauge sectors and charged matter at singular loci, and Higgs-bundle descriptions encode matter localization and interaction data through functions on an associative three-cycle [18–20]. A DeVries  $G_2$  derivation would identify  $u_-(J)$  with a deformation parameter, a localized modulus, a critical value of  $f_Q$ , or a charge-density functional. The same local geometry then has to produce the adjoint/current and order-parameter entries of the kernel.

The route comparison is:

| Route    | Source-supported scalar variable                                               | Missing DeVries map                                                                              |
|----------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Endpoint | transverse brane scalar, stretched-string separation, boundary order parameter | derive the same endpoint determinant, projection, and normalization for $u_-(J)$                 |
| Interval | boundary Higgs datum, $A_5$ holonomy, radion, boundary-condition modulus       | derive the same interval kernel, gauge fixing, scalar/eaten separation, and $\Lambda_J$          |
| $G_2$    | singularity deformation, Higgs-bundle one-form, localized modulus              | derive the local two-channel kernel and the scalar functional on the same associative-cycle data |

These maps give the negative branch a practical role in route selection. A candidate route that produces the positive-branch quotient and gives the negative eigenvector a scalar or deformation interpretation gains additional physical content. A route that assigns the negative branch to an auxiliary variable defines a narrower physical claim.

## F. Eigenvectors and branch meaning

The two roots also have eigenvectors. For the matrix in Eq. (4), an eigenvector with eigenvalue  $x$  can be represented by

$$v_x \propto \begin{pmatrix} \sqrt{J} \\ x \end{pmatrix}. \quad (68)$$

Thus the branch assignment is a statement about channel composition. The positive and negative roots are mixtures of the same two basis amplitudes. A scalar interpretation of  $x_-$  should therefore identify the basis in which one component is the vector/current side and the other is the order-parameter or boundary side.

A criterion is the following. If the same two-channel determinant produces both the vector pole ratio and the scalar instability, then the determinant should appear in an effective action, boundary condition, or self-energy equation. The corresponding theorem target is an explicit determinant with an identified second-channel field or mode. Breitenlohner–Freedman stability supplies a route caveat for AdS scalar readings: a negative mass-squared can be admissible under specified boundary conditions, while the DeVries sign still requires the source functional in Eq. (63) [35].

## VI. STRING, KALUZA–KLEIN, AND $G_2$ COMPLETION PROBLEM

The secular equation has the form of a low-level branch splitting that preserves a high-spin datum. This is the reason string and Regge language enter the construction. The original Rivero construction imposed a high-spin condition motivated by Regge behavior [1]. A string interpretation must therefore preserve the asymptotic slope while explaining why the low levels split through the two-channel determinant.

The string-theory input is concrete. Quantized strings generate towers, level matching, endpoint sectors, winding, and compactification charges. A compact circle already contains the basic Kaluza–Klein mechanism: a higher-dimensional metric reduces to a lower-dimensional metric, a gauge field, and a radius modulus; Fourier modes along the circle acquire masses  $n^2/R^2$  and charges under the gauge field descending from the metric [2]. The string worldsheet adds winding sectors, and winding modes couple to the vector descending from the antisymmetric tensor. Compactification therefore supplies more than one integer

or representation label, and different sectors can couple to different gauge fields.

Open strings add boundary data. Neumann and Dirichlet boundary conditions relate left- and right-moving oscillators differently, and D-branes localize open-string states on submanifolds [2]. Coincident branes generate nonabelian gauge fields from open-string excitations. A DeVries derivation based on endpoints or branes must identify the two channels in Eq. (69) with actual boundary degrees of freedom, endpoint charges, or brane/bulk mixing modes.

The older phenomenological string lineage is relevant here. Regge trajectories in hadron spectra motivate a rotating flux-tube picture, and the open-string Veneziano amplitude was first written to capture strong-interaction features before QCD supplied the gauge-theory description [2]. In this lineage, a low-energy spectral regularity can guide the search for a deeper string or boundary structure, provided the proposed determinant is eventually derived.

Endpoint physics gives the simplest analogy for charged vector states. In a coincident-brane system, strings stretching between branes carry off-diagonal gauge quantum numbers, and the massless excitations organize into a nonabelian gauge field [2]. The electroweak target is a broken nonabelian gauge sector coupled to a doublet order parameter. A successful endpoint model must therefore produce both the adjoint-current datum  $J = 2$  and the doublet/order-parameter datum  $J = 3/4$  inside one boundary or brane determinant.

A Kaluza–Klein or brane derivation would have to produce the determinant

$$\det \begin{pmatrix} x & -\sqrt{J} \\ -\sqrt{J} & x + J \end{pmatrix} = 0. \quad (69)$$

This can arise from a two-channel boundary problem, a mixing of brane-localized and bulk modes, or an effective orbit problem. The required output is the exact  $J$ -dependence in Eq. (69).

Extra-dimensional electroweak models show that boundary conditions can break gauge symmetry and determine vector spectra. Higgsless constructions on intervals provide a controlled language for relating boundary data, vector masses, fermion localization, and precision observables [7]. The interval viewpoint is valuable because boundary conditions can be obtained as limits of boundary dynamics. For the present construction, this gives a precise target: derive the  $J = 3/4, 2$  samples as boundary data associated with the Higgs doublet/order-parameter channel and the adjoint/current channel.

Witten’s search for realistic Kaluza–Klein theory gives the historical benchmark. Extra dimensions can generate gauge groups from geometry, including  $SU(3) \times SU(2) \times U(1)$ , while the realistic fermion quantum numbers impose severe constraints [16]. The DeVries construction inherits this lesson. A compactification mechanism must solve the electroweak spectral problem and remain compatible with chiral matter.

M-theory on  $G_2$ -holonomy manifolds supplies a separate completion map. Smooth compact  $G_2$  compactifications give four-dimensional minimal supersymmetry in the large smooth limit. Realistic charged chiral matter requires singularities supporting nonabelian gauge sectors and chiral fermions [17–19]. Local Higgs-bundle descriptions over associative three-cycles give a modern way to organize gauge sectors, localized matter, and singular transitions [20].

The completion problem can therefore be stated as a list of mechanisms and obligations.

| Mechanism                   | What it can supply                   | Required DeVries test                                     |
|-----------------------------|--------------------------------------|-----------------------------------------------------------|
| Regge/string tower          | high-spin asymptotics                | preserve slope and produce Eq. (69)                       |
| Circle KK reduction         | gauge fields and charged KK modes    | identify the charge/Casimir source of $J$                 |
| Winding sector              | second compactification charge       | test a winding-like interpretation of the second branch   |
| Interval/brane BC           | vector spectra and boundary dynamics | derive $J = 3/4, 2$ samples from boundary data            |
| Endpoint/open-string sector | localized gauge states               | identify the two channels as endpoint or brane/bulk modes |
| Singular $G_2$              | gauge/chiral sectors                 | localize the electroweak operator geometrically           |
| Higgs bundle                | matter/mass localization             | identify the two branches as spectral-cover data          |

## VII. RADICAL PLACEMENT AND SCALE ATTACHMENT

The secular equation contains the radical

$$\Delta_J^{1/2} = \sqrt{J^2 + 4J}. \quad (70)$$

This paper treats  $\Delta_J^{1/2}$  as the central object whose physical address must be identified. The algebra alone specifies a two-branch spectrum. Physics begins when the radical is assigned to a pole condition, a boundary determinant, a compactification eigenvalue, a worldsheet constraint, or a running matching condition.

### A. Placement options

The possible placements are distinct assignments of the same radical.

| Placement                    | Physical meaning                                                                    | Required derivation                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pole spectrum                | $\Delta_J^{1/2}$ enters a propagator or self-energy pole condition                  | derive the W/Z pole ratio and scalar partner branch from one spectral equation                                                                                                                                                                                                                                                                                                                    |
| GUT boundary                 | $\Delta_J^{1/2}$ fixes a high-scale weak-angle datum                                | embed the $J$ -samples in a unification representation and run to the pole scale                                                                                                                                                                                                                                                                                                                  |
| Running parameter            | $\Delta_J^{1/2}$ constrains a scheme-dependent coupling or vacuum parameter         | specify the scheme, scale, matching map, and threshold corrections                                                                                                                                                                                                                                                                                                                                |
| KK eigenvalue                | $\Delta_J^{1/2}$ comes from a compact internal wave operator                        | identify the internal operator, mode labels, degeneracies, and boundary conditions                                                                                                                                                                                                                                                                                                                |
| Open-string end-point        | $\Delta_J^{1/2}$ comes from or Chan–Paton data                                      | derive the two-channel matrix from endpoint charges and brane/bulk mixing                                                                                                                                                                                                                                                                                                                         |
| Electroweak super-connection | $\Delta_J^{1/2}$ is attached to Higgs data and a generalized gauge-Higgs connection | odd derive $P_W : \Phi_{\text{odd}} \mapsto 3/4$ , $P_Z : F_{\text{even}}^{\gamma^\perp} \mapsto 2$ , and the pole matching from one complex transverse-current data in even and the pole matching from one complex transverse-current data in even and the pole matching from one complex transverse-current data in even and the pole matching from one complex transverse-current data in even |
| Worldsheet constraint        | $\Delta_J^{1/2}$ comes from matching, weights, or a sector                          | level identify the CFT labels and show how conformal they reduce to the electroweak determinant, or a modular invariant                                                                                                                                                                                                                                                                           |

The pole-spectrum placement is the hypothesis used below. It gives the numerical clue its electroweak content because W and Z are observed as poles. It also gives the negative branch a task: the same determinant should know about the scalar/order-parameter side. This reading demands the most direct dynamical calculation, since it asks string/Kaluza–Klein structure to survive down to a low-energy pole observable.

The GUT-boundary placement gives a familiar comparison class. In  $SU(5)$  and  $Spin(10)$  settings, representation theory can give a clean weak-angle value at the high scale, with running and thresholds supplying the low-energy relation [24]. A DeVries high-scale reading would join that lineage. This paper pursues the pole reading because the negative branch and pole-ratio clue appear together in the same low-energy normalization.

## B. Scale attachment

Let  $\mathcal{O}_J$  denote the physical operator whose two-channel reduction gives

$$\det(\lambda - \mathcal{O}_J) = \lambda^2 + J\lambda - J. \quad (71)$$

The scale question is then a question about where  $\mathcal{O}_J$  lives:

$$\mathcal{O}_J \in \{\mathcal{O}_{\text{pole}}, \mathcal{O}_{\text{GUT}}, \mathcal{O}_{\text{KK}}, \mathcal{O}_{\text{brane}}, \mathcal{O}_{\text{CFT}}\}. \quad (72)$$

Each address has a different interpretation of the same algebra. A pole operator must be built from four-dimensional gauge-Higgs correlators. A GUT operator must be built from representation traces and matching. A KK operator must be a compact internal differential operator or boundary value problem. A brane operator must couple localized and bulk modes. A worldsheet operator must arise from conformal weights, current algebra, or level matching.

This list fixes the scheme dependence of the radical. A different placement changes the required derivation.

## C. Placement chain

The placement question can be recorded as a chain. A source theory  $\mathcal{S}$  supplies an operator, the operator supplies a two-channel kernel, the kernel supplies branches, and the

branches acquire physical meaning after a scheme and an assignment rule have been fixed:

$$\begin{aligned} \mathcal{S} &\longrightarrow \mathcal{O}_J \longrightarrow \mathcal{M}_J \longrightarrow \{x_+(J), x_-(J)\}, \\ \{x_+(J), x_-(J)\} &\longrightarrow \{s_{\text{pole}}^2, \mathcal{F}_{\text{sc}}, \mathcal{C}_{\text{top}}\}. \end{aligned} \quad (73)$$

Here  $\mathcal{F}_{\text{sc}}$  denotes the scalar/order-parameter functional sought in Sec. V, and  $\mathcal{C}_{\text{top}}$  denotes the global-form and representation constraints discussed in Secs. XIV and XV. The chain separates four tasks:

$$\text{operator origin: } \mathcal{S} \mapsto \mathcal{O}_J, \quad (74)$$

$$\text{two-channel reduction: } \mathcal{O}_J \mapsto \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad (75)$$

$$\text{branch reading: } \mathcal{M}_J \mapsto \{x_+(J), x_-(J)\}, \quad (76)$$

$$\text{physical assignment: } \{x_{\pm}\} \mapsto (s_{\text{pole}}^2, \mathcal{F}_{\text{sc}}, \mathcal{C}_{\text{top}}). \quad (77)$$

Each arrow has different source support. Complex-pole and scheme language belongs to the last arrow [10]. Higgs doublet, adjoint current, custodial structure, and the electroweak ray belong to the assignment arrow [8, 9]. Endpoint labels, D-brane gauge fields, and interval boundary conditions belong to the first two arrows [2, 3, 7]. Equation (76) is algebraically controlled; the other arrows are open derivation targets.

This chain also organizes the negative branch. If a candidate source theory derives  $\mathcal{M}_J$ , then  $x_-(J)$  is carried by the same light-sector basis as  $x_+(J)$ . The scalar question becomes a question about the eigenvector map,

$$u_-(J) \in \mathcal{H}_J \quad \longmapsto \quad \mathcal{F}_{\text{sc}}, \quad (78)$$

where  $\mathcal{H}_J$  is the two-dimensional light sector. A Higgs interpretation requires  $\mathcal{F}_{\text{sc}}$  to be gauge invariant. A brane interpretation requires  $\mathcal{F}_{\text{sc}}$  to be a modulus, separation, or localized boundary datum. A compactification interpretation requires  $\mathcal{F}_{\text{sc}}$  to be a normalized internal eigenmode or deformation parameter.

The chain gives a compact criterion for acceptance. A proposed mechanism should fill one of the arrows in Eq. (73) or state a new arrow with its source theory, scheme, and physical assignment.

## D. Original research route

The route developed here is:

$$\begin{aligned} \text{Regge/high-spin datum} &\longrightarrow \text{two-channel secular determinant} \\ &\longrightarrow \text{electroweak pole ratio and scalar partner branch.} \end{aligned} \tag{79}$$

The first arrow belongs to the string/dual-model tradition: spectra organize into towers with controlled asymptotics. The second arrow is the DeVries algebra. The third arrow is the remaining electroweak derivation.

A concrete model should therefore supply four ingredients:

1. an internal label whose quadratic invariant is  $J$ ;
2. a two-channel system whose off-diagonal mixing is fixed by  $\sqrt{J}$ ;
3. a diagonal trace term giving the  $-J$  entry;
4. a physical normalization that places the positive branch on the vector pole ratio and the negative branch on the scalar/order-parameter side.

The fourth ingredient is the new conceptual burden. It turns the algebraic clue into a physical claim hierarchy and fixes the relation between the pole reading and high-scale weak-angle boundary data.

## VIII. DUAL-MODEL LINEAGE AND OPEN-STRING ENDPOINT DATA

The DeVries construction enters this paper through a spectral regularity. This is why the older dual-model lineage matters. The Veneziano amplitude was introduced to capture strong-interaction regularities before its string interpretation became clear; its open-string form later supplied a world-sheet explanation of the pole tower and the associated Regge behavior [2]. The lesson for the present work is methodological and technical. A compact algebraic regularity in particle data can guide a deeper spectral construction when the amplitude or boundary problem subsequently explains the regularity.

The relevant feature is the coexistence of a low-level spectrum and a high-spin asymptotic rule. In string amplitudes, pole residues organize the spins carried by the exchanged string

states, and the tower structure follows from the quantized oscillators [2]. For the DeVries determinant, the secular equation must play an analogous role at a much lower and more constrained level:

$$\text{high-spin/Regge regularity} \quad \rightsquigarrow \quad x^2 + Jx - J = 0 \quad \rightsquigarrow \quad \{x_+(J), x_-(J)\}. \quad (80)$$

The first arrow is historical motivation until a world-sheet or boundary calculation produces the second expression. The second arrow is algebraic and already fixed. The Rivero/DeVries observation functions as a clue that selects a target determinant.

The same distinction controls the higher positive branch. Ordinary open-string Regge behavior is supplied by an oscillator level  $N_{\text{osc}}$  and a common slope; the DeVries label has to be a sector or intercept label  $j$ , with  $A_j = j(j+1)$ , if higher slots are to coexist with a straight trajectory. The conditional tower has the form

$$M_{N_{\text{osc}},j,\sigma}^2 = \mu^2 x_\sigma(A_j) + \frac{N_{\text{osc}}}{\alpha'} + \Delta_{N_{\text{osc}},j,\sigma}^{\text{Regge}}, \quad \sigma = \pm. \quad (81)$$

Here  $j$  labels the reduced DeVries sector, while  $N_{\text{osc}}$  and the trajectory spin label the string excitation. The  $j = 3/2$  positive slot is therefore a branch-assignment target. Particle identity, gauge quantum numbers, production, decay, and collider compatibility must come from the same source theory that derives the determinant.

A related brane-scaling test adds an axis map to Eq. (81). A source theory must state whether the DeVries sector label maps to a putative rotating-brane angular momentum or serves as an intercept label. The shared source package is

$$\mathfrak{R}_j = (K_j, \alpha', P_{\text{surv}}, \mathcal{B}_{\text{ND}}, \mathcal{Q}, \mathcal{T}, \chi_{\text{spin}}),$$

where  $\chi_{\text{spin}}$  is the extra axis map. The current algebraic test is summarized in Target C 13: the positive branch is bounded at large putative brane spin, while the negative magnitude grows linearly under the mass reading.

Open strings provide a more concrete mechanism than closed-string Regge language alone. An open-string state carries oscillator data in the interior and boundary data at its endpoints. D-branes give a geometric home to the endpoints, and coincident branes give nonabelian gauge fields from open-string modes [2, 3]. This suggests a precise kind of model for Eq. (69): one channel should be an adjoint or current channel associated with gauge fields, while the other should carry the order-parameter data that samples the doublet invariant.

For the DeVries determinant, the endpoint route can be written as a quadratic action for two boundary-coupled amplitudes,

$$S_J^{(2)} = \begin{pmatrix} \psi_1^\dagger & \psi_2^\dagger \end{pmatrix} \begin{pmatrix} \lambda & -\kappa_J \\ -\kappa_J & \lambda + \tau_J \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}, \quad (82)$$

with the DeVries target

$$\kappa_J^2 = J, \quad \tau_J = J. \quad (83)$$

Here  $\psi_1, \psi_2$  are placeholders for actual endpoint, brane, or localized modes. The notation is intentionally sparse. The physics is carried by the derivation of  $\kappa_J$  and  $\tau_J$ .

There are three natural endpoint readings.

1. *Chan-Paton/current reading.* The endpoint labels carry gauge indices. The diagonal entry  $\tau_J$  is a trace or current-algebra invariant, and the off-diagonal entry  $\kappa_J$  is a group-theoretic transition matrix element.
2. *Brane/bulk reading.* One amplitude is localized on a brane or boundary, while the other propagates in the bulk. The determinant comes from integrating over the bulk mode and imposing the boundary matching condition.
3. *Order-parameter reading.* One channel is an adjoint-current datum and the other is a doublet/order-parameter datum. This is the route that can explain why the electroweak samples are  $J = 2$  and  $J = 3/4$ .

Each reading must respect the same electroweak constraint. The pole ratio compares charged and neutral vector poles, while the  $J = 3/4$  datum belongs naturally to the Higgs doublet side. A successful endpoint construction therefore has to turn an order-parameter doublet invariant into the charged-vector sample in the pole quotient. This is the ordered-sampling problem in boundary language.

The open-string route also clarifies the negative branch. The two roots are eigenvalues of one two-channel system. If the positive branch is associated with the vector pole quotient, the negative branch should correspond to the partner channel in the same boundary problem. In an endpoint construction this partner may be a scalar boundary modulus, a Higgs-like order parameter, a brane separation mode, or an auxiliary channel eliminated in the low-energy action. This makes the negative branch a demand on the model: the same determinant has to explain both eigenvalues.

This places the string interpretation in a concrete form. The target is an endpoint or brane calculation whose reduced determinant has the entries in Eq. (83), whose low-energy normalization gives the W/Z pole quotient, and whose partner eigenvalue lands in the scalar/order-parameter sector.

## IX. KALUZA–KLEIN SPECTRA, BOUNDARY CONDITIONS, AND SECULAR DETERMINANTS

Kaluza–Klein theory supplies the second concrete language for the DeVries program. The original idea unifies gauge fields with geometry by expanding a higher-dimensional theory around a compact ground state. In Witten’s account, symmetries of the compact space appear as gauge symmetries in the lower-dimensional theory, and the physical spectrum is obtained from small oscillations around the compactification background [16]. Salam and Strathdee give the complementary harmonic-expansion language on compact homogeneous spaces  $G/H$ , where zero modes give gravitational plus Yang–Mills fields and higher modes give massive multiplets [33]. The same account also states the hard model-building burden: obtaining the Standard Model gauge group, fermion quantum numbers, and scalar sector is a serious representation problem, even when the gauge group itself can be engineered.

The DeVries problem asks for a narrower and sharper object inside this tradition: a compact or interval derivation of the two-channel determinant

$$\det \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} = \lambda^2 + J\lambda - J. \quad (84)$$

This determinant is the entire target of the KK route.

### A. Circle compactification and two charge channels

String compactification on a circle gives a clean source-backed example of a two-channel spectral structure. Ordinary Kaluza–Klein momentum is quantized along the circle. Closed strings add winding around the same circle. Polchinski’s circle example shows that the Kaluza–Klein vector couples to compact momentum, while the vector descending from the antisymmetric tensor couples to winding [3]. At the self-dual radius the spectrum has

enhanced current algebra, and changing the radius is described by an operator built from the same currents.

This gives a source-backed analogy:

$$\left( \begin{array}{c} \text{momentum-like channel} \\ \text{winding-like or boundary channel} \end{array} \right) \rightsquigarrow \{x_+(J), x_-(J)\}. \quad (85)$$

The analogy becomes a derivation after the two channels are reduced to Eq. (84). The self-dual-radius example also explains why a scalar modulus belongs in the discussion. The compact radius is a scalar field in the lower-dimensional theory, and gauge enhancement can occur at special values of that scalar. This is the compactification analogue of the positive/negative branch problem: a vector spectral statement and a scalar modulus can arise from the same compact sector.

A sharper test combines this channel picture with a boundary Schur complement. Let the compact or interval sector contain a momentum/current-like channel and a winding, brane-separation, or order-parameter channel. After the tower is integrated out, the reduced boundary kernel should have the form

$$K_j^{\text{eff}}(\lambda) = K_j^{\text{bdry}}(\lambda) - V_j^\dagger (\lambda - L_j)^{-1} V_j \longrightarrow \begin{pmatrix} \lambda & -\sqrt{A_j} \\ -\sqrt{A_j} & \lambda + A_j \end{pmatrix}, \quad A_j = j(j+1). \quad (86)$$

This target is the KK form of the Regge-intercept reading in Sec. VIII: the compact or boundary calculation supplies the two-channel intercept, and the oscillator or tower label supplies the large-spin growth.

## B. Interval variational data

Extra-dimensional field theory on an interval gives a more direct determinant language. Csaki–Hubisz–Meade derive boundary conditions from the variational principle for fields on an interval and show how boundary mass terms and boundary kinetic terms modify those conditions [7]. Dirichlet-Higgs and radion-stabilization models give adjacent source variables for boundary scalar profiles, radion data, and bulk Higgs stabilizers [29, 30]. A scalar example has the schematic action

$$S_{\text{int}} = \int d^4x \int_0^L dy \frac{1}{2} \left[ \partial_M \Phi \partial^M \Phi - V(\Phi) \right] + S_0[\Phi] + S_L[\Phi], \quad (87)$$

with boundary functionals  $S_0$  and  $S_L$ . The boundary variation supplies equations at  $y = 0$  and  $y = L$ . Boundary mass terms give Robin-type conditions. Boundary kinetic terms can make the boundary equation depend on the four-dimensional eigenvalue. This is exactly the structural feature needed by a secular determinant.

For a mode  $\Phi(x, y) = \phi_\lambda(x)f_\lambda(y)$ , the interval problem has the abstract form

$$\mathcal{L}_J f_\lambda(y) = \lambda f_\lambda(y), \quad 0 < y < L, \quad (88)$$

$$\mathcal{B}_{0,J}(\lambda)f_\lambda(0) = 0, \quad \mathcal{B}_{L,J}(\lambda)f_\lambda(L) = 0. \quad (89)$$

The DeVries route asks for a two-dimensional light sector  $\mathcal{H}_J$  extracted from this problem. A convenient basis is

$$\mathcal{H}_J = \text{span}\{h_J, a_J\}, \quad (90)$$

where  $h_J$  is a boundary order-parameter amplitude and  $a_J$  is an adjoint/current amplitude. The interval derivation then reduces the bulk-plus-boundary problem to an effective kernel

$$K_J^{\text{int}}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}^{\text{int}}(\lambda) & \Sigma_{ha,J}^{\text{int}}(\lambda) \\ \Sigma_{ah,J}^{\text{int}}(\lambda) & \lambda + \Sigma_{aa,J}^{\text{int}}(\lambda) \end{pmatrix}. \quad (91)$$

The DeVries target is the boundary limit

$$\Sigma_{hh,J}^{\text{int}} = 0, \quad \Sigma_{aa,J}^{\text{int}} = J, \quad \Sigma_{ha,J}^{\text{int}} \Sigma_{ah,J}^{\text{int}} = J. \quad (92)$$

This is the interval version of the endpoint target in Sec. XI.

CHM boundary data supply the interval version of this target. In the scalar example, a boundary kinetic term at  $y = 0$  has the source normalization

$$S_{\text{brane,kin}}^{(0)} = \int d^4x \frac{1}{2M_0} \partial_\mu \phi \partial^\mu \phi|_{y=0}, \quad \partial_y f_n(0) = -\frac{m_n^2}{M_0} f_n(0). \quad (93)$$

The associated product is

$$(f, g)_{\text{CHM}} = \int_0^{\pi R} f(y)g(y) dy + \sum_i \frac{1}{M_i} f(y_i)g(y_i). \quad (94)$$

For gauge fields, boundary scalar vevs give vector boundary data of the source form

$$\partial_y A_\mu|_{y_i} \mp v_i^2 A_\mu(y_i) = 0, \quad \left(\frac{m^2}{\xi_i} - v_i^2\right) \pi_i + v_i A_5|_{y_i} = 0, \quad (95)$$

with endpoint-orientation signs fixed by the variational convention. The CHM higgs-less/custodial construction also supplies the photon zero-mode and custodial boundary

accounting used by the later electroweak boundary proof. These source facts define the allowed boundary-kernel class. The equality to Eq. (92) remains the open DeVries entry target. The scalar equation makes the gauge-fixing burden explicit: the projected DeVries kernel must state the treatment of  $\xi_i$ , separate physical scalar data from eaten Goldstone data, preserve the photon zero mode, and map the interval eigenvalue into the four-dimensional transverse pole condition.

A Hodge factorization subroute gives a sharper test for the off-diagonal entry. Witten's supersymmetric quantum mechanics on forms uses the exterior derivative and its adjoint to build supercharges whose Hamiltonian is the Hodge Laplacian; the Witten deformation adds Morse-function data to the same complex [36]. If an interval or compact sector supplies normalized states  $e_J^0 \in \Omega^0$  and  $e_J^1 \in \Omega^1$  with

$$D_J e_J^0 = \sqrt{J} e_J^1, \quad D_J^\dagger e_J^1 = \sqrt{J} e_J^0, \quad (96)$$

then the de Rham part of the reduced operator supplies  $\Sigma_{ha,J}^{\text{int}} \Sigma_{ah,J}^{\text{int}} = J$ . In Target X language, Eq. (96) supplies  $Q_{\text{dR},J}$ ; the boundary or breaking datum  $B_J$  must supply the diagonal entry  $-J$ . The same interval model must derive the sign convention in the DeVries block and the ordered electroweak map. Appendix C 14 records this as a theorem target.

This gives the interval route an entry order. The Hodge/SUSY-QM calculation tests the square-root product  $\Sigma_{ha,J}^{\text{int}} \Sigma_{ah,J}^{\text{int}} = J$ . The CHM boundary calculation tests the diagonal current entry  $\Sigma_{aa,J}^{\text{int}} = J$ . A successful route must place both tests in one source Hilbert space, with one inner product, one projection to  $\mathcal{H}_J$ , and one pole-matching map. The negative branch is then read from the eigenbasis of the same reduced kernel.

### C. Dimensional Schur-complement target

The dimensional interpolation of Appendix D defines a candidate Schur-complement calculation. Let the middle-dimensional sector be split into a light boundary subspace  $\mathcal{H}_J = \text{span}\{h_J, a_J\}$  and a tower, bulk, or compact sector with operator  $L_J$ . With a boundary kernel  $K_J^{\text{bdry}}(\lambda)$ , coupling map  $V_J$ , and specified inner product, integrating out the heavy sector gives

$$K_J^{\text{eff}}(\lambda) = K_J^{\text{bdry}}(\lambda) - V_J^\dagger (\lambda - L_J)^{-1} V_J. \quad (97)$$

The required reduced kernel is

$$K_J^{\text{eff}}(\lambda) \longrightarrow \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} \quad (98)$$

in a limit where additional light channels decouple or acquire a controlled block-diagonal form. This formulation connects the  $D = 10/6$  middle line to the route kernel:  $K_J^{\text{bdry}}$  contains boundary kinetic and mass data,  $L_J$  contains the integrated tower, and  $V_J$  contains the order-parameter/current mixing. A successful interval derivation must specify these three objects, the inner product, and the normalization yielding Eq. (92).

The same target can be stated in Dirichlet-to-Neumann form. Solving the bulk mode equation with fixed boundary values gives a map from boundary traces to normal derivatives. Boundary kinetic and scalar terms add local endpoint pieces, so a source-level interval target is

$$K_J^{\text{int}}(\lambda) = P_J^\dagger (K_J^{\text{DtN}}(\lambda) + K_J^{\text{brane}}(\lambda)) P_J \longrightarrow \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix}, \quad (99)$$

where  $P_J$  is the projection to  $\mathcal{H}_J = \text{span}\{h_J, a_J\}$ . This form makes the source-data demand explicit: the Dirichlet-to-Neumann kernel, brane terms, projection, inner product, and decoupling rule must be supplied by one interval model.

The current-channel entry can be isolated after the physical boundary kernel has been converted to the dimensionless DeVries normalization. Let  $\mathcal{N}_T^\partial = \text{span}\{W_T^3, B_T\}$  carry the CHM product  $\langle \cdot, \cdot \rangle_{\text{CHM}}$ . The CHM-normalized projector package is

$$u_{\text{CHM}} \mapsto \left( \mathcal{N}_T^\partial, \langle \cdot, \cdot \rangle_{\text{CHM}}, P_{h,J}, P_{a,J}^{\gamma^\perp}, P_\gamma, K_{\gamma,J}^{\text{ref}}, \Lambda_{\text{CHM}}, \mathcal{R}_{\text{pole}} \right). \quad (100)$$

It uses the neutral trace definitions of Eqs. (46)–(49) with the source product specialized to  $\langle \cdot, \cdot \rangle_{\text{CHM}}$ . Define the unnormalized traces

$$\tilde{a}_\gamma = P_\gamma a_\gamma^{(0)}, \quad \tilde{z}_J = P_{a,J} \Pi_{\gamma^\perp}^{(T)} z^{(0)}. \quad (101)$$

The normalized traces are

$$a_\gamma = \frac{\tilde{a}_\gamma}{\sqrt{\langle \tilde{a}_\gamma, \tilde{a}_\gamma \rangle_{\text{CHM}}}}, \quad a_J^{\gamma^\perp} = \frac{\tilde{z}_J}{\sqrt{\langle \tilde{z}_J, \tilde{z}_J \rangle_{\text{CHM}}}}. \quad (102)$$

The admissible neutral CHM trace package is the following proof checklist. The projected norms are positive,

$$\langle \tilde{a}_\gamma, \tilde{a}_\gamma \rangle_{\text{CHM}} > 0, \quad \langle \tilde{z}_J, \tilde{z}_J \rangle_{\text{CHM}} > 0. \quad (103)$$

The photon-orthogonal current sample remains orthogonal to the normalized photon trace,

$$\left\langle \frac{\tilde{z}_J}{\sqrt{\langle \tilde{z}_J, \tilde{z}_J \rangle_{\text{CHM}}}}, \frac{\tilde{a}_\gamma}{\sqrt{\langle \tilde{a}_\gamma, \tilde{a}_\gamma \rangle_{\text{CHM}}}} \right\rangle_{\text{CHM}} = 0. \quad (104)$$

The source theory supplies the ordered product  $P_{a,J} \Pi_{\gamma^\perp}^{(T)}$ , or it supplies the commutation theorem

$$[P_{a,J}, \Pi_{\gamma^\perp}^{(T)}] \big|_{\mathcal{N}_T^\partial} = 0. \quad (105)$$

In the current-entry formulas below  $a_J$  abbreviates  $a_J^{\gamma^\perp}$ . The transverse kernel is

$$K_{T,J} = K_{T,J}^{\text{DtN}} + K_{T,J}^{\text{brane}}.$$

Photon-reference subtraction and the source scale are

$$P_\gamma^\dagger (K_{T,J}(0) - K_{\gamma,J}^{\text{ref}}) P_\gamma = 0, \quad \hat{\lambda} = \frac{\lambda}{\Lambda_{\text{CHM}}^2}. \quad (106)$$

With endpoint-normal signs and dimensionful brane terms fixed, define

$$\hat{K}_J^{\text{cur}}(\hat{\lambda}) = \frac{1}{\Lambda_{\text{CHM}}^2} \langle a_J, (K_{T,J}^{\text{DtN}}(\lambda) + K_{T,J}^{\text{brane}}(\lambda) - K_{\gamma,J}^{\text{ref}}(\lambda)) a_J \rangle_{\text{CHM}}, \quad \lambda = \Lambda_{\text{CHM}}^2 \hat{\lambda}. \quad (107)$$

The current kinetic slope is a theorem datum,

$$Z_J^{\text{cur}} = \partial_{\hat{\lambda}} \hat{K}_J^{\text{cur}}(0) > 0. \quad (108)$$

The canonically sloped current kernel is

$$\hat{K}_{J,\text{can}}^{\text{cur}} = (Z_J^{\text{cur}})^{-1} \hat{K}_J^{\text{cur}}, \quad \partial_{\hat{\lambda}} \hat{K}_{J,\text{can}}^{\text{cur}}(0) = 1. \quad (109)$$

The constant entry in the unrescaled normalization is equivalently

$$\hat{\Sigma}_{aa,J}^{\text{CHM}} = \frac{1}{\Lambda_{\text{CHM}}^2} [\langle a_J, K_{T,J}(0) a_J \rangle_{\text{CHM}} - \langle a_\gamma, K_{T,J}(0) a_\gamma \rangle_{\text{CHM}}]. \quad (110)$$

The source-supported CHM statement is the existence of this transverse boundary kernel from the action, gauge fixing, boundary variation, and modified product. The open DeVries-specific theorem target is

$$\hat{K}_{J,\text{can}}^{\text{cur}}(\hat{\lambda}) = \hat{\lambda} + \hat{\Sigma}_{aa,J}^{\text{CHM,can}} + O(\hat{\lambda}^2), \quad \hat{\Sigma}_{aa,2}^{\text{CHM,can}} = 2. \quad (111)$$

The target has ordered source inputs. The CHM source equations must define the neutral transverse trace space, the gauge-sector product on  $\mathcal{N}_T^\partial$ , and the boundary kernels  $K_{T,J}^{\text{DtN}}$  and

$K_{T,J}^{\text{brane}}$ . Photon reference subtraction must fix  $P_\gamma$  and  $K_{\gamma,J}^{\text{ref}}$ . The current projection  $P_{a,2}^{\gamma\perp}$ , the shared source scale  $\Lambda_{\text{CHM}}$ , and the kinetic slope  $Z_2^{\text{cur}}$  are the data required to determine

$$\left\{ P_\gamma, P_{a,2}^{\gamma\perp}, K_{\gamma,2}^{\text{ref}}, \Lambda_{\text{CHM}}, Z_2^{\text{cur}}, \langle \cdot, \cdot \rangle_{\text{CHM}} \right\} \longrightarrow \widehat{\Sigma}_{aa,2}^{\text{CHM,can}}. \quad (112)$$

The acceptance criterion is Eq. (111). The rejection criteria are

$$\widehat{\Sigma}_{aa,2}^{\text{CHM,can}} \neq 2, \quad Z_2^{\text{cur}} \leq 0, \quad \langle a_2^{\gamma\perp}, a_2^{\gamma\perp} \rangle_{\text{CHM}} \leq 0, \quad \langle a_2^{\gamma\perp}, a_\gamma \rangle_{\text{CHM}} \neq 0. \quad (113)$$

These criteria classify the interval construction in the chosen source convention. A rejection outcome leaves a neighboring boundary-kernel model with a different  $\Delta_Y^{\text{CHM}}$ , source scale, or projector normalization. This is the CHM current-entry theorem target. The corresponding source equations are the CHM boundary kinetic equations, modified product, vector Robin equations, scalar  $A_5/\pi_i$  boundary equations, and photon/custodial boundary conditions, specifically the groups recorded in Eqs. (2.11)–(2.23), (2.30)–(2.44), and (3.32)–(3.43) of Ref. [7]. Promotion of Eq. (111) requires the source definitions of  $P_{a,J}^{\gamma\perp}$ ,  $P_\gamma$ ,  $K_{\gamma,J}^{\text{ref}}$ ,  $\Lambda_{\text{CHM}}$ ,  $Z_J^{\text{cur}}$ , endpoint signs, the CHM product, and the pole map in a single convention.

The split assigns source data and theorem data separately. CHM supplies the boundary variation, modified scalar product, vector Robin equations, scalar  $A_5/\pi_i$  boundary equations, and photon/custodial accounting. The DeVries reduction supplies additional theorem data:

$$P_{h,J}, \quad P_{a,J}^{\gamma\perp}, \quad P_\gamma, \quad K_{\gamma,J}^{\text{ref}}, \quad \Lambda_{\text{CHM}}, \quad \mathcal{R}_{\text{pole}}.$$

The interval route requires derivations of the target conditions

$$\widehat{\Sigma}_{aa,2}^{\text{CHM,can}} = 2, \quad \widehat{\mathcal{I}}_{h,3/4}^{\text{CHM}} = \frac{3}{4}, \quad \Delta_Y^{\text{CHM}} = 0, \quad \Delta_W^{\text{CHM}} = 0, \quad \Delta_{\text{O1/O10}} \text{ derived.}$$

For the Target II assignment theorem this current entry has to be paired with a charged order-parameter sample in the same source space:

$$h_J = P_{h,J}(\pi_i, A_5, \delta v_i, \delta R, \alpha_H), \quad (114)$$

$$a_J^{\gamma\perp} = \frac{P_{a,J} \Pi_{\gamma\perp}^{(T)} z^{(0)}}{\sqrt{\langle P_{a,J} \Pi_{\gamma\perp}^{(T)} z^{(0)}, P_{a,J} \Pi_{\gamma\perp}^{(T)} z^{(0)} \rangle_{\text{CHM}}}}, \quad a_\gamma = \frac{P_\gamma a_\gamma^{(0)}}{\sqrt{\langle P_\gamma a_\gamma^{(0)}, P_\gamma a_\gamma^{(0)} \rangle_{\text{CHM}}}}. \quad (115)$$

Here  $\pi_i$  denotes boundary Goldstone data,  $A_5$  the fifth component of the gauge field,  $\delta v_i$  boundary scalar-vev variations,  $\delta R$  radion or interval-size variations, and  $\alpha_H$  a Wilson-line/Hosotani coordinate when the route uses one. The ordered invariant target is

$$\widehat{\mathcal{I}}_{h,J_H}^{\text{CHM}} = J_H = \frac{3}{4}, \quad \widehat{\Sigma}_{aa,J_{\text{adj}}}^{\text{CHM,can}} = J_{\text{adj}} = 2, \quad (116)$$

as target equations using one  $\Lambda_{\text{CHM}}$ , one product, and one pole map.

| Datum          | Projector  | $J$ | Hilbert factor        | Boundary datum                               | Normalization                                                     | Pole functional                  | Status                |
|----------------|------------|-----|-----------------------|----------------------------------------------|-------------------------------------------------------------------|----------------------------------|-----------------------|
| $h_J$          | $P_{h,J}$  | 3/4 | order parameter       | $A_5, \pi_i, \delta v_i, \delta R, \alpha_H$ | $\Lambda_{\text{CHM}}, \langle \cdot, \cdot \rangle_{\text{CHM}}$ | $P_W, \mathcal{R}_{\text{pole}}$ | open sample           |
| $W^\pm$ sample | $P_W$      | 3/4 | charged-sample target | charged trace coupled to $h_J$               | same as $h_J$                                                     | $M_{W,\text{pole}}$ matching     | after open map        |
| $Z$            | $P_Z$      | 2   | neutral current       | $a_J^{\gamma^\perp}$ Robin/DtN data          | same $\Lambda_{\text{CHM}}$                                       | $M_{Z,\text{pole}}$              | current target        |
| $\gamma$       | $P_\gamma$ | 0   | null direction        | photon trace                                 | zero-mode reference traction                                      | sub-zero pole                    | source-supported null |

The data still to be derived are  $P_{h,J}$ ,  $P_{a,J}^{\gamma^\perp}$ ,  $P_W$ ,  $P_Z$ ,  $K_{\gamma,J}^{\text{ref}}$ , the source scale  $\Lambda_{\text{CHM}}$ , endpoint-normal conventions, the photon projection, the ordered W/Z boundary map, and the map  $\widehat{\lambda} \mapsto m_n^2/\Lambda_{\text{CHM}}^2 \mapsto \Pi_T^{(4)}(s)$ .

#### D. Dictionary for the interval entries

The interval construction separates the entries of Eq. (84) into physically addressable data:

| Entry                                                       | Candidate source datum                                    | DeVries-specific obligation                                                                          |
|-------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| $\Sigma_{hh,J}^{\text{int}} = 0$                            | boundary scalar subtraction datum                         | reference or fix the reference by a symmetry, a brane condition, or a normalization convention       |
| $\Sigma_{aa,J}^{\text{int}} = J$                            | boundary kinetic term, or adjoint-current boundary datum  | identify the quadratic invariant that appears in the boundary equation                               |
| $\Sigma_{ha,J}^{\text{int}} \Sigma_{ah,J}^{\text{int}} = J$ | scalar-vector mixing, localized coupling                  | end-derive the invariant from a point/bulk overlap, or brane-normalized overlap or generator product |
| $(J_H, J_{\text{adj}}) = (3/4, 2)$                          | Higgs/order-parameter channel and adjoint/current channel | derive the ordered electroweak sampling rule from the coupled interval problem                       |

The table is intentionally strict. A boundary mass term by itself can produce many secular equations. The DeVries claim requires the trace and determinant data in Eq. (92) and the electroweak ordering of  $J$ -samples.

The possible origins of  $J$  are correspondingly limited:

$$J \in \{C_2(R_{\text{int}}), C_2(R_{\text{bdry}}), h_{\text{CFT}}, q_{\text{KK}}^2, \langle f_{\text{bdry}}, f_{\text{bulk}} \rangle^2\}. \quad (117)$$

Here  $C_2$  denotes a quadratic invariant,  $h_{\text{CFT}}$  a conformal weight or current-algebra datum,  $q_{\text{KK}}$  a compact charge, and the overlap term a normalized boundary/bulk integral. A derivation names one of these entries and derives the equality to  $J$ .

### E. Gauge fields, boundary breaking, and the electroweak ray

Gauge fields on an interval carry a stronger constraint than scalar toy models. Boundary conditions that break gauge symmetry have to arise from the action and preserve the consistency of the lower-dimensional gauge theory. Csaki–Hubisz–Meade use this language for electroweak symmetry breaking by boundary conditions and for Higgsless models [7]. Their framework is relevant here because it gives a controlled setting in which massive vector modes, a photon zero mode, custodial structure, and brane-localized terms can be discussed in one spectral problem.

The DeVries construction requires the boundary model to preserve the electroweak ray. The low-energy gauge-Higgs mass matrix has the form

$$\mathcal{M}_{\text{EW}}^2(v) = v^2 \mathcal{P}(g, g'), \quad (118)$$

where  $v$  is radial and  $\mathcal{P}(g, g')$  is projective. A source-derived interval theorem would produce a boundary operator whose pole condition reduces to this projective datum. Thus the interval model is required to yield a coupled charged/neutral construction:

$$\left[ \begin{array}{c} SU(2)_L \text{ charged sector} \\ SU(2)_L \times U(1)_Y \text{ neutral sector} \\ \text{photon zero mode} \end{array} \right] \longrightarrow \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{match}}^{(\text{int})}. \quad (119)$$

This requirement places the W and Z readings inside one gauge-Higgs or boundary system. The photon must remain the unbroken mode, and the massive neutral pole must carry the

adjoint/current sample. The interval theorem reaches an exact pole statement after the matching remainder  $\Delta_{\text{match}}^{(\text{int})}$  has been derived.

The corresponding proof obligation is

$$S_{5\text{D}} + S_{\text{bdry}} \longrightarrow \{\mathcal{B}_W(\lambda), \mathcal{B}_Z(\lambda), \mathcal{B}_\gamma(\lambda)\} \longrightarrow \text{pole data},$$

$$M_{\gamma, \text{pole}}^2 = \Delta_\gamma^{(\text{int})}, \quad \frac{M_{W, \text{pole}}^2}{M_{Z, \text{pole}}^2} = \frac{x_+(3/4)}{x_+(2)} + \Delta_{\text{match}}^{(\text{int})}. \quad (120)$$

The proof must include the charged and neutral boundary equations, the photon zero mode, custodial conditions, the Ward-identity target  $\Delta_\gamma^{(\text{int})} = 0$ , the reduced inner product, and the pole matching chain.

The interval version of the ordered assignment theorem is:

$$\begin{aligned} &\text{bulk gauge fields} + \text{boundary terms} + \text{order-parameter channel} \\ &\implies K_J^{\text{int}}(\lambda) \implies (J_W, J_Z) = (3/4, 2). \end{aligned} \quad (121)$$

This theorem is parallel to the endpoint theorem in Sec. XI. The two routes can eventually compete: endpoint labels emphasize matrix algebra, while interval boundary conditions emphasize variational origin and pole equations.

## F. Scalar partner from interval data

The negative branch also has a KK address. Witten's review notes that compactification naturally produces scalar degrees of freedom, including radius moduli and possible charged scalar modes tied to symmetry breaking questions [16]. Polchinski's circle compactification likewise identifies the compact radius with a lower-dimensional scalar and shows that changing the radius is tied to current algebra at enhanced-symmetry points [3]. Hosotani's compact gauge-field mechanism adds a Wilson-line scalar route [34]. In interval language, scalar data can appear as boundary values, Wilson-line-like data, brane-localized order parameters, or moduli of the compact geometry [29, 30].

The scalar-branch target is therefore

$$u_-(J) \in \mathcal{H}_J \quad \longmapsto \quad \mathcal{F}_{\text{KK}} \in \{R(x), H^\dagger H, A_5, \text{brane scalar, localized modulus}\}. \quad (122)$$

Here  $u_-(J)$  is the negative-branch eigenvector of the two-channel kernel. A derivation identifies  $\mathcal{F}_{\text{KK}}$ , states its gauge transformation properties, and explains its scheme relation to the low-energy Higgs/order-parameter variables.

The sc/int extension of this target uses a compact boundary-superconnection holonomy,

$$\mathcal{W}_{\text{sc}} = \text{P exp} \int_I (A_5 + \Phi_{\text{odd}}) dy, \quad (123)$$

and tests whether the dimensionless scalar Hessian

$$\widehat{\mathcal{F}}_{\text{sc}}^{\text{Hol/sc}} = \Lambda_J^{-2} \left. \frac{\partial^2 V_{\text{eff}}(\mathcal{W}_{\text{sc}})}{\partial \alpha_{\text{sc}}^2} \right|_{\alpha_*} \quad (124)$$

is the scalar partner of the same projected kernel that gives the hatted current entry and the photon subtraction.

### G. KK completion criterion

The KK route is fixed by the following completion chain:

$$\begin{aligned} \text{compact or interval source} &\longrightarrow \text{boundary variational problem} \longrightarrow K_J^{\text{int}}(\lambda), \\ K_J^{\text{int}}(\lambda) &\longrightarrow \lambda^2 + J\lambda - J \longrightarrow \{s_{\text{pole}}^2, \mathcal{F}_{\text{KK}}\}. \end{aligned} \quad (125)$$

The first arrow is source-backed by the KK and interval literature. The second arrow is the DeVries determinant derivation. The last arrow is the pole-placement and scalar-branch assignment. This chain gives a concrete way to compare compactification, endpoint, and  $G_2$  routes.

### X. TOY ENDPOINT-BOUNDARY OPERATOR

The previous sections list possible mechanisms. This section turns those mechanisms into a minimal toy operator. The toy model is a derivation target with named matrix entries.

Let  $h_J(x)$  denote a boundary-localized order-parameter amplitude and  $a_J(x)$  denote an adjoint/current amplitude. The quadratic light-sector action is

$$S_J^{(2)} = \int d^4x \begin{pmatrix} h_J^\dagger & a_J^\dagger \end{pmatrix} \begin{pmatrix} \lambda & -\kappa_J \\ -\kappa_J & \lambda + \tau_J \end{pmatrix} \begin{pmatrix} h_J \\ a_J \end{pmatrix}. \quad (126)$$

Here  $\lambda$  is the spectral variable after normalization. The DeVries target is

$$\tau_J = J, \quad \kappa_J^2 = J, \quad (127)$$

which gives

$$\det \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} = \lambda^2 + J\lambda - J. \quad (128)$$

The fields  $h_J$  and  $a_J$  are placeholders with fixed jobs. The  $h_J$  channel must carry the doublet/order-parameter invariant when  $J = 3/4$ . The  $a_J$  channel must carry the adjoint/current invariant when  $J = 2$ . The off-diagonal entry  $\kappa_J$  is the physical mixing amplitude between the two channels. The diagonal entry  $\tau_J$  is the trace datum of the adjoint/current side.

### A. Boundary origin

Interval field theory supplies a mechanism for such a matrix. Boundary mass and kinetic terms modify the variational boundary conditions, and brane-localized kinetic terms can make the boundary condition depend on the four-dimensional eigenvalue [7]. A schematic bulk-plus-boundary system is

$$S_{\text{bulk}} = \int d^4x \int_0^L dy \frac{1}{2} \left[ (\partial_M A_J)^2 + (\partial_M H_J)^2 \right] + S_{\text{bdry}}[A_J, H_J]. \quad (129)$$

After solving the bulk equations and restricting to the two light boundary amplitudes, the effective inverse propagator has the form

$$K_J^{\text{eff}}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}(\lambda) & \Sigma_{ha,J}(\lambda) \\ \Sigma_{ah,J}(\lambda) & \lambda + \Sigma_{aa,J}(\lambda) \end{pmatrix}. \quad (130)$$

The DeVries calculation asks for a normalization and a boundary limit in which

$$\Sigma_{hh,J} = 0, \quad \Sigma_{aa,J} = J, \quad \Sigma_{ha,J} \Sigma_{ah,J} = J. \quad (131)$$

This formulation gives each matrix entry a physical address.  $\Sigma_{hh,J}$  tests whether the order-parameter channel can be used as the zero reference.  $\Sigma_{aa,J}$  tests whether the adjoint/current channel supplies the trace  $J$ .  $\Sigma_{ha,J}$  tests whether endpoint, brane, or overlap physics supplies  $\sqrt{J}$ .

### B. Endpoint interpretation

In an open-string or D-brane reading,  $a_J$  is an endpoint current or Chan–Paton-like gauge excitation, while  $h_J$  is a boundary order-parameter mode. Coincident branes naturally give

nonabelian gauge fields from endpoint sectors [2, 3]. The toy model demands a more specific result: the endpoint algebra must reduce to the entries in Eq. (127).

The corresponding route is

$$\begin{aligned}
\text{endpoint labels} &\longrightarrow \kappa_J^2 = J \\
\text{adjoint/current trace} &\longrightarrow \tau_J = J \\
\text{boundary normalization} &\longrightarrow \Sigma_{hh,J} = 0.
\end{aligned} \tag{132}$$

This table defines the calculation. It also gives a compact way to compare candidate completions.

### C. Electroweak matching

The toy operator must match the electroweak projective data. The desired chain is

$$\begin{aligned}
\text{boundary data} &\longrightarrow \mathcal{M}_J \longrightarrow \{x_+(J), x_-(J)\} \\
&\qquad\qquad\qquad \downarrow \\
&\qquad\qquad\qquad \text{W/Z pole ratio and scalar branch.}
\end{aligned} \tag{133}$$

The positive branch should yield the pole-scheme target

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{toy}}, \quad (J_H, J_{\text{adj}}) = (3/4, 2). \tag{134}$$

Here  $\Delta_{\text{toy}}$  records every step missing from the toy model: field normalization, source scale, effective matching, and pole-scheme placement. The negative branch should identify the scalar or boundary partner in the same basis. The toy model therefore joins the three open calculations: ordered electroweak sampling, determinant derivation, and scalar-branch interpretation.

## XI. ENDPOINT GAUGE DATA AND BRANE HIGGSING

The toy operator in Sec. X becomes physical once its entries can be assigned to recognizable string or boundary data. Open strings and D-branes give one sharp language for this assignment. The route follows the original string-theory ambition: a particle spectrum is encoded in worldsheet and endpoint data, and spacetime fields arise as low-energy modes of that spectrum.

### A. Endpoint labels as a source of matrix structure

An open string ending on a stack of  $N$  branes carries endpoint labels. In modern language these labels are Chan–Paton data. The corresponding low-energy fields sit in  $N \times N$  matrices; for coincident branes, the massless excitations form a  $U(N)$  gauge field and adjoint scalar fields on the brane worldvolume [2]. Polchinski’s review states the same logic in compactification language: current algebras of the compact worldsheet theory give space-time gauge bosons, and D-brane boundary conditions give open-string gauge fields along the brane [3].

The relevant point is the matrix origin. Endpoint labels give a natural place for diagonal and off-diagonal channels:

$$|\psi; m, n\rangle, \quad m, n = 1, \dots, N, \quad (135)$$

with fields

$$A_a = (A_a)^m{}_n, \quad \Phi^I = (\Phi^I)^m{}_n. \quad (136)$$

The diagonal entries describe strings whose endpoints lie on the same brane. The off-diagonal entries describe strings stretched between distinct branes. At coincidence the off-diagonal vector modes fill out the nonabelian gauge multiplet. This is precisely the type of two-channel structure needed by Eq. (11).

The source-backed part of the argument ends here. The DeVries-specific step is sharper and remains open: the endpoint algebra must produce the two entries

$$\kappa_J^2 = J, \quad \tau_J = J, \quad (137)$$

with  $J = 3/4$  and  $J = 2$  entering in the ordered electroweak way. The endpoint labels supply a matrix arena; selecting Eq. (137) requires additional dynamics.

### B. Worldvolume Yang–Mills and adjoint scalars

The low-energy action for coincident D-branes contains the nonabelian Yang–Mills field strength and adjoint scalars [2]. Schematically,

$$S_{\text{brane}} = \int d^{p+1}\xi \, \text{Tr} \left[ -\frac{1}{4g_{\text{YM}}^2} F_{ab} F^{ab} - \frac{1}{2} D_a \Phi^I D^a \Phi^I + \frac{1}{4} [\Phi^I, \Phi^J]^2 + \dots \right]. \quad (138)$$

This schematic form is enough for the entry dictionary below. Gauge fields and scalars occupy the same matrix algebra. The covariant derivative

$$D_a \Phi^I = \partial_a \Phi^I + i[A_a, \Phi^I] \quad (139)$$

couples off-diagonal vector modes to scalar expectation values. When two branes are separated, the off-diagonal gauge fields acquire a mass proportional to the separation; equivalently, the mass is the energy of a string stretched between the branes [2]. The brane construction therefore gives a geometric Higgs mechanism with a scalar modulus and charged vector modes in one matrix system.

This structure motivates the endpoint route for the DeVries problem. The positive branch is meant to describe a vector pole-ratio clue. The negative branch asks for a scalar or order-parameter partner. A brane worldvolume action naturally contains both vector and scalar matrix entries. The target is a two-dimensional reduction normalized to

$$\mathcal{M}_J = \begin{pmatrix} 0 & \kappa_J \\ \kappa_J & -\tau_J \end{pmatrix}, \quad \kappa_J^2 = \tau_J = J. \quad (140)$$

Endpoint data therefore supply the matrix arena. The equalities  $\kappa_J^2 = \tau_J = J$ , the projection to  $\mathcal{H}_J$ , and the branch survival rule remain source-kernel theorem data.

### C. Endpoint dictionary for the DeVries entries

The endpoint hypothesis can now be stated as a dictionary. Let  $h_J$  be the boundary order-parameter amplitude and  $a_J$  the endpoint-current amplitude. Let  $\mathcal{H}_J = \text{span}\{h_J, a_J\}$  be the two-dimensional light sector. A candidate derivation must provide three pieces of data:

$$\text{zero reference: } \langle h_J | K_J | h_J \rangle = 0, \quad (141)$$

$$\text{trace entry: } \langle a_J | K_J | a_J \rangle = J, \quad (142)$$

$$\text{mixing entry: } \langle h_J | K_J | a_J \rangle \langle a_J | K_J | h_J \rangle = J. \quad (143)$$

Here  $K_J$  is the normalized quadratic kernel obtained after integrating out heavier worldvolume or bulk modes. Eq. (142) should be traced to a current algebra, adjoint Casimir, or boundary kinetic term. Eq. (143) should be traced to an endpoint overlap, a brane-localized

coupling, or a generator normalization. Eq. (141) fixes the baseline for the order-parameter channel.

The representation data suggest the following table:

| DeVries entry                      | Endpoint or brane origin                                                            | Required derivation                                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| $\tau_J = J$                       | adjoint/current trace, bound-<br>ary kinetic term, or compact<br>current algebra    | show the normalized trace is<br>the relevant quadratic invari-<br>ant                                           |
| $\kappa_J^2 = J$                   | endpoint overlap, stretched-<br>string mixing,<br>brane/order-parameter<br>coupling | show the off-diagonal product<br>or equals the same invariant                                                   |
| $\Sigma_{hh,J} = 0$                | order-parameter<br>channel                                                          | reference justify the subtraction by a<br>symmetry, boundary condi-<br>tion, or renormalization con-<br>vention |
| $(J_H, J_{\text{adj}}) = (3/4, 2)$ | doublet order parameter and<br>adjoint current                                      | derive the ordered W/Z sam-<br>pling rule                                                                       |

This dictionary is restrictive. A completion must determine the entries, the basis, and the normalization from the source theory.

#### D. Scalar branch from brane separation

The brane picture also gives a natural language for the negative branch. In the D-brane Higgs mechanism, scalar expectation values describe brane positions. Off-diagonal vector fields become massive when the branes separate. Near coincidence, the off-diagonal modes become light and the full matrix structure controls the local physics [2]. Thus a two-branch matrix can plausibly pair a vector branch with scalar-modulus data, brane-separation data, auxiliary data, or effective brane-scaling data.

For the DeVries construction, this becomes the following target. There should be a scalar

or boundary modulus  $\varphi_J$  and a normalized kernel  $K_J$  such that

$$K_J|_{\mathcal{H}_J} \sim \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad \varphi_J \text{ is read from } u_-(J). \quad (144)$$

Here  $u_-(J)$  is the eigenvector of the negative eigenvalue. The phrase “read from” is an obligation: it must become a gauge-invariant map from the eigenvector data to a scalar quantity such as  $H^\dagger H$ , a brane separation, a boundary modulus, or a Higgs-potential parameter.

The scalar branch should also respect the electroweak ray. If the compact or brane model is meant to reproduce the Standard Model Higgs mechanism, the radial vacuum parameter  $v$  controls the common mass scale while the DeVries quotient controls the projective direction. A brane-separation interpretation must therefore identify which modulus supplies the radial scale and which normalized matrix supplies the angular data.

### E. Endpoint completion criterion

The endpoint route makes the proposal more constrained. It asks for an explicit reduction:

$$\begin{aligned} \text{endpoint data} &\longrightarrow \text{worldvolume kernel} \longrightarrow \mathcal{M}_J, \\ \mathcal{M}_J &\longrightarrow \{x_+(J), x_-(J)\} \longrightarrow \text{pole ratio and scalar branch.} \end{aligned} \quad (145)$$

Each arrow has a check. The first arrow is source-supported by Chan–Paton labels, D-brane gauge fields, and adjoint scalars. The second arrow is the DeVries-specific determinant derivation. The third arrow is the exact algebra of Sec. II. The last arrow is the pole-placement and negative-branch problem. Thus the endpoint route currently supplies matrix fields and a brane-Higgsing arena. Completion requires a worldvolume quadratic kernel, a projection to  $\mathcal{H}_J$ , an inner product, and a normalization deriving  $\kappa_J^2 = \tau_J = J$  from the endpoint source theory.

This is also where the seventies lineage becomes substantive. The original dual amplitude began as a spectral object for hadron physics, and open-string amplitudes expose a tower of particle poles [2]. The technical pattern is a spectral clue, followed by the worldsheet or boundary object that produces it, followed by translation into the electroweak pole scheme. The general string-theory idea that particle data can arise from endpoint and compactification spectra is source-backed; the electroweak translation is speculative.

## XII. $G_2$ LOCALIZATION AND LOCAL SPECTRAL OPERATORS

The  $G_2$  route supplies a local language for the DeVries program. Its role in this paper is local. It gives a framework in which gauge fields, charged matter, order-parameter data, and deformation modes can live at singular loci of a compactification. M-theory on  $G_2$ -holonomy spaces is a standard route to four-dimensional  $\mathcal{N} = 1$  structure; singularities provide the gauge and matter sectors needed for particle physics [17–19]. The DeVries question is whether a localized  $G_2$  operator can reduce to the same two-channel determinant used in the pole-ratio clue.

### A. Singular support for gauge and matter sectors

The source-backed starting point is the localization of degrees of freedom. Witten explains that  $G_2$  compactifications with singularities can support abelian or nonabelian gauge groups with chiral fermions, with anomaly inflow from the bulk supplying a consistency condition [17]. Acharya and Witten describe examples where singular loci carry nonabelian gauge groups and conical or intersection points support chiral multiplets [18]. Acharya and Gukov review the broader picture: singular  $G_2$  spaces are the essential setting for nonabelian gauge groups, chiral fermions, and strongly coupled gauge dynamics in M-theory compactifications [19].

The DeVries problem asks for a local sector with three named pieces:

$$\left( \begin{array}{c} \text{singular gauge locus} \\ \text{localized charged matter or order parameter} \\ \text{deformation mode of the singularity} \end{array} \right) \rightsquigarrow \mathcal{M}_J. \quad (146)$$

The singular gauge locus is the natural address for adjoint/current data. The localized charged matter or order-parameter mode is the natural address for the doublet side. The deformation mode is the natural address for the scalar branch.

This local focus also controls the topology issue. A compact  $G_2$  model imposes global constraints, including anomaly, representation, and line-operator data. The determinant can be tested locally first. A global compactification can then be asked to support the same local operator and the Standard Model global form.

## B. Higgs-bundle variables on an associative three-cycle

Modern local  $G_2$  gauge sectors can be organized by a Higgs-bundle-like system on an associative three-cycle  $M_3$ . Braun, Cizel, Hubner, and Schafer-Nameki describe a gauge field together with a one-form Higgs field on  $M_3$ , with matter localized at critical loci and interactions controlled by gradient-flow data [20]. In the abelianized local description, the Higgs field can be written schematically as

$$\Phi = \sum_{i=1}^n t_i df_i, \quad \rho = \sum_{i=1}^n t_i \rho_i, \quad \Delta f_i = \rho_i, \quad (147)$$

where  $t_i$  are Cartan generators and the  $\rho_i$  encode charge distributions. For a representation with charge vector  $Q = (q_1, \dots, q_n)$ , the effective Morse function is

$$f_Q = \sum_{i=1}^n q_i f_i, \quad \rho_Q = \sum_{i=1}^n q_i \rho_i. \quad (148)$$

Matter in the representation  $R_Q$  localizes at points where

$$df_Q(p) = 0. \quad (149)$$

Flow lines between such critical points can generate mass terms, and trivalent flow trees can generate Yukawa couplings when the representation charges sum to zero [20]. The resulting local dictionary expresses gauge data, matter localization, mass terms, and interactions as functions of the same geometric input.

For the DeVries construction, the Higgs-bundle variables give a possible address for the local operator:

$$\mathcal{O}_J^{G_2} = \mathcal{O}_J(W, \Phi, \rho_Q, f_Q), \quad (150)$$

where  $W$  is the gauge field on  $M_3$ . The precise form of  $\mathcal{O}_J^{G_2}$  is open. The variables  $W$ ,  $\Phi$ , and  $f_Q$  already separate adjoint gauge data, scalar deformation data, and localized representation data.

The same Morse data suggest a Hodge/SUSY-QM refinement. Witten's construction uses  $d_t = e^{-th} de^{th}$ , its adjoint, and the associated deformed Laplacian to localize low modes near critical points of a Morse function [36]. In the local  $G_2$  setting,  $f_Q$  is already the source-backed Morse function for matter localization. A candidate DeVries operator may therefore begin with a finite reduction of  $(\Omega^0, \Omega^1, d_t, d_t^\dagger)$  on the singular support. The local derivation must then supply the breaking entry, the two-channel projection, and the electroweak assignment inside the same  $G_2$  model.

### C. Local two-channel kernel

Let  $a_J$  be an adjoint/current fluctuation supported along the singular gauge locus, and let  $h_J$  be a localized order-parameter fluctuation at a critical point of  $f_Q$ . Let

$$\mathcal{H}_J^{G_2} = \text{span}\{h_J, a_J\}. \quad (151)$$

After integrating out heavier localized and bulk modes, the local inverse propagator or spectral kernel has the same abstract form as the endpoint and interval kernels:

$$K_J^{G_2}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}^{G_2}(\lambda) & \Sigma_{ha,J}^{G_2}(\lambda) \\ \Sigma_{ah,J}^{G_2}(\lambda) & \lambda + \Sigma_{aa,J}^{G_2}(\lambda) \end{pmatrix}. \quad (152)$$

The DeVries target is

$$\Sigma_{hh,J}^{G_2} = 0, \quad \Sigma_{aa,J}^{G_2} = J, \quad \Sigma_{ha,J}^{G_2} \Sigma_{ah,J}^{G_2} = J. \quad (153)$$

The source-equation check for this target is explicit:

$$\Phi = \sum_i t_i df_i, \quad \rho = \sum_i t_i \rho_i, \quad \Delta f_i = \rho_i, \quad (154)$$

and

$$f_Q = \sum_i q_i f_i, \quad \rho_Q = \sum_i q_i \rho_i, \quad df_Q(p) = 0. \quad (155)$$

The current entry  $\Sigma_{aa,J}^{G_2} = J$  is tested by the following local ADE-pairing protocol. Fix a source datum

$$u_{G_2}^{\text{loc}} = (M_3, \Gamma_{\text{ADE}}, W, \Phi, \rho_i, f_i, Q, p, \gamma, A_1 \subset A_2, \Lambda_{G_2}), \quad (156)$$

where the  $A_1 \subset A_2$  enhancement records the local root-pairing test

$$\text{ad } A_2 \downarrow_{A_1 \times U(1)} = (\text{ad } A_1)_0 \oplus \mathbf{1}_0 \oplus \mathbf{2}_{+q} \oplus \mathbf{2}_{-q}. \quad (157)$$

The singular support carries the pairing

$$\langle \eta, \zeta \rangle_{G_2, u} = \int_{M_3} \text{Tr}_{\text{ADE}}(\eta \wedge *_{M_3} \zeta) + \sum_{p \in Z(df_Q)} \eta(p)^\dagger \mathbf{G}_Q(p) \zeta(p), \quad (158)$$

$$Z(df_Q) = \{p \in M_3 \mid df_Q(p) = 0\}.$$

The first term is the current-support pairing on the ADE locus. The point support term is the localized matter metric required by the later flow-overlap entry. The metric  $\mathbf{G}_Q(p)$  is

theorem data: it must be positive on the localized charge sector, gauge covariant, compatible with the charge vector  $Q$ , and reducible to the wavefunction or Hessian normalization at an isolated Morse critical point. A Morse–Bott critical component replaces the point metric by the corresponding component metric and flow-moduli convention.

The local off-diagonal target is a flow-overlap theorem. Let  $\psi_{h,J}^{G_2}$  denote the localized order-parameter wavefunction and let  $a_J^{G_2}$  denote the projected ADE current fluctuation. Their source normalizations are

$$\langle \psi_{h,J}^{G_2}, \psi_{h,J}^{G_2} \rangle_{G_Q} = 1, \quad \langle a_J^{G_2}, a_J^{G_2} \rangle_{G_2,u} = 1. \quad (159)$$

For an oriented flow line or flow-tree contribution  $\gamma$ , introduce a flow kernel  $\mathcal{K}_{\text{flow}}^{G_2}(\gamma)$ , orientation sign  $\epsilon_\gamma$ , and bilinear maps  $B_{ha}^{G_2}$  and  $B_{ah}^{G_2}$ . The hatted off-diagonal entries are theorem data:

$$\begin{aligned} \widehat{\Sigma}_{ha,J}^{G_2} &= \Lambda_{G_2}^{-1} \sum_{\gamma} \epsilon_{\gamma} B_{ha}^{G_2}(\psi_{h,J}^{G_2}, \mathcal{K}_{\text{flow}}^{G_2}(\gamma) a_J^{G_2}), \\ \widehat{\Sigma}_{ah,J}^{G_2} &= \Lambda_{G_2}^{-1} \sum_{\gamma} \epsilon_{\gamma} B_{ah}^{G_2}(a_J^{G_2}, \mathcal{K}_{\text{flow}}^{G_2}(\gamma) \psi_{h,J}^{G_2}). \end{aligned} \quad (160)$$

The flow-overlap target is

$$\widehat{\Sigma}_{ha,J}^{G_2} \widehat{\Sigma}_{ah,J}^{G_2} = J. \quad (161)$$

The source datum is the existence of localized modes at  $df_Q = 0$ , oriented flow-line mass terms, and trivalent flow-tree interaction data. The theorem specifies how the quadratic flow-line bilinear is extracted from those data. The DeVries kernel then uses the product in Eq. (161). The source-backed two-point object in Braun–Cizel–Hubner–Schafer–Nameki is the Morse–Witten mass matrix

$$M_{\text{MW}}^{ab} = \sum_{\gamma} n_{\gamma} \exp[-tq(f(p_a) - f(p_b))], \quad n_{\gamma} = \pm 1, \quad (162)$$

acting between localized chiral multiplet ground states. Its Morse–Bott extension keeps the same flow-line form for point-to-component trajectories and gives a zero matrix element for two states localized on the same critical circle. The current source package therefore supplies a narrower test,

$$M_{\text{MW}}^{ab} \rightsquigarrow (M_{ha}^{\text{MW}}, M_{ah}^{\text{MW}}) \quad \text{inside} \quad \mathcal{H}_J^{G_2} = \text{span}\{h_J^{G_2}, a_J^{G_2}\}. \quad (163)$$

The source test isolates the failed derivation step: the Braun source data lack an identification of the projected ADE current  $a_J^{G_2}$  as a Morse–Witten endpoint state in this two-channel

block. Target IIIh therefore remains active only after an additional same-basis projection theorem supplies  $M_{ha}^{\text{MW}}$ ,  $M_{ah}^{\text{MW}}$ ,  $\Lambda_{G_2}$ , and the orientation convention.

The local current test chooses a projected adjoint fluctuation  $a_J^{G_2}$  with

$$\langle a_J^{G_2}, a_J^{G_2} \rangle_{G_2, u} = 1 \quad (164)$$

and defines

$$\begin{aligned} \widehat{K}_J^{G_2, \text{cur}}(\widehat{\lambda}) &= \Lambda_{G_2}^{-2} \langle a_J^{G_2}, \mathcal{K}_{\text{cur}}^{\text{ADE}}(\lambda) a_J^{G_2} \rangle_{G_2, u}, \\ \lambda &= \Lambda_{G_2}^2 \widehat{\lambda}. \end{aligned} \quad (165)$$

Here  $\mathcal{K}_{\text{cur}}^{\text{ADE}}$  denotes the projected quadratic ADE-current operator induced on the singular gauge locus, including source scale, projection, and possible singular-support boundary terms. Its detailed construction from the local  $G_2$  gauge theory is part of the theorem data. The canonical current kernel is fixed by

$$Z_J^{G_2} = \partial_{\widehat{\lambda}} \widehat{K}_J^{G_2, \text{cur}}(0) > 0, \quad \widehat{K}_J^{G_2, \text{cur}, \text{can}} = (Z_J^{G_2})^{-1} \widehat{K}_J^{G_2, \text{cur}}. \quad (166)$$

The target is

$$\widehat{K}_J^{G_2, \text{cur}, \text{can}}(\widehat{\lambda}) = \widehat{\lambda} + \widehat{\Sigma}_{aa, J}^{G_2, \text{can}} + O(\widehat{\lambda}^2), \quad \widehat{\Sigma}_{aa, 2}^{G_2, \text{can}} = 2. \quad (167)$$

This local test precedes the off-diagonal flow-overlap product and the electroweak pole map. Compact global completion, anomaly inflow, the charge lattice, hypercharge embedding, and pole matching remain theorem data. Appendix C 7 states the same protocol as Target IIIg.

#### D. Dictionary for the $G_2$ entries

The local  $G_2$  dictionary is:

| DeVries entry                                 | $G_2$ origin                                                                                                                                                                                             | Derivation target |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| $\Sigma_{aa,J}^{G_2} = J$                     | ADE singular gauge sector with derive the canonical current $A_1 \subset A_2$ enhancement and entry $\widehat{\Sigma}_{aa,2}^{G_2,\text{can}} = 2$<br>singular-support pairing                           |                   |
| $\Sigma_{ha,J}^{G_2} \Sigma_{ah,J}^{G_2} = J$ | localized matter at $df_Q$ = derive the flow-overlap prod-<br>0, flow-line coupling, bifun-uct from the point-support<br>damental vev, or singularity- metric and local wavefunc-<br>deformation mixing  | tions             |
| $\Sigma_{hh,J}^{G_2} = 0$                     | reference order-parameter chan- justify the zero entry through<br>nel at a critical point of $f_Q$ local normalization or sym-<br>metry of the critical-point<br>problem                                 |                   |
| $(J_H, J_{\text{adj}}) = (3/4, 2)$            | localized doublet/order- derive $P_W, P_Z, P_\gamma$ , hyper-<br>parameter data, adjoint/current charge, charge lattice, and<br>data, and photon-orthogonal pole map in the same com-<br>completion data | pact model        |

This table turns  $G_2$  geometry into a determinant problem. Each geometric object must provide one entry of  $K_J^{G_2}$ , one assignment rule, or one consistency check. A  $G_2$  route that supports the pole-ratio claim has to derive at least one of  $\Sigma_{aa,J}^{G_2}$ ,  $\Sigma_{ha,J}^{G_2} \Sigma_{ah,J}^{G_2}$ , or  $\Sigma_{hh,J}^{G_2}$  from Higgs-bundle or localized-singularity data with a stated inner product. A  $G_2$  route enters the dimensional interpolation after the local operator is embedded into a compact model whose chiral spectrum, anomaly inflow, global form, and electroweak charge normalization are stated.

The normalization target includes the local inner product on the two-channel space,

$$\mathcal{H}_J^{G_2} = \text{span}\{h_J^{G_2}, a_J^{G_2}\}, \quad \langle h_J^{G_2}, a_J^{G_2} \rangle_{G_2}, \quad (168)$$

where the pairing is the singular-support form of Eq. (158). The current-entry derivation identifies the gauge-field fluctuation  $W$  along the singular gauge locus and then evaluates

$$W|_{M_3} \rightsquigarrow \widehat{K}_J^{G_2,\text{cur},\text{can}} \rightsquigarrow \widehat{\Sigma}_{aa,2}^{G_2,\text{can}} = 2. \quad (169)$$

The rejection outputs are

$$\widehat{\Sigma}_{aa,2}^{G_2,\text{can}} \neq 2, \quad Z_2^{G_2} \leq 0, \quad \langle P_{a,2}^{G_2} a_2, P_{a,2}^{G_2} a_2 \rangle_{G_2,u} \leq 0, \quad (170)$$

or an unresolved extra current channel. These outputs classify the local ADE pairing before a compactification or pole claim is made. The off-diagonal-product derivation identifies the localized matter and Higgs-bundle data,

$$(\Phi, \rho_Q, f_Q, \psi_{h,J}^{G_2}, \mathcal{K}_{\text{flow}}^{G_2}, \epsilon_\gamma) \rightsquigarrow \widehat{\Sigma}_{ha,J}^{G_2} \widehat{\Sigma}_{ah,J}^{G_2} = J, \quad (171)$$

through the bilinear definitions of Eq. (160). Trivalent flow trees supply interaction terms; the quadratic kernel requires the two-point flow-line bilinear and its same-basis normalization. The rejection outputs are

$$\widehat{\Sigma}_{ha,J}^{G_2} \widehat{\Sigma}_{ah,J}^{G_2} \neq J, \quad \mathbf{G}_Q \leq 0, \quad (172)$$

orientation sum lacks a source convention,      extra coupled light channel.

The order-parameter reference is derived from the critical-point normalization of  $f_Q$ ,

$$df_Q(p) = 0 \rightsquigarrow \Sigma_{hh,J}^{G_2} = 0. \quad (173)$$

This is the local  $G_2$  analogue of the interval Schur-complement target: the source theory must provide the three named entries before the singularity language has derivational force.

### E. Higgsing as singular deformation

Acharya and Witten give a pattern directly relevant here: in examples where singular loci meet, gauge groups live on the singular components, chiral multiplets can be supported at the meeting point, and Higgsing corresponds to a geometric deformation or smoothing of the singularity [18]. In their brane-related examples, a vev for a bifundamental field can break a product gauge symmetry to a diagonal subgroup, while the local singular geometry changes accordingly.

For DeVries, this suggests the deformation chain

$$\begin{aligned} \text{intersecting singular loci} &\longrightarrow \text{localized chiral/order-parameter mode} \\ &\longrightarrow \text{deformed singularity} \longrightarrow K_J^{G_2}(\lambda). \end{aligned} \quad (174)$$

The scalar branch would then be read from the deformation direction. If  $u_-(J)$  is the negative-branch eigenvector in  $\mathcal{H}_J^{G_2}$ , the target scalar map is

$$\begin{aligned} u_-(J) &\longmapsto \mathcal{F}_{G_2}, \\ \mathcal{F}_{G_2} &\in \left\{ \begin{array}{l} \text{deformation parameter, } f_Q(p), \rho_Q, \\ \text{bifundamental vev, localized modulus} \end{array} \right\}. \end{aligned} \quad (175)$$

The map must also respect the electroweak ray. The deformation parameter can supply the scalar side, while the pole-ratio branch supplies the projective vector side.

### F. Anomaly and global checks

The  $G_2$  route has consistency tests. Witten’s anomaly-inflow discussion links local singularity data to four-dimensional chiral matter and anomaly cancellation [17]. Therefore a local DeVries operator has to pass the following checks:

1. the singular gauge locus supports the required adjoint sector;
2. localized charged matter has the desired representation content;
3. anomaly inflow or its local analogue is compatible with the charged fields;
4. the global gauge-group form of Sec. XV is compatible with the local representations;
5. flavor labels remain separate from compactification topology until a representation map is derived.

These checks can reject a candidate completion even when the two-channel determinant is obtained locally.

### G. $G_2$ completion criterion

The  $G_2$  route has the same completion chain as the endpoint and interval routes:

$$\begin{aligned} \text{singular } G_2 \text{ neighborhood} &\longrightarrow (W, \Phi, \rho_Q, f_Q) \longrightarrow K_J^{G_2}(\lambda), \\ K_J^{G_2}(\lambda) &\longrightarrow \lambda^2 + J\lambda - J \longrightarrow \{s_{\text{pole}}^2, \mathcal{F}_{G_2}, \mathcal{C}_{\text{top}}\}. \end{aligned} \tag{176}$$

The first arrow is source-backed by singular  $G_2$  compactification and Higgs-bundle literature. The second arrow is the DeVries determinant derivation. The final arrow is the pole-placement, scalar-branch, and global-form assignment. The local determinant is the first compatibility condition; compact global data then supply anomaly, representation, and line-operator tests.

### XIII. COMPARISON OF THE THREE LOCAL ROUTES

The endpoint, interval, and  $G_2$  sections introduce three ways to place the DeVries determinant inside a theory with localized gauge and order-parameter data. The section casts the three routes in a common reduction format. Each route is tested by the same normalized two-channel kernel, electroweak assignment, branch interpretation, and global compatibility conditions. A derivation in any route must supply a physical origin for the matrix entries and turn the string, extra-dimensional, or singular-geometric address into a calculation.

#### A. Common reduction problem

Let  $\mathcal{S}_r$  denote one of the three route descriptions,

$$r \in \{\text{end}, \text{int}, G_2\}, \quad \mathcal{S}_r = \begin{cases} \mathcal{S}_{\text{end}} & \text{open-string endpoint or D-brane sector,} \\ \mathcal{S}_{\text{int}} & \text{interval Kaluza-Klein boundary sector,} \\ \mathcal{S}_{G_2} & \text{localized singularity sector.} \end{cases} \quad (177)$$

The desired reduction chooses a two-dimensional light subspace

$$\mathcal{H}_{J,r} = \text{span}\{h_{J,r}, a_{J,r}\}, \quad (178)$$

where  $h_{J,r}$  is the order-parameter channel and  $a_{J,r}$  is the adjoint/current channel. In this basis the reduced spectral kernel has the form

$$K_{J,r}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}^r(\lambda) & \Sigma_{ha,J}^r(\lambda) \\ \Sigma_{ah,J}^r(\lambda) & \lambda + \Sigma_{aa,J}^r(\lambda) \end{pmatrix}. \quad (179)$$

The DeVries target is the simultaneous limit and normalization

$$\Sigma_{hh,J}^r = 0, \quad \Sigma_{aa,J}^r = J, \quad \Sigma_{ha,J}^r \Sigma_{ah,J}^r = J. \quad (180)$$

The available source equations select a candidate entry sequence. The interval/CHM route begins with the admissible neutral trace package of Eqs. (103)–(108). That package supplies positive projected norms, photon-orthogonal normalization, source-ordered or commuting projectors, photon-reference subtraction, a shared  $\Lambda_{\text{CHM}}$ , and the current-slope datum  $Z_J^{\text{cur}}$ . Its first theorem target is

$$\widehat{\Sigma}_{aa,2}^{\text{CHM,can}} = 2. \quad (181)$$

Sec. IX states the corresponding acceptance criteria in terms of the CHM source equations and projector data. The same source package then carries

$$\Delta_W^{\text{CHM}}, \quad \Delta_Y^{\text{CHM}}, \quad \Sigma_{ha,J} \Sigma_{ah,J} = J, \quad \Delta_{\text{O1/O10}}. \quad (182)$$

The Hodge/SUSY-QM route supplies the companion square-root theorem target. The  $G_2$  route provides Target IIIg, a local ADE-pairing current test whose first acceptance equation is  $\widehat{\Sigma}_{aa,2}^{G_2,\text{can}} = 2$ , and Target IIIh, a local flow-overlap product target for  $\widehat{\Sigma}_{ha,J}^{G_2} \widehat{\Sigma}_{ah,J}^{G_2} = J$ . Its electroweak projectors, charge lattice, and pole map remain open obligations. The endpoint/Chan–Paton route supplies a matrix arena whose DeVries use requires a world-volume quadratic kernel, projection, and normalization. Equation (180) is stronger than a qualitative mechanism. It assigns one geometric, endpoint, or boundary object to each matrix entry. It also fixes the algebraic route to

$$\det K_{J,r}(\lambda) = \lambda^2 + J\lambda - J, \quad \lambda \in \{x_+(J), x_-(J)\}. \quad (183)$$

The route is fully specified when one source package supplies the reduced space, source product, entry kernels, scalar functional, and pole map, and derives the route-indexed arrows in

$$\begin{array}{ccc} (u_r, \mathcal{H}_{J,r}) & \longrightarrow & (\langle \cdot, \cdot \rangle_r, \Lambda_r, P_{h,J}^r, P_{a,J}^r, K_{\text{cur}}^r, K_{\text{mix}}^r) \\ \downarrow & & \downarrow \\ (\mathcal{F}_{\text{sc}}^r, \mathcal{R}_{\text{pole}}^r) & \longrightarrow & (\widehat{\Sigma}_{aa,J}^r, \widehat{\Sigma}_{ha,J}^r, \widehat{\Sigma}_{ah,J}^r, P_W^r, P_Z^r, P_\gamma^r, \Delta_{\text{match}}^{(r)}). \end{array} \quad (184)$$

The current proof obligations include the canonical CHM neutral-current entry  $\widehat{\Sigma}_{aa,2}^{\text{CHM,can}} = 2$ , the  $G_2$  ADE current entry  $\widehat{\Sigma}_{aa,2}^{G_2,\text{can}} = 2$ , the same-basis Hodge/SUSY-QM or  $G_2$  flow-overlap product  $\widehat{\Sigma}_{ha,J}^r \widehat{\Sigma}_{ah,J}^r = J$ , and the ordered projector package  $P_W^r, P_Z^r, P_\gamma^r$  with its pole remainder.

The reduction has three candidate arenas. The endpoint arena is supported by the open-string fact that endpoint labels and coincident branes generate matrix-valued gauge fields, with brane positions supplying scalar data [2, 3]. The interval arena is supported by extra-dimensional models in which boundary conditions and brane-localized terms set vector spectra [7]. The  $G_2$  arena is supported by singular compactifications that localize gauge sectors and chiral matter, together with Higgs-bundle variables organizing local matter and interaction data [17–20]. These sources support the listed ingredients. Selection of the two-channel subspace, inner product, normalization, and entries in Eq. (180) remains the derivation target.

The interval arena supplies source-derived boundary data. The CHM source equations fix the boundary kinetic normalization, the corresponding modified product, vector Robin data from boundary scalar vevs, scalar  $A_5/\pi_i$  boundary equations, and photon/custodial accounting. Sec. IX states the convention data. In compact route notation,

$$\partial_y f_\lambda = -(\lambda/M_i)f_\lambda, \quad (f, g)_{\text{CHM}} = \int fg + \sum_i M_i^{-1} f(y_i)g(y_i), \quad \partial_y A_\mu \mp v_i^2 A_\mu = 0. \quad (185)$$

Thus the source-backed statement is

$$\text{CHM boundary data} \implies K_{\text{int}}(\lambda) \text{ exists,} \quad K_{\text{int}}(\lambda) \implies \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} \text{ is open.} \quad (186)$$

Equation (185) supplies boundary ingredients; the projection to  $\mathcal{H}_J$  and the  $J$ -normalization are theorem data. The interval route has explicit vector and scalar boundary equations for an entry-level source target. The entry tests have two linked forms: Hodge/SUSY-QM factorization tests  $\Sigma_{ha,J}\Sigma_{ah,J} = J$ , and the canonical CHM boundary-current test  $\hat{\Sigma}_{aa,2}^{\text{CHM,can}} = 2$ . Sec. IX C defines the neutral trace package, photon subtraction,  $Z_J^{\text{cur}}$ , canonical current kernel, and target equation. The CHM equations supply the boundary variation, product, vector Robin data, scalar  $A_5/\pi_i$  equations, custodial accounting, and photon-zero accounting. The projection  $P_{a,J}$ , reference subtraction, source scale  $\Lambda_{\text{CHM}}$ , endpoint-normal conventions, ordered W/Z boundary map, and pole map remain theorem data. The parent KK-fixing notes sharpen the same issue into a single-source test: the dimensional middle line, negative-branch scalar reading, and electromagnetic endpoint can be combined when one source variable supplies

$$u \mapsto (t_{\text{dim}}(u), t_{\text{EW}}(u), K_J(u, \lambda), \mathcal{F}_{\text{sc}}(u; J)). \quad (187)$$

The scalar and electromagnetic readings are downstream of the current entry, ordered W/Z assignment, source-to-pole calculation, and photon-zero normalization. Target C 11 records the O17 endpoint coupling package after those data are supplied.

The common reduction is filtered by three admissibility tests. First, trace and coefficient statements stay inside the matrix where they are derived until a common source operator and field normalization identify the trace spaces. Second, Wigner–Eckart or Clebsch routes must specify the parent representation, operator tensor type, projection, and normalization. The tested doublet-parent route is removed by the ordinary selection rule. The triplet axial route

gives the wrong parent-Casimir datum for Eq. (180). Third, an  $\text{SO}(32)$ -flavor completion must show how its representation sector couples to the electroweak kernel, with extra-state projection and anomaly constraints stated before any flavor claim is used in the mass-ratio argument.

For the superconnection/CHM route, these filters have concrete meanings. The odd Higgs trace space, the even transverse-current trace space, and the interval boundary scalar product stay separated until one source operator defines their common inner product. The Wigner–Eckart route is a failed subroute, so the superconnection route must supply its own projection. The  $\text{SO}(32)$  material remains a flavor-boundary sector until a derived coupling inserts that representation sector into the electroweak kernel.

## B. Entry-by-entry comparison

The first entry is the diagonal current contribution,

$$a_{J,r} \rightsquigarrow \Sigma_{aa,J}^r = J. \quad (188)$$

The candidate source objects are a Chan–Paton or brane gauge amplitude, a boundary value of a bulk gauge mode, or a fluctuation along a singular gauge locus.

The second entry is the off-diagonal product,

$$h_{J,r} \leftrightarrow a_{J,r}, \quad \Sigma_{ha,J}^r \Sigma_{ah,J}^r = J. \quad (189)$$

The candidate source objects are a stretched open string or endpoint overlap, a boundary overlap between an order parameter and a bulk gauge profile, or a localized matter/deformation coupling.

The third entry is the order-parameter reference contribution,

$$h_{J,r} \rightsquigarrow \Sigma_{hh,J}^r = 0. \quad (190)$$

The candidate source objects are a boundary reference channel, a normalized boundary order-parameter self-energy, or a critical-point normalization of a localized mode. Appendix C 7 records the entry-by-entry proof obligations.

The following table records the three entries in a common dictionary.

| Kernel entry                 | Endpoint/D-brane route                             | Interval KK route                                                                                                                                                                                              | $G_2$ route                                                       |
|------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| $\Sigma_{aa} = J$            | Chan–Paton current or CHM<br>brane gauge amplitude | boundary Target IIIg:<br>current-kernel target $A_2$ ADE current pairing<br>with source conven- with $\widehat{\Sigma}_{aa,2}^{G_2,\text{can}} = 2$<br>tions; $J$ -normalization<br>open                       | $A_1 \subset$                                                     |
| $\Sigma_{ha}\Sigma_{ah} = J$ | stretched string or end-<br>point overlap          | boundary scalar vev, Target IIIh:<br>$A_5/\pi_i$ equation, and matter at $df_Q = 0$ , ori-<br>modified product; open ended flow kernel, bi-<br>product normalization linear maps, and point-<br>support metric | localized                                                         |
| $\Sigma_{hh} = 0$            | order-parameter refer-<br>ence channel             | boundary scalar refer-<br>ence term; open sub-<br>traction rule                                                                                                                                                | critical-point normal-<br>ization with local-to-<br>pole matching |

### C. Electroweak assignment as a route-independent target

The same electroweak assignment target applies to every route. The low-energy field-theory data contain an  $SU(2)_L$  Higgs doublet and adjoint gauge currents. Their available quadratic invariants give

$$J_H = \frac{3}{4}, \quad J_{\text{adj}} = 2. \quad (191)$$

The target map is

$$\mathcal{A}_r : (\mathcal{S}_r, \mathcal{H}_{J,r}) \longrightarrow (J_W, J_Z) = (J_H, J_{\text{adj}}). \quad (192)$$

The ordered assignment is a theorem target. The observed W and Z vector bosons are gauge-sector particles after symmetry breaking. The quotient assigns the charged comparison to the order-parameter invariant and the neutral comparison to the adjoint/current invariant. The required derivation must produce this ordered pair from the gauge-Higgs coupling structure, endpoint boundary labels attached to the Higgsing field, interval boundary data that distinguish charged and neutral sectors, or a localized singularity model where doublet and

adjoint data meet. The Standard Model mass matrix supplies the projective ratio and the representation data; the ordered sampling maps remain additional theorem data.

If the ordered assignment and pole-placement theorem hold, the pole-ratio reading has one route-labeled form:

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{match}}^{(r)}. \quad (193)$$

Here  $r$  labels the endpoint, interval, or  $G_2$  route. The exact pole reading requires a derived  $\Delta_{\text{match}}^{(r)} = 0$  in the chosen pole scheme, or a computed remainder whose size and sign are fixed by the source-to-EFT-to-pole chain. A compactification-scale or unification-scale derivation makes the quotient a boundary datum; the pole claim then requires the matching chain.

#### D. Branch interpretation and scalar data

Every route diagonalizes the same two-channel kernel. Therefore each route has to account for both eigenvectors. Let

$$u_{\pm,J,r} = \alpha_{\pm,J,r} h_{J,r} + \beta_{\pm,J,r} a_{J,r} \quad (194)$$

be normalized eigenvectors of the route kernel in the DeVries limit. The positive branch supplies the pole-ratio clue. The negative branch supplies the scalar-side question,

$$u_{-,J,r} \rightsquigarrow \mathcal{F}_{\text{sc},r}. \quad (195)$$

The functional  $\mathcal{F}_{\text{sc},r}$  depends on the route. Endpoint, interval, and  $G_2$  sources provide candidate scalar addresses: brane or boundary moduli, localized scalar modes, and singularity-deformation data. These addresses acquire physical content after a gauge-invariant functional, normalization, and electroweak scheme are specified. Appendix C 8 states the same-source condition: the first five entries of the scalar package must agree with the vector pole package, including  $\mathcal{H}_J$ , the inner product, the projection,  $\Lambda_J$ , and  $\mathcal{R}_{\text{pole}}$ .

The scalar map has to be tied to the electroweak ray. The gauge-Higgs system has a radial vacuum scale and a projective vector-mass ratio. The two-root structure is physically restrictive because a determinant used for the pole clue also supplies the partner branch. A source route must identify how the positive branch fixes the projective vector datum and how the negative branch enters the radial or scalar sector.

## E. Scale placement

The route comparison separates three scale placements:

$$\text{pole scale,} \quad \text{compactification or boundary scale,} \quad \text{unification scale.} \quad (196)$$

The construction attaches the electroweak clue to the pole scale. The attachment gives the clue its direct electroweak content. The GUT lineage supplies the comparison class: a clean weak-angle value can be a high-scale structural datum, then acquire low-energy meaning through running and thresholds. In the present construction the clean value is attached to the pole ratio, so the scale-placement theorem is logically prior to any compactification geometry.

The common matching theorem has the form

$$\mathcal{S}_r \longrightarrow K_{J,r}^{\text{bare}}(\lambda) \longrightarrow \Gamma_{\text{eff}}^{(4)}[H, W, B] \longrightarrow \Delta_{V,T}^{-1}(s)|_{s=s_{W,Z}}. \quad (197)$$

Equation (197) is the pole-placement theorem target. The first arrow is the route reduction. The second arrow matches the source kernel into a four-dimensional gauge-Higgs effective action. The third arrow places the result in the dressed transverse pole convention of Sec. III. Endpoint, interval, and  $G_2$  routes differ in the first arrow; they share the last two arrows when the pole reading is retained. Appendix C 4 gives the equivalent transverse self-energy form of the same theorem.

For endpoint and interval routes, the central obligation is the relation between a boundary determinant and a four-dimensional pole. Boundary conditions define a spectrum before loop corrections and before conversion to pole observables. The derivation specifies how the determinant survives matching to the pole definition, or how the pole reading emerges from a dressed boundary self-energy.

For the  $G_2$  route, a local singularity model lives at a compactification or high scale. The local kernel can organize representations and couplings. A pole-spectrum reading requires a low-energy matching chain. The resulting target is a two-stage theorem:

$$\mathcal{S}_{G_2} \longrightarrow K_{J,G_2} \longrightarrow \Delta_{V,T}^{-1}(s) \quad \text{with} \quad s = s_{W,Z}. \quad (198)$$

The first arrow is local geometry. The second arrow is four-dimensional matching. Both arrows remain open obligations.

## F. Route outcome

A route succeeds when it supplies the common kernel target, the ordered electroweak assignment, the pole-matching chain, the scalar-branch map, and the global compatibility checks from one stated source theory. These criteria turn endpoint, interval, and  $G_2$  language into comparable derivation problems. Extra light channels, a different trace, a different mixing product, a high-scale placement, or an incompatible global form would classify the DeVries relation inside a larger family of spectral constructions.

## XIV. FLAVOR BOUNDARY AND REPRESENTATION ALGEBRA

The electroweak mass-ratio problem and the flavor problem carry separate obligations in this paper. The boundary condition is:

$$\begin{aligned} &\text{DeVries electroweak determinant} \\ &\quad \text{and generation/flavor organization} \\ &\text{are linked only by an explicit model.} \end{aligned} \tag{199}$$

This condition requires any compactification or endpoint picture to give an explicit equation linking flavor data to the electroweak determinant.

Representation theory remains essential. In the Standard Model and in the familiar grand-unification lineage, fermions live in representations of the internal gauge group, while gauge bosons live in the adjoint sector [23, 24].  $SU(5)$ ,  $Spin(10)$ , Pati–Salam, and  $E_6$  show how a larger algebra can unify the Standard Model representation data. This is the correct role of the GUT material in the present paper: it supplies representation discipline and a comparison class for structural weak-angle statements.

The DeVries construction uses a different representation question. The electroweak determinant asks why one sample is associated with the Higgs/order-parameter doublet invariant and the other with the adjoint/current invariant:

$$J_H = T_H(T_H + 1) = 3/4, \quad J_{\text{adj}} = T_{\text{adj}}(T_{\text{adj}} + 1) = 2. \tag{200}$$

Flavor representations can constrain a completion. The missing step is a derived map from endpoint, compactification, or localized matter data into the two-channel electroweak operator.

The  $\text{SO}(32)$ -flavor idea is therefore an organizing boundary sector in this paper. In an endpoint language it may label external or internal flavor degrees of freedom, Chan–Paton-like data, or a representation algebra for fermion multiplicities. The representation data contain the proposed branching data

$$16 = (5, 3) + (1, 1), \quad \wedge^2(5, 3) = (15, \bar{3}) + (\bar{10}, 6). \quad (201)$$

These equations currently function as representation constraints. Their use requires a primary source or a derived embedding in which the endpoint or orientifold construction supplies the gauge group, projection, massless spectrum, and coupling to the electroweak operator. Its required checks are concrete:

1. identify the representation space carrying the  $\text{SO}(32)$ -flavor labels;
2. map the Standard Model chiral fermions into that space;
3. specify orientifold, tadpole, or endpoint constraints if a Type-I-like  $\text{SO}(32)$  setting is used;
4. project, lift, or decouple extra states from Eq. (201);
5. show anomaly cancellation, hypercharge normalization, and global-form compatibility;
6. couple the flavor sector to the electroweak determinant through a derived operator.

This keeps the flavor idea as a compatibility sector. The electroweak ratio concerns  $W/Z$  pole structure and the Higgs/order-parameter channel. Generation structure concerns chiral matter, Yukawa textures, and possible endpoint or compactification multiplicities. A string/Kaluza–Klein completion may eventually bind them together through an equation or a local geometric construction.

The compactification version has an additional caveat. Geometry can generate gauge symmetries, chiral localization, and multiple wavefunctions with the same gauge quantum numbers [16, 18, 20]. That makes topology relevant to flavor. This paper treats generation counting as a separate task because the DeVries determinant is a two-channel electroweak object. The decisive test is whether the same local model that yields generations also yields

$$\det(\lambda - \mathcal{O}_J) = \lambda^2 + J\lambda - J \quad (202)$$

with the ordered electroweak samples in Eq. (200).

The practical rule for later sections is therefore simple. Flavor material may enter as a constraint on endpoints, compactification, anomaly cancellation, and global form. It may motivate a local operator. The W/Z mass-ratio argument rests on the electroweak determinant and its pole placement.

## XV. GLOBAL FORM OF THE STANDARD MODEL GAUGE GROUP

The local Standard Model gauge algebra admits several global forms. The gauge group may be written as

$$G_{\text{SM}} = \frac{SU(3)_C \times SU(2)_W \times U(1)_Y}{\Gamma}, \quad \Gamma \subset \mathbb{Z}_6, \quad (203)$$

with  $\Gamma = 1, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_6$  as the standard possibilities [21, 22]. Local perturbative fields see the Lie algebra. Line operators, magnetic sectors, theta-angle periodicities, and topological probes see the quotient.

This is relevant to the DeVries program for two reasons. First, any compactification, endpoint, or  $G_2$  completion has to specify the global group it realizes. Second, a pole-spectrum determinant for the electroweak sector has to coexist with the same line-operator spectrum as the low-energy Standard Model. Thus a proposed  $\mathcal{O}_J$  comes with a global-form compatibility condition:

$$\mathcal{O}_J \text{ is defined in a theory with } G_{\text{SM}} = (SU(3) \times SU(2) \times U(1))/\Gamma. \quad (204)$$

Tong’s line-operator analysis classifies how the quotient changes the electric, magnetic, and dyonic line sectors [21]. The quotient restricts admissible electric line operators and changes the allowed magnetic line operators through Dirac quantization. It also changes theta-angle periodicities. After electroweak symmetry breaking, the line sectors descend to electromagnetic probes, so  $\Gamma$  leaves a low-energy topological signature.

Hsin and collaborators propose a transport-style probe of this global form through a fractional topological response coefficient [22]. The important point here is structural: the global form is, in principle, observable through nonlocal or topological response data. It supplies a completion constraint for the DeVries program, because a string or compactification model generally determines more than a local Lie algebra.

Grand unification provides the comparison class.  $SU(5)$  and  $Spin(10)$  naturally package Standard Model representations in a way that favors the  $\mathbb{Z}_6$  quotient, and the subgroup  $S(U(2) \times U(3))$  realizes the quotient structure inside  $SU(5)$  [23, 24]. This places the GUT lineage beside the DeVries pole reading: high-scale structural placement is already known in unified models, while the DeVries construction assigns its clean value to the low-energy pole ratio and therefore needs a separate pole-placement derivation.

The global-form constraint can be expressed as a commutative diagram requirement. If a compactification or endpoint model gives an ultraviolet group  $G_{UV}$ , then the low-energy electroweak sector should fit into

$$\begin{array}{ccc} G_{UV} & \longrightarrow & G_{SM} \\ \downarrow & & \downarrow \\ \mathcal{O}_{UV} & \longrightarrow & \mathcal{O}_J, \end{array} \quad (205)$$

where the upper arrow fixes  $\Gamma$  and the lower arrow reduces the ultraviolet spectral operator to the DeVries two-channel operator. A model that supplies  $\mathcal{O}_J$  while leaving  $\Gamma$  unspecified has an underdetermined topological sector.

The mass-ratio argument itself should use local electroweak fields and pole positions. The global-form material acts as a filter on completions. A viable string/Kaluza–Klein/ $G_2$  derivation should state:

1. which quotient  $\Gamma$  appears;
2. which line operators survive compactification and electroweak breaking;
3. how the Higgs/order-parameter field transforms under the chosen global group;
4. how the DeVries operator respects the same quotient.

## XVI. CONCLUSION AND OUTLOOK

The construction begins with an exact two-branch algebra. The first assumption A1 fixes

$$Q(J) = \mu^2 \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad J = s(s+1),$$

whose dimensionless eigenvalues obey  $x^2 + Jx - J = 0$ . The quotient

$$1 - \frac{x_+(3/4)}{x_+(2)}$$

is therefore a rigid consequence of  $A1$  once the ordered samples  $(3/4, 2)$  are chosen. The physical interpretation additionally requires  $A2$  and  $A3$ .

The placement hypothesis  $A2$  uses the complex-pole mass ratio

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}.$$

This choice gives the spectral language a low-energy meaning and separates the proposal from  $\overline{\text{MS}}$  running angles, effective weak angles, Breit–Wigner line-shape masses, and high-scale GUT boundary values. The pole placement is an assumption awaiting a dynamical pole condition. Appendix C 4 states the corresponding theorem target.

The Standard Model gauge-Higgs sector supplies the projective datum

$$\frac{M_W^2}{M_Z^2} = \frac{g^2}{g^2 + g'^2}$$

and the broken-to-unbroken vacuum ray. It also supplies two representation data relevant to the proposed assignment: the Higgs/order-parameter doublet with  $J_H = 3/4$ , and the adjoint/current channel with  $J_{\text{adj}} = 2$ . The assignment hypothesis  $A3$  is the ordered map

$$(J_W, J_Z) = (J_H, J_{\text{adj}}) = (3/4, 2)$$

from gauge-Higgs, boundary, endpoint, or compactification data. Target II is a derivation of this map from the same source data that define the two-channel operator.

The electroweak ray adds a separate admissibility condition. A source route must preserve one radial gauge-Higgs parameter, the photon null direction, and the simultaneous collapse of the charged and neutral massive vector poles at fixed projective direction. Appendix C 6 records this rank-and-ray condition. Coupling-space paths and custodial-breaking shifts are separate deformation problems.

The pole convention, electroweak ray, theorem-target structure, source protocol, flavor boundary, and global-form filter serve as fixed acceptance conditions. They define the scheme, deformation class, source discipline, and completion tests that every route derivation has to satisfy. The DeVries determinant remains the dynamical target.

The negative branch belongs to the same diagonalization as the positive branch. Its scalar interpretation remains conditional on a gauge-invariant functional, a normalization, and a scheme. Candidate targets include a Higgs-potential parameter, a scalar pole, a boundary modulus, a brane separation mode, or a compactification deformation. Any source theory

that derives the positive-branch determinant must also account for the partner eigenvalue. The same condition governs any electromagnetic-coupling claim. The construction leaves  $\alpha$  as a Standard Model input: the scalar normalization, the  $U(1)_{\text{em}}$  generator and charge lattice, and the matching pair  $(Q_\alpha, \Delta_{\text{th}})$  remain open source-theory data. The electroweak content of the DeVries clue is the pole ratio  $s_{\text{pole}}^2$ .

Target III is the route-kernel derivation. Endpoint, interval, and  $G_2$  sources give addresses for a two-channel kernel. In common notation the required reduction is

$$K_J(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J} & \Sigma_{ha,J} \\ \Sigma_{ah,J} & \lambda + \Sigma_{aa,J} \end{pmatrix}, \quad \Sigma_{hh,J} = 0, \quad \Sigma_{aa,J} = J, \quad \Sigma_{ha,J} \Sigma_{ah,J} = J.$$

Open-string endpoint data, interval boundary conditions, and localized  $G_2$  gauge sectors provide candidate arenas for such a kernel. The exact DeVries entries remain to be derived from one source theory with a specified field basis and inner product.

The route analysis imposes three necessary consistency conditions. Trace statements stay inside the reduced matrices where they are derived until a shared source operator identifies the trace spaces. The tested Wigner–Eckart/Clebsch parent routes obey their selection rules and miss the parent data required by Eq. (180). An  $\text{SO}(32)$ -flavor sector can constrain a completion after its projection, anomaly, hypercharge, global-form, and electroweak-coupling obligations are supplied.

A source route that preserves a common Regge slope can place the DeVries roots as intercepts in

$$M_{N_{\text{osc}},j,\sigma}^2 = \mu^2 x_\sigma(j(j+1)) + \frac{N_{\text{osc}}}{\alpha'} + \Delta_{N_{\text{osc}},j,\sigma}^{\text{Regge}}.$$

The first higher positive sector is  $j = \frac{3}{2}$ , an algebraic assignment datum. A particle interpretation requires  $\mathfrak{R}_{3/2}$ : survival projection, gauge representation, chirality or vectorlike status, anomaly constraints, mass-generation rule, production and decay channels, widths, and collider-source check. Appendix C 12 records this filter.

The interval route carries source-derived boundary machinery: variational boundary conditions, modified inner products, vector Robin data, photon zero-mode accounting, and scalar  $A_5/\pi_i$  equations. The  $G_2$  route carries localization and compact-completion machinery, with a scale-placement burden before it reaches pole observables. The electroweak superconnection route packages odd Higgs data and even current data inside one gauge-Higgs

complex. The unresolved entry equations in one reduced basis are

$$\Sigma_{ha,J}\Sigma_{ah,J} = J, \quad \widehat{\Sigma}_{aa,J}^{\text{CHM}} = J,$$

with the source scale, reference subtraction, photon projection, and pole map fixed from the same interval data.

The route-kernel comparison combines these two routes. A boundary electroweak superconnection on a CHM interval should give a hatted kernel

$$\widehat{K}_J^{\text{sc/int}} = \Lambda_J^{-2} P_{\text{sc}}^\dagger [K_T^{\text{DtN}} + K_T^{\text{brane}} + \langle \mathcal{F}_{\text{sc}}, \mathcal{F}_{\text{sc}} \rangle_{\text{sc}} - K_\gamma^{\text{ref}}] P_{\text{sc}}.$$

The source-completion conditions are

$$P_W(\Phi_{\text{odd}}) = \frac{3}{4}, \quad P_Z(F_{\text{even}}^{\gamma\perp}) = 2, \quad P_\gamma(F_{\text{even}}^\gamma) = 0,$$

extra-channel decoupling, the negative-branch scalar functional, and the complex-pole remainder  $\Delta_{\text{O1/O10}}$ .

The resulting route tests are explicit. Target I gives the pole-placement and high-scale separation test. Target II gives the ordered-assignment source test. Target III gives the common source-kernel entry tests. Target IV gives the scalar or auxiliary interpretation test for the partner branch. Target IX gives the brane-scaling admissibility diagnostic. Target X gives the Hodge/superconnection square-root and breaking-operator test. The W/Z input set is fixed in Sec. IIIB. Appendix C collects the remaining analytical work as theorem targets.

Target IV is the scalar interpretation of the partner branch. The present paper establishes the algebraic clue, the pole-placement hypothesis, and candidate arenas for the determinant. A dynamical claim for the DeVries quotient requires the two-channel determinant, the ordered electroweak assignment, ray admissibility, the pole-matching theorem, and the controlled scalar interpretation of the negative branch. Global-form and representation checks then constrain any compact completion. Appendix C collects these requirements in theorem-target form.

## Appendix A: Algebraic details

We record the elementary algebra. From Eq. (5),

$$x_\pm(J) = \frac{-J \pm \sqrt{J(J+4)}}{2}. \quad (\text{A1})$$

The positive branch has the large- $J$  expansion

$$x_+(J) = 1 - \frac{1}{J} + \frac{2}{J^2} + O(J^{-3}), \quad (\text{A2})$$

while

$$x_-(J) = -J - 1 + \frac{1}{J} + O(J^{-2}). \quad (\text{A3})$$

Thus the positive branch approaches a fixed asymptotic value while the negative branch carries the large negative partner required by the trace.

For  $J = 3/4$ ,

$$x_+\left(\frac{3}{4}\right) = \frac{-3/4 + \sqrt{57}/4}{2} = \frac{\sqrt{57} - 3}{8}. \quad (\text{A4})$$

For  $J = 2$ ,

$$x_+(2) = \frac{-2 + \sqrt{12}}{2} = \sqrt{3} - 1. \quad (\text{A5})$$

Therefore

$$s_{\text{dV}}^2 = 1 - \frac{\sqrt{57} - 3}{8(\sqrt{3} - 1)}. \quad (\text{A6})$$

## Appendix B: Numerical reproducibility

The script ‘calculations/devries\_spectrum.py’ computes the branch roots and the seed arithmetic value from a supplied  $M_Z$ . The central functions are

```
x_plus(J) = (sqrt(J*J + 4*J) - J)/2
sin2_devries = 1 - x_plus(3/4)/x_plus(2)
MW = MZ * sqrt(1 - sin2_devries)
```

The test file ‘calculations/tests/test\_devries\_spectrum.py’ checks the root identities, the numerical value of  $s_{\text{dV}}^2$ , and the seed value obtained from  $M_Z = 91.1880 \text{ GeV}$ .

## Appendix C: Theorem targets for a derivation

This appendix states the analytical obligations in theorem form. Each target names its hypotheses, conclusion, source basis, and failure mode. A physical derivation proves one of the targets below, replaces it with a stronger statement, or demotes the associated claim.

## 1. Common format

The targets share a common data structure. Let  $\mathcal{S}$  be a source theory or effective description. It may be a four-dimensional electroweak theory, a boundary effective action from an interval compactification, an open-string endpoint sector, or a localized  $G_2$  system. Let  $\mathcal{H}_J$  be the two-dimensional light subspace that is meant to carry the order-parameter and current amplitudes. The desired reduction has the form

$$\mathcal{S} \longrightarrow \mathcal{H}_J \longrightarrow \mathcal{M}_J = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix} \longrightarrow \{x_+(J), x_-(J)\}. \quad (\text{C1})$$

The reduction has physical content after the normalization of  $\mathcal{H}_J$  and the scheme in which the resulting eigenvalues are read have been specified. The pole-observable section fixes the electroweak comparison as a complex-pole statement, with Breit-Wigner/on-shell and running quantities treated as translated inputs [10, 14]. The electroweak section fixes the low-energy field-theory data that any assignment theorem must respect [5, 8, 9]. The string and Kaluza–Klein sections list candidate sources of the two-channel determinant [2, 3, 7, 33].

For each target below, “proof” means an explicit derivation from a stated action, boundary condition, operator algebra, or compactification mode equation. A heuristic argument may motivate the target, yet the paper’s final physical claim must track the proven statement.

## 2. Target 0: unified source-to-pole matching

*Hypotheses.* Let  $r \in \{\text{end}, \text{int}, G_2\}$  label an endpoint, interval, or localized-singularity source description  $\mathcal{S}_r$ . The source theory supplies a two-dimensional light subspace

$$\mathcal{H}_{J,r} = \text{span}\{h_{J,r}, a_{J,r}\}, \quad \langle \cdot, \cdot \rangle_r, \quad P_{J,r}, \quad (\text{C2})$$

where the inner product and projection map  $P_{J,r}$  define the reduced field basis. Extra light channels are absent, integrated out, or block-diagonalized by the same projection. The route reduction has Schur form

$$K_{J,r}^{\text{eff}}(\lambda) = K_{J,r}^{\text{bdry}}(\lambda) - V_{J,r}^\dagger(\lambda - L_{J,r})^{-1}V_{J,r}. \quad (\text{C3})$$

For the interval/CHM realization, Target 0 includes the neutral admissibility lemma of Sec. IX C. The source route must prove positive projected CHM norms, photon-orthogonality

preservation, the projector-order or commutation condition, photon-reference subtraction, the shared  $\Lambda_{\text{CHM}}$ , and the kinetic-slope condition

$$Z_J^{\text{cur}} = \partial_{\hat{\lambda}} \hat{K}_J^{\text{cur}}(0) > 0. \quad (\text{C4})$$

The comparison with  $J$  uses the canonical current entry obtained by dividing by  $Z_J^{\text{cur}}$ .

*Conclusion.* In the reduced basis of Eq. (C2),

$$K_{J,r}^{\text{eff}}(\lambda) = \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix}. \quad (\text{C5})$$

The same field basis carries the electroweak sampling maps

$$P_W : h_{J,r} \mapsto C_2(\mathbf{2}) = \frac{3}{4}, \quad P_Z : a_{J,r}^{\gamma\perp} \mapsto C_2(\text{adj}) = 2, \quad P_\gamma : a_{\gamma,r} \mapsto 0, \quad (\text{C6})$$

and the source kernel contributes to the four-dimensional transverse self-energy in the pole scheme,

$$K_{J,r}^{\text{eff}}(\lambda) \longrightarrow \Pi_{T,V}^{(4)}(s; J, r) \longrightarrow \Delta_{V,T}^{-1}(s_V; J, r) = 0. \quad (\text{C7})$$

Thus one theorem supplies the source reduction, the reduced inner product, the DeVries entries, the ordered W/Z sampling maps, and the pole-placement chain. For the CHM electroweak realization, the sampling maps have the refined form

$$P_W(h_{J,r}) = \frac{3}{4} + \Delta_W^r, \quad P_Z(a_{J,r}^{\gamma\perp}) = 2 + \Delta_Y^r, \quad P_\gamma(a_{\gamma,r}) = 0, \quad (\text{C8})$$

with  $\Delta_W^r = \Delta_Y^r = 0$  as the zero-remainder outcome in the selected source convention.

*Source basis.* The endpoint, interval, and  $G_2$  sections supply candidate arenas for the three route types. The interval route has the most explicit boundary operator language, while the  $G_2$  route supplies localization language for gauge and matter data. The unified theorem is the common acceptance test shared by the routes.

*Failure mode.* The target fails if the source theory yields extra coupled light channels, a route-dependent inner product, a different trace or mixing product, an unordered electroweak sampling map, or a high-scale quantity whose matching to complex poles changes the determinant. Such an outcome classifies the source as a neighboring spectral construction.

### 3. Target 0a: admissibility filters for route reductions

*Hypotheses.* Assume a proposed source route supplies a reduced two-channel matrix, a parent representation route for the ordered  $J$ -assignment, or an  $\text{SO}(32)$ -flavor boundary sector. Let  $Q(J)$ ,  $K_{\text{seed}}^{(1)}$ , and  $K_-^{\sigma_3}$  denote distinct reduced matrices until a common source operator, projection, field basis, and normalization identify their vector spaces.

*Conclusion.* The route must pass three filters before it can support the DeVries pole-ratio claim:

1. any trace, determinant, or coefficient transfer among  $Q(J)$ ,  $K_{\text{seed}}^{(1)}$ , and  $K_-^{\sigma_3}$  is derived from a common source operator;
2. any Wigner–Eckart or Clebsch derivation specifies the parent representation, operator tensor type, projection, and normalization;
3. any  $\text{SO}(32)$ -flavor sector states its extra-state projection, anomaly and hypercharge checks, global-form compatibility, and electroweak-kernel coupling.

*Source basis.* The trace-space separation follows from the fact that the matrices above act on different reduced variables in the present construction. The natural Wigner–Eckart trials have been tested at the level needed for route triage: the doublet-parent route is removed by the selection rule, and the triplet axial route gives a parent-Casimir datum different from the DeVries target. The  $\text{SO}(32)$  sector supplies representation data and a possible string boundary arena.

*Failure mode.* The target fails for a route if a coefficient is imported across trace spaces, if the ordered  $J$ -assignment depends on a parent representation already excluded by the selection rule or parent-Casimir datum, or if flavor data are used before the completion checks are supplied.

### 4. Target I: pole placement

*Hypotheses.* Assume a source theory  $\mathcal{S}$  gives a two-branch spectral equation with roots  $x_{\pm}(J)$ . Assume the electroweak comparison is made with transverse gauge-boson complex poles,

$$\det \Delta_{VT}^{-1}(s_V) = 0, \quad s_V = M_{V,\text{pole}}^2 - iM_{V,\text{pole}}\Gamma_{V,\text{pole}}. \quad (\text{C9})$$

Assume the mass and width symbols are in the pole parameterization of Eq. (C9). If an input uses the alternative parameterization  $s_{\text{pole}} = (M - i\Gamma/2)^2$ , first translate it to Eq. (C9). If W/Z inputs are quoted in a common variable-width Breit–Wigner line-shape convention, convert

$$M_{V,\text{pole}} = \frac{M_{V,\text{BW}}}{\sqrt{1 + \gamma_V^2}}, \quad \Gamma_{V,\text{pole}} = \frac{\Gamma_{V,\text{BW}}}{\sqrt{1 + \gamma_V^2}}, \quad \gamma_V = \frac{\Gamma_{V,\text{BW}}}{M_{V,\text{BW}}}. \quad (\text{C10})$$

This gives

$$M_{V,\text{BW}}^2 = M_{V,\text{pole}}^2 + \Gamma_{V,\text{pole}}^2. \quad (\text{C11})$$

Assume common experimental mass conventions are converted into the pole convention before forming

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}. \quad (\text{C12})$$

*Conclusion.* The theorem target is a pole-spectrum identification:

$$s_{\text{pole}}^2 = 1 - \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{match}} \quad (\text{C13})$$

with a stated matching remainder. The theorem either derives  $\Delta_{\text{match}} = 0$  in the pole scheme or computes the controlled radiative, threshold, and scheme contribution. The theorem must also explain why the assigned scale is the low-energy pole spectrum, with high-scale boundary placements handled by a separate derivation. For any compactification, endpoint, interval, or local  $G_2$  source, the scale-placement statement has the matching form

$$\mathcal{S} \longrightarrow K_J^{\text{bare}}(\lambda) \longrightarrow \Gamma_{\text{eff}}[H, W, B] \longrightarrow \Delta_{V,T}^{-1}(s)|_{s=s_{W,Z}}. \quad (\text{C14})$$

The first arrow is the source reduction. The second arrow is four-dimensional effective matching. The third arrow is the dressed transverse pole condition. The pole reading requires all three arrows in one scheme. Target 0 packages this requirement with the reduced field basis and electroweak sampling maps. Equivalently, the source kernel must produce the relevant pole self-energy term,

$$K_J^{\text{bare}}(\lambda) \longrightarrow \Pi_{T,V}^{(4)}(s; J), \quad \Delta_{V,T}^{-1}(s_V; J) = 0, \quad (\text{C15})$$

with the same pole parameterization and field basis used in Eq. (C9).

*Source basis.* Section III separates complex poles, Breit–Wigner/on-shell masses,  $\overline{\text{MS}}$  running angles, effective weak angles, GUT boundary values, and compactification eigenvalues. Martin–Robertson supply the scheme language used there. The CDF-II input remains outside the pole-ratio theorem until it is converted to the same mass convention and uncertainty treatment.

*Implication.* Target I fixes the scale-placement classification for the present construction. Pole placement is a theorem target with a declared matching remainder  $\Delta_{\text{match}}$ . High-scale, compactification-scale, running, and effective-angle readings are separate boundary-data problems until a source route supplies the running, threshold, and scheme map to Eq. (C12).

*Failure mode.* The target fails if the derived clean value lives at a high-scale boundary, a running  $\overline{\text{MS}}$  angle, an effective leptonic angle, or a compactification eigenvalue with a different translation law. Then the DeVries quotient becomes a boundary datum whose relation to W/Z pole data requires a separate running, threshold, and matching calculation.

## 5. Target II: electroweak assignment

*Hypotheses.* Let  $H$  be the Standard Model Higgs doublet and let  $J_H = 3/4$  be its  $SU(2)_L$  quadratic invariant. Let  $J_{\text{adj}} = 2$  be the corresponding adjoint invariant for  $SU(2)_L$  currents. The gauge-Higgs sector contains the usual mass terms

$$M_W^2 = \frac{g^2 v^2}{4}, \quad M_Z^2 = \frac{(g^2 + g'^2) v^2}{4}, \quad M_\gamma^2 = 0, \quad (\text{C16})$$

with the custodial structure and  $\rho = 1$  tree-level relation discussed in Sec. IV.

*Conclusion.* Target II states the low-energy electroweak mass-map data and the ordered invariants. Target C 4 carries complex-pole matching and scheme conversion. The required Target II theorem is the existence of a source package

$$\mathcal{D}_{\text{EW}} : (H, J_\mu^a, B_\mu) \longmapsto (\mathcal{H}_J, \langle \cdot, \cdot \rangle, \Lambda_J, P_W, P_Z, P_\gamma, \mathcal{R}_{\text{pole}}), \quad (\text{C17})$$

with

$$P_W(h_J) = J_H = \frac{3}{4}, \quad P_Z(a_J^{\gamma^\perp}) = J_{\text{adj}} = 2, \quad P_\gamma(a_\gamma) = 0. \quad (\text{C18})$$

The pole-level conclusion is then

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{O1}}. \quad (\text{C19})$$

The derivation must say why the charged comparison samples the order-parameter invariant and why the neutral comparison samples the adjoint/current invariant. The allowed mathematical channels for the charged sample are: a Higgs-representation contribution inside the gauge-Higgs mass map, a scalar contribution to the  $W$  transverse self-energy, or a source/boundary label coupled to the Higgsing field.

The acceptance conditions are:

1. the same  $\mathcal{H}_J$ , inner product, and  $\Lambda_J$  support  $h_J$  and  $a_J^{\gamma^\perp}$ ;
2.  $P_\gamma$  fixes the photon null reference;
3. the neutral projector  $P_Z$  is defined after the  $\gamma^\perp$  projection in the  $W^3$ – $B$  sector;
4.  $\Delta_{O1}$  is derived in the pole scheme or retained as a stated source remainder;
5. the negative branch uses the same source package through the scalar functional recorded in Target C 8.

*Source basis.* The Higgs doublet and adjoint invariants are fixed representation-theoretic facts. The electroweak mass matrix, photon null direction, custodial relation, and projective reading of  $g : g'$  are standard low-energy data [5, 8, 9]. CHM interval sources supply boundary kinetic terms, vector Robin data, scalar  $A_5/\pi_i$  equations, custodial boundary data, and photon-zero-mode accounting [7]. Coquereaux supplies the generalized-connection grading vocabulary for a gauge-Higgs source complex [39]. The source-completion theorem is the ordered assignment from these data.

*Failure mode.* The target fails if the assignment uses observed vector spin or an arbitrary post-selection of  $J$  values. A successful derivation must arise from the gauge-Higgs variables, a boundary operator coupled to them, or a compactification mechanism whose low-energy reduction yields the ordered pair.

## 6. Target IIa: electroweak ray admissibility

*Hypotheses.* Let the electroweak gauge-Higgs system be varied along the radial family

$$H_t = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ tv + h \end{pmatrix}, \quad g(t) = g, \quad g'(t) = g', \quad 0 \leq t \leq 1. \quad (\text{C20})$$

At tree level this gives

$$M_W^2(t) = t^2 \frac{g^2 v^2}{4}, \quad M_Z^2(t) = t^2 \frac{(g^2 + g'^2) v^2}{4}, \quad M_\gamma^2(t) = 0. \quad (\text{C21})$$

The ratio  $g : g'$ , the photon null direction, and the custodial  $\rho = 1$  relation are held fixed at the stated tree-level matching point before loop and pole translations are applied.

*Conclusion.* The DeVries assignment must commute with the ray:

$$\mathcal{D}_{\text{EW}}(\mathcal{E}_t) = (J_H, J_{\text{adj}}) \quad \text{for } 0 < t \leq 1, \quad \frac{M_W^2(t)}{M_Z^2(t)} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{OI}}^{\text{ray}}(t). \quad (\text{C22})$$

The ray fixes the admissible deformation family. The DeVries quotient is the source-route target, encoded by the derived remainder  $\Delta_{\text{OI}}^{\text{ray}}(t)$  and its complex-pole continuation. For any source route in Target C 2, the source family must fit into the ray-admissibility square

$$\begin{array}{ccc} \mathcal{S}_r(t_{\text{EW}}) & \longrightarrow & K_{J,r}^{\text{eff}}(t_{\text{EW}}, \lambda) \\ \downarrow & & \downarrow \\ \Gamma_{\text{eff}}^{(4)}[H, W, B; t_{\text{EW}}] & \longrightarrow & \Delta_{T,V}^{-1}(s; t_{\text{EW}}). \end{array} \quad (\text{C23})$$

The same radial coordinate controls the charged and neutral massive vector sectors, while the photon remains the null mode.

*Source basis.* Sec. IV derives the tree-level mass matrix, the photon null direction, the projective ratio, and the radial broken-to-unbroken ray from standard gauge-Higgs data. The source routes already require the same low-energy gauge-Higgs sector after matching.

*Failure mode.* The target fails if a source route changes the photon null direction, changes the electroweak rank pattern, changes  $g : g'$  along the radial path, or gives separate charged and neutral deformation parameters. Such a route represents a distinct coupling-space or custodial-breaking deformation and must be analyzed as a separate model.

## 7. Target III: two-channel boundary determinant

*Hypotheses.* Let  $h_J$  be an order-parameter amplitude and  $a_J$  an adjoint/current amplitude in a light boundary subspace

$$\mathcal{H}_J = \text{span}\{h_J, a_J\}$$

with a stated inner product, projection map from the source Hilbert space, and normalization rule. Extra channels are integrated out or block-diagonalized before the two-by-two kernel is compared with the DeVries target. Let the normalized quadratic kernel be

$$K_J(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J} & \Sigma_{ha,J} \\ \Sigma_{ah,J} & \lambda + \Sigma_{aa,J} \end{pmatrix}. \quad (\text{C24})$$

The source may be an interval theory with brane-localized terms, an open-string endpoint sector, or a localized singularity construction. Boundary mass and kinetic terms can alter

eigenvalue conditions in extra-dimensional models [7]. Dirichlet-Higgs, radion, and compact gauge-field sources supply scalar variables for the interval route [29, 30, 34]. Open-string endpoints and D-brane gauge sectors supply a separate endpoint language [2, 3].

For the interval route, the admissible source data include CHM boundary kinetic terms, the modified scalar product, boundary scalar vevs, vector Robin conditions,  $A_5/\pi_i$  scalar boundary equations, radion variables, Dirichlet-Higgs boundary profiles, and Wilson-line phases. A sharpened interval subtarget is

$$K_J^{\text{int}}(\lambda) = P_J^\dagger (K_J^{\text{DtN}}(\lambda) + K_J^{\text{brane}}(\lambda)) P_J, \quad \mathcal{H}_J = \text{span}\{h_J, a_J\}. \quad (\text{C25})$$

Here  $K_J^{\text{DtN}}$  is the bulk Dirichlet-to-Neumann kernel,  $K_J^{\text{brane}}$  contains boundary kinetic and scalar terms, and  $P_J$  encodes the two-channel projection. The theorem must derive the projection, the modified inner product, the removal or block form of extra modes, and the normalization of the three entries below. Within this target, Eq. (C25) is the current source-theory route test. CHM data supply the boundary kinetic product, boundary scalar vev terms, vector Robin data, scalar  $A_5/\pi_i$  equations, custodial constraints, and photon zero-mode accounting. The proof obligation is the projection to  $\mathcal{H}_J$ , the product normalization, the ordered W/Z boundary map, and the source-to-pole matching chain.

*CHM current-entry convention subtarget.* The first entry-level subtarget is the normalized transverse-current extraction. The CHM equations fix the source product, endpoint sign convention, vector Robin equations,  $A_5/\pi_i$  scalar equations, and photon/custodial accounting. Let  $P_\gamma$  select the photon reference trace. The reference subtraction obeys

$$P_\gamma^\dagger (K_{T,J}^{\text{DtN}} + K_{T,J}^{\text{brane}} - K_{\gamma,J}^{\text{ref}}) P_\gamma = 0. \quad (\text{C26})$$

The source equations are CHM Eqs. (2.11)–(2.23), (2.30)–(2.44), and (3.32)–(3.43), covering the boundary kinetic condition, modified product, vector Robin data, scalar boundary equations, and photon/custodial boundary conditions [7]. Let  $\Lambda_{\text{CHM}}$  be the shared source scale that makes the spectral variable dimensionless,

$$\hat{\lambda} = \frac{\lambda}{\Lambda_{\text{CHM}}^2}. \quad (\text{C27})$$

The admissibility lemma requires positive projected norms, preserved photon-orthogonality, a source-ordered or commuting projector product, the photon-reference subtraction of Eq. (C26), and the current-slope datum

$$Z_J^{\text{cur}} = \partial_{\hat{\lambda}} \hat{K}_J^{\text{cur}}(0) > 0. \quad (\text{C28})$$

The current-channel kernel is

$$\begin{aligned}\widehat{K}_J^{\text{cur}}(\widehat{\lambda}) &= \frac{1}{\Lambda_{\text{CHM}}^2} \langle a_J, (K_{T,J}^{\text{DtN}}(\lambda) + K_{T,J}^{\text{brane}}(\lambda) - K_{\gamma,J}^{\text{ref}}(\lambda)) a_J \rangle_{\text{CHM}}, \\ \lambda &= \Lambda_{\text{CHM}}^2 \widehat{\lambda}, \quad \langle a_J, a_J \rangle_{\text{CHM}} = 1.\end{aligned}\tag{C29}$$

The canonical current kernel is

$$\widehat{K}_J^{\text{cur,can}} = (Z_J^{\text{cur}})^{-1} \widehat{K}_J^{\text{cur}}, \quad \partial_{\widehat{\lambda}} \widehat{K}_J^{\text{cur,can}}(0) = 1.\tag{C30}$$

The conclusion sought from this extraction is

$$\widehat{K}_J^{\text{cur,can}}(\widehat{\lambda}) = \widehat{\lambda} + \widehat{\Sigma}_{aa,J}^{\text{CHM,can}} + O(\widehat{\lambda}^2), \quad \widehat{\Sigma}_{aa,2}^{\text{CHM,can}} = 2.\tag{C31}$$

Acceptance requires endpoint signs, dimensions, product normalization, photon/reference subtraction,  $Z_J^{\text{cur}}$ , and the shared source scale as prerequisites for using  $\widehat{\Sigma}_{aa,2}^{\text{CHM,can}} = 2$ . The source-equation protocol is

$$\left\{ P_\gamma, P_{a,2}^{\gamma\perp}, K_{\gamma,2}^{\text{ref}}, \Lambda_{\text{CHM}}, Z_2^{\text{cur}}, \langle \cdot, \cdot \rangle_{\text{CHM}} \right\} \longrightarrow \widehat{\Sigma}_{aa,2}^{\text{CHM,can}},\tag{C32}$$

where the arrow denotes the required source-equation map from the CHM boundary kinetic conditions, vector Robin equations, scalar boundary equations, photon zero-mode condition, and custodial boundary data. Acceptance requires

$$\widehat{\Sigma}_{aa,2}^{\text{CHM,can}} = 2\tag{C33}$$

with positive projected norms, preserved photon orthogonality, and  $Z_2^{\text{cur}} > 0$ . The rejection criteria are

$$\widehat{\Sigma}_{aa,2}^{\text{CHM,can}} \neq 2, \quad Z_2^{\text{cur}} \leq 0, \quad \langle a_2^{\gamma\perp}, a_2^{\gamma\perp} \rangle_{\text{CHM}} \leq 0, \quad \langle a_2^{\gamma\perp}, a_\gamma \rangle_{\text{CHM}} \neq 0.\tag{C34}$$

These criteria give a source-convention classification of the interval current entry and determine the associated  $\Delta_Y^{\text{CHM}}$ , scale, or projector normalization for a noncanonical value. Here  $K_{\gamma,J}^{\text{ref}}$ ,  $P_\gamma$ ,  $P_{a,J}$ ,  $\Lambda_{\text{CHM}}$ , the endpoint-normal convention, the wavefunction normalization of  $a_J$ ,  $Z_J^{\text{cur}}$ , and the dimensionless  $J$ -normalization are part of the theorem data. Missing source definitions for  $K_{\gamma,J}^{\text{ref}}$ ,  $P_\gamma$ ,  $P_{a,J}$ ,  $Z_J^{\text{cur}}$ , or  $\Lambda_{\text{CHM}}$  leave the route as an admissible CHM boundary-kernel address with an open DeVries entry. Target C 14 supplies the complementary square-root entry test; this target supplies the hatted current diagonal test.

*Local  $G_2$  ADE-pairing subtarget (Target IIIg).* The  $G_2$  route has a local current target parallel to the CHM current target. The source datum is

$$u_{G_2}^{\text{loc}} = (M_3, \Gamma_{\text{ADE}}, W, \Phi, \rho_i, f_i, Q, p, \gamma, A_1 \subset A_2, \Lambda_{G_2}), \quad (\text{C35})$$

with Higgs-bundle source equations

$$\Phi = \sum_i t_i df_i, \quad \rho = \sum_i t_i \rho_i, \quad \Delta f_i = \rho_i, \quad f_Q = \sum_i q_i f_i, \quad df_Q(p) = 0. \quad (\text{C36})$$

The local enhancement test is

$$\text{ad } A_2 \downarrow_{A_1 \times U(1)} = (\text{ad } A_1)_0 \oplus \mathbf{1}_0 \oplus \mathbf{2}_{+q} \oplus \mathbf{2}_{-q}. \quad (\text{C37})$$

The singular-support pairing is

$$\langle \eta, \zeta \rangle_{G_2, u} = \int_{M_3} \text{Tr}_{\text{ADE}}(\eta \wedge *_{M_3} \zeta) + \sum_{p \in Z(df_Q)} \eta(p)^\dagger \mathbf{G}_Q(p) \zeta(p), \quad (\text{C38})$$

$$Z(df_Q) = \{p \in M_3 \mid df_Q(p) = 0\}.$$

The hatted current kernel is

$$\begin{aligned} \widehat{K}_J^{G_2, \text{cur}}(\widehat{\lambda}) &= \Lambda_{G_2}^{-2} \langle a_J^{G_2}, \mathcal{K}_{\text{cur}}^{\text{ADE}}(\lambda) a_J^{G_2} \rangle_{G_2, u}, \\ \lambda &= \Lambda_{G_2}^2 \widehat{\lambda}, \quad \langle a_J^{G_2}, a_J^{G_2} \rangle_{G_2, u} = 1. \end{aligned} \quad (\text{C39})$$

Here  $\mathcal{K}_{\text{cur}}^{\text{ADE}}$  is a theorem datum: the projected quadratic ADE-current operator on the singular gauge locus, including source scale, local projection, and any singular-support boundary terms. The canonical form uses

$$Z_J^{G_2} = \partial_{\widehat{\lambda}} \widehat{K}_J^{G_2, \text{cur}}(0) > 0, \quad \widehat{K}_J^{G_2, \text{cur}, \text{can}} = (Z_J^{G_2})^{-1} \widehat{K}_J^{G_2, \text{cur}}. \quad (\text{C40})$$

The local current-entry acceptance equation is

$$\widehat{K}_J^{G_2, \text{cur}, \text{can}}(\widehat{\lambda}) = \widehat{\lambda} + \widehat{\Sigma}_{aa, J}^{G_2, \text{can}} + O(\widehat{\lambda}^2), \quad \widehat{\Sigma}_{aa, 2}^{G_2, \text{can}} = 2. \quad (\text{C41})$$

The rejection outputs are

$$\widehat{\Sigma}_{aa, 2}^{G_2, \text{can}} \neq 2, \quad Z_2^{G_2} \leq 0, \quad \langle P_{a, 2}^{G_2} a_2, P_{a, 2}^{G_2} a_2 \rangle_{G_2, u} \leq 0, \quad (\text{C42})$$

or a surviving extra current channel in the local block. The same local source package then carries the flow-overlap target below.

*Local  $G_2$  flow-overlap subtarget (Target IIIh).* Target IIIh asserts the off-diagonal product in the same local source package as Target IIIg. Let the critical support be either an isolated-Morse critical set or a Morse–Bott component with a stated component metric and flow-moduli orientation. Let  $\psi_{h,J}^{G_2}$  be the localized order-parameter wavefunction and let  $a_J^{G_2}$  be the projected ADE current mode. Their normalization is

$$\langle \psi_{h,J}^{G_2}, \psi_{h,J}^{G_2} \rangle_{G_Q} = 1, \quad \langle a_J^{G_2}, a_J^{G_2} \rangle_{G_2,u} = 1. \quad (\text{C43})$$

For each oriented flow contribution  $\gamma$ , introduce a flow kernel  $\mathcal{K}_{\text{flow}}^{G_2}(\gamma)$ , an orientation sign  $\epsilon_\gamma$ , and bilinear maps  $B_{ha}^{G_2}$  and  $B_{ah}^{G_2}$ . The hatted off-diagonal entries are

$$\begin{aligned} \widehat{\Sigma}_{ha,J}^{G_2} &= \Lambda_{G_2}^{-1} \sum_{\gamma} \epsilon_\gamma B_{ha}^{G_2}(\psi_{h,J}^{G_2}, \mathcal{K}_{\text{flow}}^{G_2}(\gamma) a_J^{G_2}), \\ \widehat{\Sigma}_{ah,J}^{G_2} &= \Lambda_{G_2}^{-1} \sum_{\gamma} \epsilon_\gamma B_{ah}^{G_2}(a_J^{G_2}, \mathcal{K}_{\text{flow}}^{G_2}(\gamma) \psi_{h,J}^{G_2}). \end{aligned} \quad (\text{C44})$$

The local acceptance equation is

$$\widehat{\Sigma}_{ha,J}^{G_2} \widehat{\Sigma}_{ah,J}^{G_2} = J. \quad (\text{C45})$$

The source data are the Braun–Cizel–Hubner–Schafer–Nameki flow construction: localized modes at  $df_Q = 0$ , flow-line mass terms between critical points, orientation signs, possible cancellations among multiple flow contributions, and trivalent flow-tree interaction data. Target IIIh uses the quadratic flow-line bilinear as the kernel candidate. The trivalent data enter as interaction terms and as a constraint on charge-sector compatibility. The source test isolates the actual Braun two-point object:

$$M_{\text{MW}}^{ab} = \sum_{\gamma} n_\gamma \exp[-tq(f(p_a) - f(p_b))], \quad n_\gamma = \pm 1. \quad (\text{C46})$$

The derivation attempt fails at the map

$$M_{\text{MW}}^{ab} \rightsquigarrow \left( \widehat{\Sigma}_{ha,J}^{G_2}, \widehat{\Sigma}_{ah,J}^{G_2} \right), \quad (\text{C47})$$

because the source matrix acts between localized chiral multiplet ground states, while Target IIIh requires a closed two-channel  $(h_J^{G_2}, a_J^{G_2})$  block containing a projected ADE current mode. The narrowed acceptance test is

$$\widehat{\Sigma}_{ha,J}^{G_2} = \Lambda_{G_2}^{-1} M_{ha}^{\text{MW}}, \quad \widehat{\Sigma}_{ah,J}^{G_2} = \Lambda_{G_2}^{-1} M_{ah}^{\text{MW}}, \quad (\text{C48})$$

after same-basis compatibility with Target IIIg, positive local metric, orientation convention, and extra-channel control are supplied. Rejection outputs are

$$\begin{aligned} \widehat{\Sigma}_{ha,J}^{G_2} \widehat{\Sigma}_{ah,J}^{G_2} &\neq J, & \mathbf{G}_Q &\leq 0, \\ \sum_{\gamma} \epsilon_{\gamma}(\cdots) &\text{ lacks a source convention,} & \mathcal{H}_J^{G_2} &\text{ fails closure under the flow kernel.} \end{aligned} \quad (\text{C49})$$

The flow target is then combined with the completion projectors

$$P_W(h_J) = \frac{3}{4} + \Delta_W^{G_2}, \quad P_Z(a_J^{\gamma^\perp}) = 2 + \Delta_Y^{G_2}, \quad P_\gamma(a_J) = 0, \quad (\text{C50})$$

and the pole-chain equation

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(3/4)}{x_+(2)} + \Delta_{\text{match}}^{G_2}. \quad (\text{C51})$$

Compact completion supplies the charge lattice, Standard Model global form, hypercharge embedding, anomaly inflow constraints, line-operator data, and the source-to-pole map. The local ADE current test precedes these completion data.

*Conclusion.* A successful source route must exhibit a controlled field basis, inner product, projection, and normalization in which

$$\Sigma_{hh,J} = 0, \quad \Sigma_{aa,J} = J, \quad \Sigma_{ha,J} \Sigma_{ah,J} = J, \quad (\text{C52})$$

so that

$$\det K_J(\lambda) = \lambda^2 + J\lambda - J. \quad (\text{C53})$$

Equivalently, the physical derivation fixes the two named entries

$$\tau_J = J, \quad \kappa_J^2 = J. \quad (\text{C54})$$

*Source basis.* Sec. X gives the target action and the required entries. CHM support eigenvalue-dependent boundary conditions, modified inner products, vector boundary data from scalar vevs, and scalar boundary remnants as interval ingredients. Salam–Strathdee support the harmonic-expansion and massive-tower language, Haba–Oda and Bucci support boundary scalar/radion variables, and Hosotani supports compact gauge-field scalar data. Endpoint and D-brane sources support the separate endpoint arena. The exact DeVries entries remain conjectural. A sharper interval target is an eaten-Goldstone boundary kernel:

$$S_{5D}[A_M, \Phi_i] \xrightarrow{\delta S, \mathcal{G}_\xi} (K_T^{\partial I}[A_\mu], K_{A_5\pi}^{\partial I}) \xrightarrow{P_J^\dagger(\cdot)P_J} K_J^{\text{eaten}}(\lambda). \quad (\text{C55})$$

Here  $h_J$  is tested as the boundary Higgs/Goldstone variable and  $a_J$  as the transverse adjoint-current boundary value. The same boundary Higgsing datum would then source the vector kernel and the scalar partner functional. The corresponding source-to-pole square has explicit boundary variables:

$$\begin{aligned}
(v_0, v_L, R_0, \theta_H) &\longrightarrow (K_T^{\partial I}, K_{A_5\pi}^{\partial I}) \\
\downarrow &\downarrow P_J^\dagger(\cdot)P_J \\
(t_{\text{EW}}, \mathcal{F}_{\text{sc}}) &\longrightarrow K_J^{\text{eaten}}(\lambda) \longrightarrow \Pi_{T,V}^{(4)}(s; J).
\end{aligned} \tag{C56}$$

Here  $v_0$  and  $v_L$  denote boundary Higgs data,  $R_0$  denotes a radion or interval-size datum, and  $\theta_H$  denotes a Hosotani/Wilson-line phase. The theorem must supply the gauge-fixing parameter treatment, the physical-scalar/eaten-mode separation, photon zero-mode preservation,  $\lambda \mapsto m_n^2 \mapsto \Pi_T(s)$ , and the first entry-level derivation, preferably  $\Sigma_{aa,J} = J$  from the boundary operator and inner product.

*Failure mode.* The target fails if boundary variation gives a different trace, a different off-diagonal product, or an additional channel that changes the determinant class. The result in that case would be a larger family of secular relations containing the DeVries matrix as a limiting or approximate case.

*Implication.* Target III fixes the source-kernel comparison for this paper. It reduces each route to the same entry tests in Eq. (C52), with named CHM, endpoint,  $G_2$ , and Hodge/superconnection subtargets. A physical derivation still has to prove the entries from a stated source action, boundary condition, or local operator in the same reduced basis.

## 8. Target IV: negative branch as scalar data

*Hypotheses.* Assume Targets I–III identify a physical two-channel matrix whose eigenvalues are  $x_\pm(J)$ . Let  $u_\pm(J)$  be normalized eigenvectors. The negative branch is then part of the same diagonalization as the positive branch:

$$\mathcal{M}_J u_\pm(J) = x_\pm(J) u_\pm(J). \tag{C57}$$

The scalar side of the electroweak theory contains the gauge-invariant radial datum, the Higgs pole, and the parameters of the symmetry-breaking potential [8, 9]. Candidate source theories also contain boundary scalar variables,  $A_5$  or Wilson-line holonomies, radions, brane-separation data, and  $G_2$  deformation data.

*Conclusion.* There exists a same-source scalar package

$$u \longmapsto (\mathcal{H}_J, \langle \cdot, \cdot \rangle_u, P_J, \Lambda_J, \mathcal{R}_{\text{pole}}, \mathcal{F}_{\text{sc}}) \quad (\text{C58})$$

whose first five entries agree with the source package used for the vector pole quotient. The scalar functional is gauge-invariant and has a stated mass dimension  $d_{\text{sc}}$ . It obeys

$$\begin{aligned} \widehat{\mathcal{F}}_{\text{sc}}(u; J_\star) &\equiv \Lambda_J^{-d_{\text{sc}}} \mathcal{F}_{\text{sc}}(H^\dagger H, (D_\mu H)^\dagger D^\mu H, V_{\text{eff}}, \dots; s_{\text{ren}}, u) \\ &= C_{\text{sc}}(s_{\text{ren}}, u) |x_-(J_\star)|. \end{aligned} \quad (\text{C59})$$

for a stated  $J_\star$ , normalization  $C_{\text{sc}}$ , scheme  $s_{\text{ren}}$ , scale  $\Lambda_J$ , observable type, and pole-vs-potential status. Candidate readings include a Higgs-potential parameter, the radial vacuum scale, the Higgs pole, a Wilson-line curvature, a boundary modulus, a brane-separation mode, or a  $G_2$  deformation eigenvalue.

*Acceptance subconditions.* The target requires:

1. the same reduced space  $\mathcal{H}_J$ , inner product, and projection  $P_J$  used in Targets I–III;
2. a declared scale  $\Lambda_J$  and mass dimension  $d_{\text{sc}}$ ;
3. a source-fixed  $J_\star$ ;
4. a gauge-invariant scalar variable or pole observable;
5. a scheme and matching prescription for potential parameters;
6. an auxiliary-branch outcome if  $u_-(J)$  projects outside the physical scalar sector.

A prioritized interval realization is the Wilson-line curvature target

$$\mathcal{F}_{\text{sc}}^{\text{Hol}} = \left. \frac{\partial^2 V_{\text{eff}}(\alpha_H; \Lambda, s_{\text{ren}})}{\partial \alpha_H^2} \right|_{\alpha_H = \alpha_\star}, \quad \alpha_H \sim g_5 \int_I dy A_5, \quad (\text{C60})$$

with the same interval kernel and boundary data that generate the transverse vector determinant. In the boundary-superconnection realization this is refined to Eq. (C106), where the Wilson-line variable is replaced by the compact holonomy of  $A_5 + \Phi_{\text{odd}}$  and the same  $P_{\text{sc}}$ ,  $\Lambda_J$ , and photon reference used in Target X.

*Source basis.* Sec. V states the scalar target table, the same-source package, the holonomy-curvature option, and the eigenvector interpretation. The negative branch is treated as an analytical partner of the pole clue, with its Higgs/order-parameter meaning open. Breitenlohner–Freedman supply an AdS stability caveat: scalar mass-squared values below zero can be compatible with positive energy under specified boundary conditions [35]. The DeVries sign still requires its own source theorem.

*Implication.* Target IV fixes the interpretation used in the present paper. The branch is retained as a same-source scalar or auxiliary theorem target. The present construction makes no Higgs-scale prediction from  $x_-(J)$  before the functional in Eq. (C59) and its source package are derived.

*Failure mode.* The target fails if the negative branch has auxiliary meaning in the derived source theory. The determinant problem may still be motivated by the positive branch; scalar claims then leave the physical conclusion.

## 9. Target V: global-form and localization compatibility

*Hypotheses.* Assume a string, brane, Kaluza–Klein, or  $G_2$  source derives Targets II–IV. The low-energy gauge group has global form

$$G_{\text{SM}} = \frac{SU(3) \times SU(2) \times U(1)}{\Gamma}, \quad \Gamma \subseteq \mathbb{Z}_6. \quad (\text{C61})$$

Line operators and global identifications provide tests of the Standard Model gauge group beyond the Lie algebra [21, 22]. Singular  $G_2$  spaces can localize gauge and chiral matter sectors in M-theory constructions [17–20].

*Conclusion.* The derived operator  $\mathcal{M}_J$  is compatible with the allowed line operators, representation assignments, anomaly constraints, and localized matter data of the chosen completion. A successful completion must specify whether flavor labels are endpoint data, compactification data, representation-sector data, or a combination constrained by the source geometry.

*Source basis.* Secs. XIV and XV state the boundary conditions: the  $\text{SO}(32)$ -flavor idea is a representation-sector clue, and global-form data are completion constraints. This paper avoids identifying generations with compactification topology until a representation map is

derived. The representation data contain the proposed boundary data

$$16 = (5, 3) + (1, 1), \quad \wedge^2(5, 3) = (15, \bar{3}) + (\bar{10}, 6), \quad (\text{C62})$$

which require an orientifold or endpoint construction, extra-state projection or decoupling, anomaly and hypercharge checks, global-form checks, and a derived coupling to the electroweak  $K_J$ .

*Failure mode.* The target fails if the determinant arises in a gauge or flavor sector incompatible with Standard Model line operators, anomaly conditions, or observed representations. A failure here would be decisive for a proposed completion even if the local two-channel determinant is mathematically correct.

## 10. Target VI: dimensional interpolation

*Hypotheses.* Let  $t_{\text{EW}}$  denote the gauge-Higgs radial coordinate of Target C6. Let  $t_{\text{dim}}$  denote compact, boundary, brane, or singular data in a source theory. A source derivation may identify the two through a map

$$\chi : t_{\text{dim}} \mapsto t_{\text{EW}}, \quad (\text{C63})$$

or through a joint source variable

$$u \mapsto (t_{\text{dim}}(u), t_{\text{EW}}(u), v(u), m_h^2(u), \Lambda_J(u), \langle \cdot, \cdot \rangle_u, \\ P_J(u), \hat{K}_J(u, \hat{\lambda}), \mathcal{D}_J^{\text{extra}}(u), \mathcal{F}_{\text{sc}}(u; J), \mathcal{Y}_{\text{top}}(u), \mathcal{R}_{\text{pole}}(u)), \quad (\text{C64})$$

with  $\hat{\lambda} = \lambda/\Lambda_J(u)^2$ . The symbol  $u$  denotes a joint source coordinate until a route identifies it as a compact modulus, boundary parameter, brane datum, singularity deformation, or mixed source coordinate. The full-gauge endpoint source trail is

$$(D, n_{\text{KK}}) = (11, 7) : SU(3) \times SU(2) \times U(1). \quad (\text{C65})$$

The five-extra-dimensional endpoint remains a reconstruction target until its electroweak embedding identifies the surviving  $U(1)$  with  $U(1)_{\text{em}}$ :

$$(D, n_{\text{KK}}) = (9, 5) : SU(3) \times U(1)_{\text{geom}}. \quad (\text{C66})$$

The colourless electroweak dictionary is

$$(D, n_{\text{KK}}^{\text{ew}}) = (7, 3) : SU(2) \times U(1), \\ (D, n_{\text{KK}}^{\text{ew}}) = (5, 1) : U(1)_{\text{geom}} \rightarrow U(1)_{\text{em}} \quad \text{after embedding theorem.} \quad (\text{C67})$$

The  $D = 9$  and colourless  $D = 5$  endpoint obligations split into a compact isometry or gauge generator, a Standard Model electromagnetic charge lattice, and a four-dimensional kinetic normalization. At the middle point  $u = u_\star$ , assume a light subspace

$$\mathcal{H}_J(u_\star) = \text{span}\{h_J, a_J\} \quad (\text{C68})$$

with  $h_J$  an order-parameter channel and  $a_J$  an adjoint/current channel, together with  $\langle \cdot, \cdot \rangle_{u_\star}$ ,  $P_J(u_\star)$ ,  $\Lambda_J(u_\star)$ , and a decoupling or block-reduction rule  $\mathcal{D}_J^{\text{extra}}(u_\star)$ . A top-sensitive extension also supplies a Yukawa or boundary datum  $\mathcal{Y}_{\text{top}}(u_\star)$  attached to the same source variable.

*Conclusion.* The middle-dimensional sector passes the dimensional-interpolation test when the normalized projection gives the DeVries block,

$$\Lambda_J(u_\star)^{-2} P_J(u_\star)^\dagger \mathcal{O}_J(u_\star, \lambda) P_J(u_\star) \longrightarrow \widehat{K}_J(u_\star, \widehat{\lambda}) = \begin{pmatrix} \widehat{\lambda} & -\sqrt{J} \\ -\sqrt{J} & \widehat{\lambda} + J \end{pmatrix}, \quad (\text{C69})$$

with all additional light channels eliminated or block-diagonalized by  $\mathcal{D}_J^{\text{extra}}$ . This equation is the acceptance target for the  $(D, n_{\text{KK}}) = (10, 6)$  and colourless  $(D, n_{\text{KK}}^{\text{ew}}) = (6, 2)$  middle lines:

$$(\mathcal{B}(u), \langle \cdot, \cdot \rangle_u, P_J(u), \Lambda_J(u), \mathcal{D}_J^{\text{extra}}(u)) \mapsto \widehat{K}_J(u, \widehat{\lambda}) \stackrel{u=u_\star}{=} \begin{pmatrix} \widehat{\lambda} & -\sqrt{J} \\ -\sqrt{J} & \widehat{\lambda} + J \end{pmatrix}. \quad (\text{C70})$$

After Target C 5 is derived, the same sector would yield the ordered electroweak samples

$$(J_W, J_Z) = (J_H, J_{\text{adj}}) = \left( \frac{3}{4}, 2 \right). \quad (\text{C71})$$

The full-gauge middle line is then  $(D, n_{\text{KK}}) = (10, 6)$ . The colourless middle line is  $(D, n_{\text{KK}}^{\text{ew}}) = (6, 2)$ . The reconstruction target also supplies the pole-matching chain in Eq. (C14).

An optional top-sector extension of the same construction supplies

$$\mathcal{Y}_{\text{top}}(u_\star) \rightarrow \left( y_t(u_\star), m_t = \frac{y_t(u_\star)v(u_\star)}{\sqrt{2}}, \Pi_{VV}^{(t)}(s; u_\star) \right), \quad (\text{C72})$$

with the top representation, localization or boundary data, anomaly/global-form checks, and pole-scheme matching stated in the same field basis. The top extension passes when  $\mathcal{Y}_{\text{top}}$  is derived from the same  $u_\star$ ,  $\langle \cdot, \cdot \rangle_{u_\star}$ , projection, and pole map as Eq. (C69).

*Source basis.* Witten’s Kaluza–Klein analysis supplies the source trail for the seven-extra-dimensional  $SU(3) \times SU(2) \times U(1)$  endpoint and the associated fermion quantum-number obstruction [16]. The same paper contains a five-sphere  $SU(3) \times U(1)$  symmetry address. The six-dimensional string sources support a landscape of six-dimensional string settings and dualities [25–28]. Boundary scalar vevs on intervals modify vector boundary conditions and can leave scalar remnants, giving a concrete template for an active-channel middle line [7]. The same interval source records the third-generation localization and boundary-mixing pressure associated with a heavy top and  $Zb\bar{b}$  constraints. This gives a CHM interval source test,

$$\begin{aligned}
u_{\text{int}} = (R, v_i, Z_i, \theta_H, c_{Q_3}, c_{t_R}, M_{\text{mix}}, \dots) &\longrightarrow (K_T^{\partial I}, K_{A_{5\pi}}^{\partial I}, \mathcal{Y}_{\text{top}}) \\
\downarrow &\downarrow \\
(t_{\text{dim}}, t_{\text{EW}}, v) &\longrightarrow \Lambda_J(u_{\text{int}})^{-2} P_J^\dagger (K_J^{\text{DtN}} + K_J^{\text{brane}} - K_J^{\text{ref}}) P_J \\
&\downarrow \\
&\begin{pmatrix} \hat{\lambda} & -\sqrt{J} \\ -\sqrt{J} & \hat{\lambda} + J \end{pmatrix} \\
&\mathcal{Y}_{\text{top}}(u_\star) \rightarrow (y_t, m_t, \Pi_{VV}^{(t)}) .
\end{aligned} \tag{C73}$$

Local  $G_2$  Higgs-bundle models give a separate template for top-Yukawa source data [20]. Henkel–Lauret’s Hodge-Laplacian calculation on homogeneous three-spheres gives an explicit one-form spectral matrix with representation mixing and curvature-controlled diagonal data [40]. This source supports a sharpened middle-line test:

$$\Delta_{S_{\text{Berger}}^3(\rho)}^{(1)} \longrightarrow \hat{K}_J^{\text{geom}}(\rho, \hat{\lambda}) \longrightarrow P_J^\dagger (K_J^{\text{geom}} + K_J^{\text{CHM}}) P_J \longrightarrow \begin{pmatrix} \hat{\lambda} & -\sqrt{J} \\ -\sqrt{J} & \hat{\lambda} + J \end{pmatrix} . \tag{C74}$$

The scalar side of this test has a BF-admissibility check,

$$-\frac{9}{4L_{\text{AdS}}^2} \leq \mu^2 x_-(J) < 0, \quad u_-(J) \mapsto \mathcal{F}_{\text{sc}}(u; J), \tag{C75}$$

with  $P_J$ ,  $\rho$ , the inner product, and extra-channel decoupling still open. Appendix D converts these facts into a source-support table and a middle-line reconstruction target.

*Failure mode.* The target fails if the middle-dimensional sector produces unresolved extra light channels, a kernel beyond the DeVries determinant class, a high-scale datum with a different matching law, a dimensional family lacking a map to the electroweak ray,

an ordered assignment other than  $(J_H, J_{\text{adj}}) = (3/4, 2)$ , chirality or anomaly/global-form conflict, missing electromagnetic embedding, or route-dependent normalization. Such an outcome classifies the dimensional idea as an organizing analogy for a different source-theory reduction.

## 11. Target VII: electromagnetic endpoint coupling

*Hypotheses.* Assume Targets C 2, C 5, C 4, C 8, and C 10 have supplied a common source route. The same route must identify the  $D = 9$  endpoint factor with  $U(1)_{\text{em}}$ , fix its generator normalization, and give a gauge kinetic term

$$-\frac{1}{4e_{\text{geom}}^2} \int d^4x F_{\mu\nu} F^{\mu\nu} \quad (\text{C76})$$

after reduction. Assume also that the scalar branch has a vacuum-ray normalization

$$v^2 = C_v \mu^2 |x_-(J_v)|, \quad M_W^2 = C_W \mu^2 x_+(J_H), \quad (\text{C77})$$

where  $C_v$ ,  $C_W$ ,  $J_v$ , the inner product, and the matching scheme are outputs of the source theorem.

*Route outputs.* Target VII has three route-level output packages. An endpoint route supplies the compact generator and charge lattice. An interval route supplies the photon projection and common source scale. A local  $G_2$  route supplies the same data through the singular-locus gauge generator and four-dimensional kinetic normalization:

*Conclusion.* The endpoint theorem defines a construction coupling from fixed scalar and vector normalizations:

$$g_{\text{sec}}^2 = \frac{4C_W x_+(J_H)}{C_v |x_-(J_v)|}. \quad (\text{C78})$$

The electromagnetic comparison has two matching layers:

$$\sin^2 \theta_S(Q_\alpha) = s_{\text{dV}}^2 + \Delta_\theta(Q_\alpha), \quad (\text{C79})$$

$$\alpha_S(Q_\alpha) = \frac{g_{\text{sec}}^2 \sin^2 \theta_S(Q_\alpha)}{4\pi} + \Delta_{\text{th}}(Q_\alpha), \quad (\text{C80})$$

where the theorem must identify the observable, the scale  $Q_\alpha$ , the threshold prescription, and the running convention. The special reading  $J_v = J_{\text{adj}} = 2$ ,  $C_v = 2$ , and  $C_W = 1$  is a subtarget of Eq. (C77); it remains conjectural until derived from the source route.

TABLE I. Route outputs required by Target VII.

| Route        | Required source outputs                                                                                        | Acceptance datum                                                       |
|--------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Endpoint/KK  | $\xi_{\text{em}}$ , charge lattice, generator normalization, $g_D$ , compact metric, threshold map             | $e_{\text{geom}}$ , $Q_\alpha$ , $\Delta_{\text{th}}$ in one scheme    |
| Interval/CHM | $P_\gamma$ , $P_W$ , $P_Z$ , $\Lambda_{\text{CHM}}$ , $\mathcal{F}_{\text{sc}}$ , photon-reference subtraction | $\widehat{\Sigma}_{aa,J}$ , scalar normalization, and pole/running map |
| Local $G_2$  | singular-locus $U(1)$ generator, harmonic representative, charge lattice, local kinetic pairing                | same $U(1)_{\text{em}}$ normalization and matching pair                |

*Source basis.* The Standard Model identities  $M_W^2 = g^2 v^2/4$  and  $e = g \sin \theta_W$  supply the low-energy algebraic bridge [8, 9]. Martin–Robertson give the scheme language for running electroweak parameters [10]. Jegerlehner supplies a dedicated running- $\alpha$  source through the Adler function and hadronic vacuum polarization [32]. The PDG constants table is kept in the source inventory for constants; this comparison still needs a dedicated source check before selecting an intermediate hadronic matching point [11]. Witten’s Kaluza–Klein discussion supports the qualitative mechanism: gauge fields arise from compact geometry and their couplings are controlled by the compact metric once the dynamics of that metric is known [16]. These sources support the theorem target. The scalar normalization, electromagnetic generator normalization, charge lattice, and matching pair remain external to the derived construction, so  $\alpha$  stays a Standard Model input.

*Failure mode.* The target fails from any missing element in Eq. (C77), from a  $D = 9$  endpoint with unspecified  $U(1)_{\text{em}}$  charge normalization, or from a matching pair  $(Q_\alpha, \Delta_{\text{th}})$  supplied as external measured input. That outcome leaves the electromagnetic coupling as a Standard Model input and preserves the pole-ratio clue as the construction’s electroweak content.

## 12. Target VIII: Regge-intercept survival and higher branch slots

The Rivero/DeVries source note uses high-spin language as motivation for the quadratic. A string or boundary completion has to preserve ordinary Regge growth while explaining why the low intercept is governed by the DeVries kernel. Separate the sector label from the trajectory labels by writing

$$A_j = j(j+1), \quad x_\sigma(A_j) = \frac{-A_j + \sigma\sqrt{A_j^2 + 4A_j}}{2}, \quad \sigma = \pm. \quad (\text{C81})$$

The source-theory target is

$$\mathcal{S}_{\text{Regge/KK}} \longrightarrow K_j(\lambda) \longrightarrow \{x_\pm(A_j)\} \longrightarrow M_{N_{\text{osc}},j,\sigma}^2 = \mu^2 x_\sigma(A_j) + \frac{N_{\text{osc}}}{\alpha'} + \Delta_{N_{\text{osc}},j,\sigma}^{\text{Regge}}. \quad (\text{C82})$$

Here  $j$  is a DeVries-sector label,  $N_{\text{osc}}$  is the oscillator or tower level, and the physical trajectory spin is supplied by the source theory. The local string source supports the common-slope template  $M^2 = (N-1)/\alpha'$  with  $J_{\text{max}} = \alpha'M^2 + 1$  [2]; the DeVries contribution can therefore enter as an intercept through a source-derived  $K_j$ .

Targets VIII and IX use one open-string/D-brane source package,

$$\mathfrak{R}_j = (K_j, \alpha', P_{\text{surv}}, \mathcal{B}_{\text{ND}}, \mathcal{Q}, \mathcal{T}, \chi_{\text{spin}}), \quad (\text{C83})$$

where  $P_{\text{surv}}$  is the projection or BRST/GSO survival rule,  $\mathcal{B}_{\text{ND}}$  is the boundary-condition map,  $\mathcal{Q}$  records charges,  $\mathcal{T}$  records tensions, and  $\chi_{\text{spin}}$  is the possible map from the DeVries sector label to a putative rotating-brane angular momentum. The common acceptance diagram is

$$\mathcal{S}_{\text{open/D}} \longrightarrow \mathfrak{R}_j \longrightarrow M_{N_{\text{osc}},j,\sigma}^2 \longrightarrow \begin{cases} j = \frac{3}{2} \text{ assignment datum,} \\ x_+ x_- = -j(j+1) \text{ branch-duality test.} \end{cases} \quad (\text{C84})$$

*Hypotheses.*

1. A world-sheet, Chan–Paton, interval, compactification, or  $G_2$  source produces a reduced two-channel kernel  $K_j(\lambda)$  with determinant  $\lambda^2 + A_j\lambda - A_j$ .
2. The electroweak pole quotient fixes the ordered low sectors  $j = \frac{1}{2}, 1$  through Targets C 4 and C 5.

3. A common normalization fixes  $\mu$  across the electroweak slots and any higher slot being discussed.
4. The source projection specifies which positive and negative intercepts survive, and supplies the stability or BRST/GSO rule for negative-branch data.
5. Gauge representation, chirality, anomaly cancellation, production channel, decay channel, width, and collider limits are supplied as part of any particle assignment.

*First higher slot.* The first sector beyond the electroweak pair is  $j = \frac{3}{2}$ . Target VIII asks the source theory to decide whether the positive intercept

$$M_{0,3/2,+}^2 = \mu^2 x_+ (15/4) + \Delta_{0,3/2,+}^{\text{Regge}} \quad (\text{C85})$$

is projected out, assigned to a neutral resonance, assigned to a charged or coloured sector, or retained as an algebraic branch datum. Eq. (C85) supplies the branch value; the assignment requires the source-theory quantum numbers and a source-checked collider analysis. Biekotter–Heinemeyer–Weiglein provide one phenomenology source for a neutral 95.4 GeV diphoton reference [31]; the branch identity, gauge representation, and survival rule remain source-theory targets.

*Assignment filter.* The  $j = \frac{3}{2}$  root is an algebraic Regge-intercept slot with the conditional mass formula in Eq. (C85). A particle interpretation requires  $\mathfrak{R}_{3/2}$ : the source kernel, common slope, survival projection, charge assignments, chirality or vectorlike status, anomaly constraints, production and decay maps, width estimate, and collider-source check. The slot supplies a source-completion test.

*Failure mode.* The target fails from any missing entry in  $\mathfrak{R}_j$ , from a route-dependent slope, or from absent higher-sector quantum numbers. The higher positive roots then remain algebraic slots in the secular construction, and phenomenology uses a source-checked assignment.

### 13. Target IX: branch scaling and brane-duality admissibility

*Hypotheses.* Assume Target C 12 has supplied the source package  $\mathfrak{R}_j$ . The brane-scaling test adds an axis map

$$\chi_{\text{spin}} : j \longmapsto s_{\text{br}}(j), \quad J_{\text{br}} = s_{\text{br}}(s_{\text{br}} + 1),$$

where  $s_{\text{br}}$  is a putative rotating-brane angular momentum. Identifying  $s_{\text{br}}$  with the DeVries sector label is an additional source hypothesis. Assume a mass reading  $M_\sigma^2 = \mu^2 x_\sigma(J_{\text{br}})$  or a scalar-functional version with the same large- $J_{\text{br}}$  magnitude data. The branch exchange is the algebraic involution

$$x \mapsto -\frac{J}{x}, \quad x_+(J)x_-(J) = -J. \quad (\text{C86})$$

*Conclusion.* The source theory must account for the large-label behavior

$$x_+(J_{\text{br}}) = 1 + O(s_{\text{br}}^{-2}), \quad |x_-(J_{\text{br}})| = s_{\text{br}}(s_{\text{br}} + 1) + O(1), \quad (\text{C87})$$

and hence for the mass-scaling diagnostic

| branch | mass behavior                                 | brane-scaling test       | diagnostic label                     |
|--------|-----------------------------------------------|--------------------------|--------------------------------------|
| +      | $M_+(s_{\text{br}}) \rightarrow \mu$          | $M \sim s_{\text{br}}^0$ | $p = 0$ point/D0                     |
| −      | $ M_-(s_{\text{br}})  \sim \mu s_{\text{br}}$ | $M \sim s_{\text{br}}^1$ | $p \rightarrow \infty$ space filling |

(C88)

whenever a rotating  $p$ -brane scaling law  $M \sim s^{p/(p+1)}$  is part of the source route. The D0 and space-filling labels are admissibility diagnostics that require a primary source for the rotating-brane exponent and a boundary-condition map. A literal T-duality reading requires a compact coordinate, boundary conditions, charges, tensions, and a map that exchanges the branch sectors through the Neumann–Dirichlet transformation. Tong’s string notes support the D0, space-filling, and T-duality boundary-condition vocabulary [2]; the rotating-brane exponent and the branch map remain source-theory obligations.

*Source basis.* The source basis is algebraic plus source-address data. The involution in Eq. (C86) follows from the secular quadratic. The separation of the sector label from the ordinary oscillator level is already enforced by Target C12. The endpoint and brane sections supply a matrix arena with D-brane gauge fields, adjoint scalars, and boundary conditions. A worldvolume derivation supplies the branch sectors, the mass reading for  $x_-$ , the projection rule, and the duality map.

*Implication.* Target IX fixes the branch-scaling diagnostic. The D0 and space-filling labels are diagnostic outputs of Eq. (C88) when a source route supplies  $\chi_{\text{spin}}$ , a mass reading, and a rotating-brane scaling law. A physical duality claim requires the Neumann–Dirichlet map, charges, tensions, and negative-branch reading listed above.

*Failure mode.* The target fails from a source package lacking  $\chi_{\text{spin}}$ , from a negative root assigned to auxiliary data outside brane-scaling content, or from a branch exchange implemented outside boundary-condition duality. The retained result is then Target VIII's ordinary Regge behavior through  $N_{\text{osc}}/\alpha'$ , with Target IX recorded as a rejected brane-duality interpretation.

#### 14. Target X: Hodge/SUSY-QM origin of the DeVries block

*Hypotheses.* Assume the source theory contains a  $\mathbb{Z}_2$ -graded Hilbert space

$$\mathcal{H}_J^{\text{SQM}} = E_J^0 \oplus E_J^1 = \text{span}\{e_J^0, e_J^1\}, \quad (\text{C89})$$

with a differential  $D_J : E_J^0 \rightarrow E_J^1$ , adjoint  $D_J^\dagger$ , and a chosen inner product. The de Rham/Hodge model of supersymmetric quantum mechanics supplies the source template:  $d + d^\dagger$  is a supercharge, the Hamiltonian is the Hodge Laplacian, and a Witten deformation couples the complex to Morse data [36]. The reduced basis is required to obey

$$D_J e_J^0 = \sqrt{J} e_J^1, \quad D_J^\dagger e_J^1 = \sqrt{J} e_J^0, \quad (\text{C90})$$

with acceptance conditions

$$D_J^\dagger D_J e_J^0 = J e_J^0, \quad D_J D_J^\dagger e_J^1 = J e_J^1, \quad J = \lambda_J = s(s+1), \quad (\text{C91})$$

in a normalized basis with a fixed sign convention and absence of extra light channels. The same finite basis must support the electroweak samples and the negative-branch scalar functional. Under these conditions, the projected de Rham operator is

$$Q_{\text{dR},J} = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & 0 \end{pmatrix}. \quad (\text{C92})$$

Witten's deformed complex supplies the source equations

$$d_t = e^{-ht} d e^{ht}, \quad d_t + d_t^\dagger, \quad H_t = (d_t + d_t^\dagger)^2, \quad (\text{C93})$$

Witten's paper supplies  $Q_1 = d + d^*$ ,  $Q_2 = i(d - d^*)$ ,  $H = dd^* + d^*d$ ,  $d_t$ , and  $H_t$ ; page 666 expands  $H_t$  with gradient and Hessian terms. These equations are prerequisites for elevating the Hodge route beyond a theorem target. The source theory must also contain a breaking or boundary operator  $B_J$  acting on the same projected basis.

*Conclusion.* The Hodge/SUSY-QM route succeeds if the projected operator has the form

$$Q_{\text{red},J} = Q_{\text{dR},J} + B_J, \quad B_J = \begin{pmatrix} 0 & 0 \\ 0 & -J \end{pmatrix}, \quad (\text{C94})$$

in the same field basis used by the ordered electroweak map and the negative-branch scalar functional. Equivalently,

$$\begin{array}{ccc} (\Omega_J^0, \Omega_J^1, D_J, D_J^\dagger) & \longrightarrow & \mathcal{H}_J^{\text{SQM}} & \longrightarrow & Q_{\text{red},J} \\ & & \downarrow & & \downarrow \\ & & (P_W, P_Z, u_-) & & \{x_+(J), x_-(J)\}. \end{array} \quad (\text{C95})$$

The side data  $P_W$ ,  $P_Z$ , and  $u_-$  denote, respectively, the charged pole sample, the neutral pole sample, and the scalar/order-parameter branch map.

*Target II compatibility.* The electroweak-superconnection version of this target requires the same projected complex to supply the ordered assignment:

$$\begin{array}{ccc} \Phi_{\text{odd}} & \xrightarrow{P_W} & J_H = \frac{3}{4} \\ \downarrow \mathcal{D}_J + \mathcal{D}_J^\dagger & & \downarrow \\ F_{\text{even}}^{\gamma^\perp} & \xrightarrow{P_Z} & J_{\text{adj}} = 2. \end{array} \quad (\text{C96})$$

Equation (C96) is a compatibility target. The cited sources supply grading and generalized-connection vocabulary; the ordered projectors, branch map, and pole quotient remain theorem data. Coquereaux gives the finite-algebra version of this compatibility target. In the local source, odd matrices  $\Omega_\pm, \Omega'_\pm$  generate a  $\mathbb{Z}_2$ -graded associative algebra; even anticommutators supply the electroweak generators  $I_3, Y, Q$ ; the generalized connection  $\mathcal{A}_{\text{Coq}}$  contains gauge one-forms and scalar zero-forms; and the curvature is  $\mathcal{F}_{\text{Coq}} = d\mathcal{A}_{\text{Coq}} + \mathcal{A}_{\text{Coq}} \odot \mathcal{A}_{\text{Coq}}$  [39]. The corresponding source-to-target map is

$$\mathcal{A}_{\text{Coq}} \longmapsto (P_{\text{sc}}, \langle \cdot, \cdot \rangle_{\text{sc}}, P_W, P_Z, P_\gamma, B_J, \Delta_{\text{sc}}^{\text{op}}), \quad (\text{C97})$$

where  $P_{\text{sc}}$ , the common inner product, the ordered projectors, the operator  $B_J$ , and the pole remainder are the DeVries theorem data. The pole comparison then has the form

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{sc}}. \quad (\text{C98})$$

The definition of  $\Delta_{\text{sc}}$  begins with a translation of all source entries to the complex-pole mass scheme. The interval version tests the same assignment against a CHM Dirichlet-to-

Neumann and brane-boundary kernel:

$$\begin{array}{ccc}
(\Phi_{\text{odd}}, F_{\text{even}}^{\gamma\perp}) & \xrightarrow{\mathcal{A}_{\text{sc}}} & \mathcal{H}_J = \text{span}\{h_J, a_J\} \\
\downarrow & & \downarrow P_J \\
(v_0, v_L, R, \theta_H) & \xrightarrow{K_J^{\text{DtN}} + K_J^{\text{brane}}} & K_J^{\text{sc/int}}(\lambda)
\end{array} \tag{C99}$$

The reduced target is

$$K_J^{\text{sc/int}}(\lambda) = P_J^\dagger (K_J^{\text{DtN}}(\lambda) + K_J^{\text{brane}}) P_J \longrightarrow \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix}. \tag{C100}$$

The finite-extension version of Target X is a self-adjoint-extension problem: derive  $P_{\text{sc}}$ ,  $B_J = \text{diag}(0, -J)$ , and the ordered electroweak samples from one interval, brane,  $G_2$ , or abstract SQM source operator. Witten's deformation supplies an analogy for source-theory diagonal terms through its Hessian contribution to  $H_t$ ; the DeVries entry  $-J$  still requires its own source-theory derivation. The route succeeds when  $P_W$ ,  $P_Z$ , the inner product,  $B_J$ , and  $\Delta_{\text{sc}}$  all come from one gauge-Higgs superconnection reduction. Equivalently, the Coquereaux graded-curvature projection must produce

$$P_{\text{sc}}^\dagger \langle \mathcal{F}_{\text{Coq}}, \mathcal{F}_{\text{Coq}} \rangle_{\text{sc}} P_{\text{sc}} = \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} + \Delta_{\text{sc}}^{\text{op}}(\lambda). \tag{C101}$$

The active test can be packaged as a boundary electroweak superconnection on a CHM interval. Let  $u_{\text{sc/int}}$  denote the source datum containing the finite superconnection boundary data and the interval bulk data. The theorem target is

$$\begin{aligned}
\widehat{K}_J^{\text{sc/int}}(\widehat{\lambda}) &= \Lambda_J^{-2} P_{\text{sc}}^\dagger \left[ K_T^{\text{DtN}}(\lambda) + K_T^{\text{brane}}(\lambda) + \langle \mathcal{F}_{\text{sc}}, \mathcal{F}_{\text{sc}} \rangle_{\text{sc}} - K_\gamma^{\text{ref}}(\lambda) \right] P_{\text{sc}} \\
&= \begin{pmatrix} \widehat{\lambda} & -\sqrt{J} \\ -\sqrt{J} & \widehat{\lambda} + J \end{pmatrix} + \Delta_J^{\text{sc/int}}(\widehat{\lambda}), \quad \lambda = \Lambda_J^2 \widehat{\lambda}.
\end{aligned} \tag{C102}$$

The associated diagram is

$$\begin{array}{ccc}
u_{\text{sc/int}} & \longrightarrow & (\mathcal{A}_{\text{sc}}, \mathcal{F}_{\text{sc}}, K^{\text{DtN}}, \langle \cdot, \cdot \rangle) \xrightarrow{P_{\text{sc}}, P_\gamma, \Lambda_J} \widehat{K}_J(\widehat{\lambda}) \\
\downarrow & & \downarrow \\
(t_{\text{EW}}, \mathcal{F}_{\text{scalar}}) & \longrightarrow & (x_+(3/4), x_+(2), x_-(J_\star)) \longrightarrow (M_W/M_Z, \mathcal{F}_{\text{sc}}, \Delta_{\text{pole}}).
\end{array} \tag{C103}$$

The source-completion conditions are

$$\begin{aligned}
P_W(\Phi_{\text{odd}}) &= \frac{3}{4} + \Delta_W^{\text{CHM}}, \quad P_Z(F_{\text{even}}^{\gamma\perp}) = 2 + \Delta_Y^{\text{CHM}}, \\
P_\gamma(F_{\text{even}}^\gamma) &= 0, \quad \Delta_J^{\text{sc/int}} \text{ is derived in the pole scheme.}
\end{aligned} \tag{C104}$$

The zero-remainder outcome sets  $\Delta_W^{\text{CHM}} = \Delta_Y^{\text{CHM}} = 0$  in the selected source convention. The scalar extension of the same target introduces a compact boundary-superconnection holonomy

$$\mathcal{W}_{\text{sc}} = \text{P exp} \int_I (A_5 + \Phi_{\text{odd}}) dy, \quad (\text{C105})$$

and asks whether its effective-potential Hessian supplies the negative-branch functional,

$$\widehat{\mathcal{F}}_{\text{sc}}^{\text{Hol/sc}} = \Lambda_J^{-2} \left. \frac{\partial^2 V_{\text{eff}}(\mathcal{W}_{\text{sc}})}{\partial \alpha_{\text{sc}}^2} \right|_{\alpha_\star} = C_{\text{sc}} |x_-(J_\star)| + \Delta_{\text{Hol/sc}}. \quad (\text{C106})$$

The same construction requires a source derivation of  $\Delta_{\text{Hol/sc}}$  in the same projected basis, photon reference, source scale, and pole convention as the vector branch. The corresponding acceptance diagram is

$$\begin{array}{ccc} u_{\text{sc}/\text{Hol}} & \longrightarrow & (\mathcal{A}_{\text{sc}}, \mathcal{W}_{\text{sc}}, K_T^{\text{DtN}}, K_T^{\text{brane}}) \longrightarrow (\widehat{K}_J, P_W, P_Z, P_\gamma) \\ \downarrow & & \downarrow \\ (t_{\text{EW}}, \alpha_{\text{sc}}) & \longrightarrow & (x_+(3/4), x_+(2), x_-(J_\star)) \longrightarrow (M_W/M_Z, \widehat{\mathcal{F}}_{\text{sc}}^{\text{Hol/sc}}, \Delta_{\text{pole}}). \end{array} \quad (\text{C107})$$

The source-internal weak-angle values, including the  $SU(2|1)$  trace normalizations and the 3/8 value in the extended-family discussion, are part of the algebraic normalization. The pole-spectrum quotient in Eq. (C98) enters after the projection and complex-pole matching steps. For the interval implementation, Target C7 is the diagonal partner of this factorization target: the Hodge/SUSY-QM complex supplies the candidate square-root entry, and the hatted CHM current kernel supplies the candidate  $\Sigma_{aa,J} = J$  entry after the source scale  $\Lambda_J$  and reference subtraction are fixed. The Target X square-root product is tested after the CHM admissibility lemma has fixed the neutral current trace and canonical diagonal entry; this keeps the Hodge product,  $P_W/P_Z/P_\gamma$ , and pole remainder in the same reduced basis.

*Source basis.* The square-root entry has a concrete mathematical source: on normalized Laplacian eigenspaces, a differential maps a zero-form channel to an exact one-form channel with norm controlled by the eigenvalue. Witten’s Morse-theory paper supplies the de Rham/Hodge and Witten-deformation template, including the rendered-page formulas quoted above [36]. Fayet supplies electroweak supersymmetry vocabulary in which spin-zero BEH fields sit in massive gauge multiplets with  $W$  and  $Z$  bosons [37]. Gates–Rana supply a worldline supersymmetry source for spinning-particle systems [38]. Coquereaux supplies an  $SU(2|1)$  superconnection source in which Higgs fields enter a generalized connection for electroweak symmetry breaking [39]. The local source pages record scalar-product and gauge-rescaling freedoms; these freedoms enter the theorem target as normalization data. These

sources give addresses for the factorization, spin, and electroweak grading. The DeVries breaking operator, finite projection, and pole-chain matching remain theorem obligations.

*Implication.* Target X fixes the Hodge/SUSY-QM comparison. The accepted source use is the normalized square-root template in Eq. (C90), the finite superconnection arena, and the boundary-superconnection interval package in Eq. (C102). The source derivation still has to produce  $B_J$ , the projection, extra-channel decoupling, ordered electroweak samples, the scalar partner, and the pole-map compatibility in one self-adjoint reduced basis.

*Failure mode.* The target fails if the projected differential yields a different normalization, if  $B_J$  gives a different diagonal entry, if additional light channels alter the determinant class, or if the electroweak projection conflicts with Targets C 5 and C 8. The surviving result would then be a Hodge factorization analogy for the off-diagonal entry.

## 15. Claim hierarchy and acceptance criteria

The theorem targets separate four kinds of statements:

| Level            | Paper content                                                                                                                                                                                                                    | Required basis                                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Derived          | algebra of $Q(J)$ , roots, branch identities, trace and determinant form                                                                                                                                                         | equation-level derivation in Sec. II                                                                                              |
| Source-supported | electroweak mass matrix, cited local source fragments and pole/running scheme distinctions, endpoint/brane, KK, $G_2$ , and Hodge/SUSY-QM background                                                                             | distinct references                                                                                                               |
| Conjectural      | pole placement of the DeVries quotient, negative-branch scalar reading, route-specific kernel entries, broken Hodge/SUSY-QM reduction                                                                                            | theorem target with assumptions and failure mode                                                                                  |
| Open             | ordered $J$ -assignment, electroweak ray admissibility, full determinant derivation, scalar functional, global-gebra, localized operator, or reform compatibility of a compact completion, sourced $B_J$ breaking mechanics term | proof from a stated action, boundary condition, endpoint algebra, localized operator, or reduced supersymmetric quantum mechanics |

A minimal derivation has the chain

$$\mathcal{S} \longrightarrow \mathcal{H}_J \longrightarrow K_J(\lambda) \longrightarrow (J_H, J_{\text{adj}}) \longrightarrow (s_{\text{pole}}^2, \mathcal{F}_{\text{sc}}). \quad (\text{C108})$$

If Target C 11 is added, the derivation has a conditional corollary

$$(s_{\text{pole}}^2, \mathcal{F}_{\text{sc}}, U(1)_{\text{em}}) \longrightarrow \alpha_{\mathcal{S}}(Q_{\alpha}), \quad (\text{C109})$$

with the weak-angle and threshold remainders of Eqs. (C79) and (C80). If Target C 12 is added, the source theory must also supply the intercept corollary

$$K_j(\lambda) \longrightarrow M_{N_{\text{osc},j,\sigma}}^2 = \mu^2 x_{\sigma}(j(j+1)) + \frac{N_{\text{osc}}}{\alpha'} + \Delta_{N_{\text{osc},j,\sigma}}^{\text{Regge}}. \quad (\text{C110})$$

The source theory  $\mathcal{S}$  may be a boundary effective action, an endpoint sector, a localized  $G_2$  system, or a Hodge/SUSY-QM reduction. The light subspace  $\mathcal{H}_J$  must contain an order-parameter channel and an adjoint/current channel. The kernel  $K_J(\lambda)$  must reduce to the

DeVries determinant in a stated normalization. The ordered pair  $(J_H, J_{\text{adj}}) = (3/4, 2)$  must arise from gauge-Higgs or source-theory data. The final arrow must attach the positive branch to the pole observable and the negative branch to a defined scalar functional. The same derivation must preserve the electroweak ray of Target C6.

The common acceptance criterion is

$$\begin{aligned}
\det K_J(\lambda) &= \lambda^2 + J\lambda - J, \\
M_\gamma^2(t) &= 0, \quad \frac{M_W^2(t)}{M_Z^2(t)} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{O1}}^{\text{ray}}(t), \\
(J_W, J_Z) &= (3/4, 2), \\
u_-(J) &\mapsto \mathcal{F}_{\text{sc}}, \\
K_J^{\text{bare}} &\rightarrow \Pi_{T,V}^{(4)}(s; J) \rightarrow \Delta_{V,T}^{-1}(s_V; J) = 0.
\end{aligned} \tag{C111}$$

Partial results still sharpen the construction when they prove one arrow in Eq. (C108). An interval model may derive the determinant class, an endpoint model may derive the ordered representation pair, and a  $G_2$  model may supply a localization mechanism. Each result changes the status of a corresponding theorem target.

## 16. Acceptance conditions

The theorem targets define the following acceptance conditions:

1. State the source theory  $\mathcal{S}$  and the light subspace  $\mathcal{H}_J$ .
2. Specify the normalization that makes  $\Sigma_{hh,J} = 0$ ,  $\Sigma_{aa,J} = J$ , and  $\Sigma_{ha,J}\Sigma_{ah,J} = J$ .
3. Derive the ordered assignment  $(J_H, J_{\text{adj}}) = (3/4, 2)$  from low-energy gauge-Higgs or boundary data.
4. Prove ray admissibility: one radial electroweak parameter, persistent photon null direction, and simultaneous W/Z mass collapse.
5. Give the pole-scheme translation for every quoted W/Z mass input.
6. Identify the negative branch as scalar data, auxiliary data, or a derived sector with another physical role.

7. If the electromagnetic endpoint is used, derive the  $U(1)_{\text{em}}$  gauge kinetic normalization, the scalar/vector normalization constants, and the matching pair  $(Q_\alpha, \Delta_{\alpha, \text{match}})$ .
8. If the Regge-intercept extension is used, separate the DeVries sector label  $j$ , the oscillator level  $N_{\text{osc}}$ , and the physical trajectory spin, then derive the projection rule for the  $j = 3/2$  positive intercept.
9. If the branch-scaling interpretation is used, derive Eq. (C88), the negative-branch mass reading, and the boundary-condition duality map from the source theory.
10. If the Hodge/SUSY-QM route is used, derive Eq. (C94) from normalized 0/1-form channels, the source of  $J$ , the finite projection, the breaking operator  $B_J$ , and the same pole/scalar maps used in the electroweak acceptance criterion.
11. Check global-form, anomaly, and representation compatibility.

These conditions define the acceptance test for a source derivation in this paper.

#### **Appendix D: Dimensional interpolation target**

This appendix records a dimensional theorem target for the Higgs-sector problem. Its role is to organize the Kaluza–Klein, string, Higgs, and top-quark questions through a common source package.

The source support has four levels. Witten’s seven-extra-dimensional  $SU(3) \times SU(2) \times U(1)$  endpoint is source-backed. The five-extra-dimensional  $SU(3) \times U(1)_{\text{geom}}$  endpoint is a geometric endpoint target whose electromagnetic embedding remains to be derived. The  $D = 10$  and colourless  $D = 6$  middle lines are reconstruction labels for a source sector that must supply  $\mathcal{B}$ ,  $\mathcal{H}_J$ ,  $\langle \cdot, \cdot \rangle_u$ ,  $P_J(u)$ ,  $\Lambda_J(u)$ ,  $\widehat{K}_J$ , a decoupling rule for extra channels, and  $\mathcal{R}_{\text{pole}}$ . The scalar branch and top-sector readings require the same variable to control the gauge-Higgs ray, the reduced kernel, and the relevant Yukawa or boundary datum. The source claims in this appendix rest on Witten, CHM, the six-dimensional string sources, and local  $G_2$  references.

## 1. Full-gauge count

The full-gauge count keeps colour inside the compactification symmetry. Witten’s realistic Kaluza–Klein analysis gives the source-backed endpoint: a compact space with  $SU(3) \times SU(2) \times U(1)$  symmetry requires seven extra dimensions in the class considered there, giving total dimension  $D = 11$  [16]. The same source records the associated fermion quantum-number obstruction for that seven-dimensional compactification.

The proposed interpolation is

$$\begin{aligned}
 D = 11, \quad n_{\text{KK}} = 7 & : SU(3) \times SU(2) \times U(1) \quad \text{unbroken electroweak endpoint } v = 0, \\
 D = 10, \quad n_{\text{KK}} = 6 & : \text{DeVries interpolation sector,} \\
 D = 9, \quad n_{\text{KK}} = 5 & : SU(3) \times U(1)_{\text{geom}} \quad \text{geometric endpoint target.}
 \end{aligned} \tag{D1}$$

The arrows in Eq. (D1) denote a theorem map to be supplied by a source derivation. A physical interpolation requires a single compactification, boundary, or singular source family that supplies the deformation parameter and the four-dimensional matching map. The heavy- $W/Z$  endpoint is a geometric or effective endpoint target outside the electroweak radial ray. Its compatibility with fixed  $g : g'$  and the photon null direction requires a source theorem. The source support of the three lines is uneven, and the distinction is part of the claim. The source-backed data and reconstruction targets are:

| Line                        | Source-backed content                                                                                                                                               | DeVries obligation                                                                         |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| $D = 11, n_{\text{KK}} = 7$ | Witten’s $SU(3) \times SU(2) \times U(1)$ define the unbroken endpoint and KK endpoint and its fermion its Higgs/order-parameter channel quantum-number obstruction | relation to the electroweak ray                                                            |
| $D = 10, n_{\text{KK}} = 6$ | source theory open; Schur-complement reconstruction label                                                                                                           | identify the six-dimensional internal object, its light two-channel sector, and its kernel |
| $D = 9, n_{\text{KK}} = 5$  | Witten’s five-sphere discussion contains an $SU(3) \times U(1)$ symmetry address                                                                                    | derive the electroweak embedding, generator normalization, and charge lattice              |

Witten’s five-sphere material supplies the  $SU(3) \times U(1)$  symmetry address. Electromagnetic normalization and the electroweak embedding remain part of the theorem target.

The same interpolation can be written as a factorized active-channel problem:

$$D = 4 + n_c + n_{\text{ew}}, \quad n_c = 4, \quad n_{\text{ew}} : 3 \rightarrow 2 \rightarrow 1. \quad (\text{D2})$$

The full-gauge and colour-spectator diagrams are then

$$\begin{array}{ccccc} \mathcal{C}_4 \times \mathcal{E}_3 & \xrightarrow{u} & \mathcal{C}_4 \times \mathcal{E}_2(u_\star) & \xrightarrow{u} & \mathcal{C}_4 \times \mathcal{E}_1 \\ D = 11 & & D = 10 & & D = 9 \\ \mathcal{E}_3 & \xrightarrow{u} & \mathcal{E}_2(u_\star) & \xrightarrow{u} & \mathcal{E}_1 \\ D = 7 & & D = 6 & & D = 5. \end{array} \quad (\text{D3})$$

Here  $\mathcal{C}_4$  denotes the colour compact factor and  $\mathcal{E}_k$  denotes the active electroweak compact or boundary channel. The middle object must supply

$$\mathcal{E}_2(u_\star) \rightarrow \mathcal{H}_J \rightarrow K_J \rightarrow (J_H, J_{\text{adj}}) \rightarrow \Delta_{T,V}^{-1}.$$

The  $D = 10$  middle line is a Schur-complement reconstruction label. A derivation must identify the six-dimensional internal object, its boundary or compact operator, the light two-channel sector, the ordered electroweak assignment, and the pole-matching rule.

## 2. Colourless electroweak count

There is a reduced count in which colour is treated as a spectator sector. The electroweak part of the endpoint symmetry then changes as

$$\begin{aligned} D = 7, \quad n_{\text{KK}}^{\text{ew}} = 3 : SU(2) \times U(1), \\ D = 6, \quad n_{\text{KK}}^{\text{ew}} = 2 : \text{DeVries electroweak interpolation sector}, \\ D = 5, \quad n_{\text{KK}}^{\text{ew}} = 1 : U(1)_{\text{geom}} \rightarrow U(1)_{\text{em}} \quad \text{after embedding theorem.} \end{aligned} \quad (\text{D4})$$

This version ties the six-dimensional middle line to six-dimensional string sources. Type IIA compactified on K3 and the heterotic string compactified on  $T^4$  provide a six-dimensional string-string duality setting [25, 26]; heterotic small instantons give nonperturbative six-dimensional dynamics [27]; and six-dimensional anomaly and string-universality constraints give consistency conditions [28]. These sources support a six-dimensional source arena. The colourless  $D = 6$  line remains a Schur-complement reconstruction label whose DeVries use requires the data in Eq. (D8).

### 3. Interpolation parameter

The interpolation parameters must be stated in field-theory and geometric terms. The electroweak field theory supplies a radial order parameter  $t_{\text{EW}}$  through the Higgs vacuum expectation value. The compactification descriptions supply a separate coordinate  $t_{\text{dim}}$  through radii, boundary conditions, brane-localized terms, or singularity deformations. A source derivation must supply a map

$$\chi : t_{\text{dim}} \longmapsto t_{\text{EW}} \quad (\text{D5})$$

or a joint source variable refined below in Eq. (D7). The dimensional family has source operators

$$\mathcal{O}_J(t_{\text{dim}}), \quad t_{\text{dim}} = 0 : SU(3) \times SU(2) \times U(1), \quad t_{\text{dim}} = \infty : SU(3) \times U(1)_{\text{geom}}, \quad (\text{D6})$$

or the corresponding colourless electroweak endpoints. The  $U(1)_{\text{em}}$  identification is an electroweak embedding obligation. A DeVries point needs a joint source datum. Until a route supplies it,  $u$  is a formal joint source coordinate. Candidate realizations include a compact modulus, an interval/boundary parameter, a brane datum, or a singularity deformation. The required package is

$$\begin{aligned} u \longmapsto & (t_{\text{dim}}(u), t_{\text{EW}}(u), v(u), m_h^2(u), \Lambda_J(u), \langle \cdot, \cdot \rangle_u, \\ & P_J(u), \hat{K}_J(u, \hat{\lambda}), \mathcal{D}_J^{\text{extra}}(u), \mathcal{F}_{\text{sc}}(u; J), \\ & \mathcal{Y}_{\text{top}}(u), \mathcal{R}_{\text{pole}}(u)), \end{aligned} \quad (\text{D7})$$

where  $v(u)$  follows the allowed electroweak ray,  $m_h^2(u)$  records scalar stiffness along the same family,  $\mathcal{F}_{\text{sc}}$  is the scalar or negative-branch functional,  $\hat{\lambda} = \lambda/\Lambda_J(u)^2$ ,  $\mathcal{D}_J^{\text{extra}}$  records decoupling or block reduction of all additional modes, and  $\mathcal{Y}_{\text{top}}$  is a possible top-sector datum. Target C 14 gives an optional Hodge/superconnection route for the source of  $K_J$ . The middle-dimensional data required at a DeVries point are

$$\mathfrak{I}_J(u_\star) = (\mathcal{B}(u_\star), \mathcal{H}_J(u_\star), \langle \cdot, \cdot \rangle_{u_\star}, P_J(u_\star), \Lambda_J(u_\star), \hat{K}_J(u_\star, \hat{\lambda}), \mathcal{D}_J^{\text{extra}}(u_\star), \mathcal{R}_{\text{pole}}(u_\star)), \quad (\text{D8})$$

where  $\mathcal{B}(u_\star)$  is the compact, boundary, or singular object,  $\mathcal{H}_J(u_\star) = \text{span}\{h_J, a_J\}$  is the light order-parameter/current subspace,  $P_J$  is the projection into that subspace,  $\Lambda_J$  is the common source scale,  $\hat{K}_J$  is the dimensionless reduced kernel, and  $\mathcal{R}_{\text{pole}}$  is the matching

rule to the transverse pole propagator. At the DeVries point the light two-channel reduction must take the form

$$\Lambda_J(u_\star)^{-2} P_J(u_\star)^\dagger \mathcal{O}_J(u_\star, \lambda) P_J(u_\star) \longrightarrow \hat{K}_J(u_\star, \hat{\lambda}) = \begin{pmatrix} \hat{\lambda} & -\sqrt{J} \\ -\sqrt{J} & \hat{\lambda} + J \end{pmatrix}, \quad (\text{D9})$$

with  $\mathcal{D}_J^{\text{extra}}$  eliminating or block-diagonalizing all additional light channels. Setting  $\hat{\lambda} = 0$  gives the source operator target

$$\begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}. \quad (\text{D10})$$

After a successful pass of Eq. (D9), the positive branch is assigned to the pole-ratio target, and the negative branch is assigned to a scalar, boundary, brane, or compactification datum. The current  $u_{\text{sc/int}}$  package gives this middle-dimensional test a source comparison:

| Datum in $u_{\text{sc/int}}$                       | Source control                             | Unresolved map                                                                                                     |
|----------------------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $K_T^{\text{DtN}} + K_T^{\text{brane}}$            | CHM interval                               | boundary hatted current entry $\hat{\Sigma}_{aa,2} = 2$ equations                                                  |
| $\mathcal{A}_{\text{sc}}, \mathcal{F}_{\text{sc}}$ | Coquereaux $SU(2 1)$ su-                   | $P_W, P_Z, P_\gamma$ after kinetic normal-perconnection source ization                                             |
| $D_J, D_J^\dagger$                                 | Witten Hodge/SUSY-QM                       | square-root entry in the same re-temple duced basis                                                                |
| $\mathcal{W}_{\text{sc}}$                          | Hosotani Wilson-line                       | negative-branch Hessian $\hat{\mathcal{F}}_{\text{sc}}^{\text{Hol/sc}}$ route plus superconnec- tion boundary data |
| $G_2$ localized data                               | Acharya–Witten Higgs-bundle sources        | and compact completion, charge lat- tice, anomaly, pole map                                                        |
| six-dimensional string heterotic/type-II arena     | heterotic/type-II dimensional source trail | six- source object for the colourless $D = 6$ middle line                                                          |

The compatibility diagram is

$$\begin{aligned} (D, n_{\text{KK}}) &= (11, 7) \text{ or } (7, 3) \xrightarrow{t_{\text{dim}}} (D, n_{\text{KK}}) = (9, 5) \text{ or } (5, 1), \\ (D, n_{\text{KK}}) &= (10, 6) \text{ or } (6, 2) : \quad \mathcal{O}_J(u_\star)|_{\mathcal{H}_J} \rightarrow \hat{K}_J(u_\star, \hat{\lambda}), \\ \hat{K}_J(u_\star, \hat{\lambda}) &\longrightarrow (J_W, J_Z) = (J_H, J_{\text{adj}}), \\ (J_W, J_Z) &\longrightarrow \Delta_{V,T}^{-1}(s_{W,Z}). \end{aligned} \quad (\text{D11})$$

Thus the dimensional interpolation contributes through a kernel derivation, a map from dimensional data to the electroweak ray, an ordered assignment theorem, and a pole-matching theorem. The associated source conditions are

$$\text{Target VI} \equiv \{u, \mathcal{H}_J, P_J, \Lambda_J, \widehat{K}_J, \mathcal{D}_J^{\text{extra}}, \mathcal{R}_{\text{pole}}\} \longrightarrow \begin{cases} \text{Target II : } (J_W, J_Z) = (3/4, 2), \\ \text{Target IV : } x_-(J) \rightarrow \mathcal{F}_{\text{sc}}, \\ \text{Target X : } D_J + D_J^\dagger \rightarrow \sqrt{J}, \\ \text{Target VII : } U(1)_{\text{geom}} \rightarrow U(1)_{\text{em}}. \end{cases} \quad (\text{D12})$$

The same obligation can be drawn as a single-source interpolation diagram:

$$\begin{array}{ccc} u = 0 & u = u_\star & u \rightarrow \infty \\ v = 0 & \widehat{K}_J(u_\star, \widehat{\lambda}) \text{ passes Eq. (D9)} & U(1)_{\text{geom}} \text{ endpoint} \\ m_h^2(u) \text{ specified} & x_+(J) \rightarrow \text{pole ratio, } x_-(J) \rightarrow \mathcal{F}_{\text{sc}} & \text{embedding theorem} \end{array} \quad (\text{D13})$$

$$\mathcal{Y}_{\text{top}}(u_\star) \rightarrow y_t$$

The first column is field-theory ray data, the middle column is the DeVries kernel target, and the last column is the endpoint embedding target.

#### 4. Electromagnetic endpoint normalization

The  $D = 9$  endpoint also carries a coupling-normalization target. The endpoint theorem has three named subtargets:

1. identify the compact isometry or gauge generator  $\xi_{\text{em}}(y; t_{\text{dim}})$ ;
2. derive the Standard Model electromagnetic charge lattice and generator normalization;
3. match the reduced four-dimensional kinetic term in the pole/running convention used by Target C 11.

Once the surviving geometric factor has been identified as  $U(1)_{\text{em}}$ , the compact source must determine the four-dimensional kinetic term

$$\frac{1}{e_{\text{geom}}^2} \sim \frac{1}{g_D^2} \int_{\mathcal{B}_5(t_{\text{dim}})} d^5 y \sqrt{g_5} |\xi_{\text{em}}(y; t_{\text{dim}})|^2, \quad (\text{D14})$$

with a specified generator normalization and charge lattice. This is the geometric form of the electromagnetic part of Target C 11. Witten's Kaluza–Klein discussion supports the general

mechanism that gauge couplings are determined by compact metric data when the metric dynamics is known [16]. The DeVries endpoint problem asks for the same source variable to supply the  $U(1)_{\text{em}}$  endpoint, the scalar branch normalization, and the matching scale  $Q_\alpha$  in Eq. (C80).

The pass/fail data for Eq. (D14) are

$$\mathfrak{E}_\alpha(u) = (\xi_{\text{em}}(u), \Gamma_Q(u), N_{\text{gen}}(u), g_D(u), g_5(u), \mathcal{T}_\alpha(u), Q_\alpha(u)),$$

where  $\Gamma_Q$  is the charge lattice,  $N_{\text{gen}}$  is the generator normalization, and  $\mathcal{T}_\alpha$  is the threshold prescription. A successful endpoint theorem maps  $\mathfrak{E}_\alpha(u)$  to

$$(e_{\text{geom}}^2, \sin^2 \theta_S(Q_\alpha), \Delta_{\text{th}}(Q_\alpha))$$

in the same scheme used for the vector-pole comparison and the scalar normalization.

The dependency chain is

$$\text{O17} \Leftarrow \text{Target II} + \text{Target IV} + \text{Target VI} + \mathcal{R}_{\text{pole/running}}. \quad (\text{D15})$$

Equivalently, the photon-zero endpoint route must supply the map

$$\begin{aligned} u \longrightarrow (P_W, P_Z, P_\gamma, \Lambda_J, \mathcal{F}_{\text{sc}}, \xi_{\text{em}}) &\longrightarrow (s_{\text{dV}}^2, g_{\text{sec}}^2, e_{\text{geom}}^2) \\ &\downarrow \\ &\alpha_S(Q_\alpha) + \Delta_{\text{th}}(Q_\alpha). \end{aligned} \quad (\text{D16})$$

The assignments  $v = \sqrt{2} M_-(2)$ ,  $J_v = 2$ ,  $C_v = 2$ , and  $C_W = 1$  are separate subtargets. Their use in an electromagnetic coupling statement requires a derived scalar normalization, charged-vector normalization, generator normalization, charge lattice, matching scale, and threshold map. When the  $U(1)_{\text{geom}}$  endpoint lacks the Standard Model electromagnetic charge lattice, Target VI remains a geometric endpoint target and O17 has only a geometric endpoint conclusion. This paper therefore leaves  $\alpha$  as an external low-energy input.

The branch-to-coupling obligations can be recorded as

$$\begin{aligned} Q(J) &\longrightarrow x_+(3/4), x_+(2) &\longrightarrow s_{\text{dV}}^2 \\ \downarrow & & \downarrow \\ x_-(2) &\longrightarrow v_\star = \sqrt{2} \mu \sqrt{|x_-(2)|} &\longrightarrow g_\star^2, \alpha_\star \\ \downarrow & & \downarrow \\ \mathcal{B}_5(t_{\text{dim}}) &\longrightarrow U(1)_{\text{em}} \text{ normalization} &\longrightarrow Q_\alpha, \Delta_{\alpha, \text{match}}. \end{aligned} \quad (\text{D17})$$

Each arrow is a theorem target. The first row belongs to the pole-ratio construction. The second row requires the negative-branch scalar theorem. The third row requires a compactification or boundary theorem for the electromagnetic gauge kinetic term and its low-energy matching.

## 5. Top-quark scale

The dimensional interpolation also creates a top-sector question. In the Standard Model, fermion masses are generated by Yukawa couplings to the Higgs vacuum expectation value,

$$m_f = \frac{y_f v}{\sqrt{2}}, \quad g_{hff} = \frac{m_f}{v}. \quad (\text{D18})$$

The top quark has the largest direct fermionic coupling to the electroweak order parameter. This fact is visible in Higgs phenomenology, in electroweak corrections with strong top-mass sensitivity, and in vacuum-stability analyses [5, 8, 9]. Local  $G_2$  Higgs-bundle constructions also provide a source-theory template: flow-tree data can engineer a top Yukawa coupling in an  $SU(5)$ -type local model [20]. In the interval route, CHM supplies a separate top-sector lesson: a heavy top pushes the third-generation localization and boundary mixing, while the observed bottom coupling to the  $Z$  restricts the same data [7]. A DeVries claim requires a map from that localization or boundary datum to  $\mathcal{Y}_{\text{top}}(u)$ .

The DeVries question is sharper: can the six-dimensional interior sector select the top quark as the fermion attached most directly to the order parameter? The source-theory target is

$$\mathcal{Y}_{\text{top}}(u_\star) \longrightarrow (y_t(u_\star), m_t(u_\star), \Pi_{VV}^{(t)}(s; u_\star)), \quad (\text{D19})$$

with the top representation, localization or boundary data, anomaly and global-form checks, and pole-scheme matching stated in the same basis as  $\widehat{K}_J$ . The CHM interval route gives

an active test package:

$$\begin{array}{ccc}
u_{\text{int}} = (R, v_i, Z_i, \theta_H, c_{Q_3}, c_{t_R}, M_{\text{mix}}, \dots) & \longrightarrow & (K_T^{\partial I}, K_{A_5\pi}^{\partial I}, \mathcal{Y}_{\text{top}}) \\
\downarrow & & \downarrow \\
(t_{\text{dim}}, t_{\text{EW}}, v) & \longrightarrow & \Lambda_J(u_{\text{int}})^{-2} P_J^\dagger (K_J^{\text{DtN}} + K_J^{\text{brane}} - K_J^{\text{ref}}) P_J \\
& & \downarrow \\
& & \begin{pmatrix} \hat{\lambda} & -\sqrt{J} \\ -\sqrt{J} & \hat{\lambda} + J \end{pmatrix}.
\end{array} \tag{D20}$$

The companion compatibility target is

$$\begin{array}{ccc}
x_-(J) & \longrightarrow & \mathcal{F}_{\text{sc}}(u; J) \\
\downarrow & & \downarrow \\
t_{\text{EW}}(u) & \longrightarrow & v(u) \\
\downarrow & & \downarrow \\
\mathcal{Y}_{\text{top}}(u) & \longrightarrow & (y_t(u), m_t(u), \Pi_{VV}^{(t)}(s; u)).
\end{array} \tag{D21}$$

Here  $\Pi_{VV}^{(t)}$  denotes the top-loop contribution to electroweak vector self-energies in the matched four-dimensional theory. This is an open target. This paper records the source-backed Standard Model fact, the local  $G_2$  template for Yukawa data, and the required form of a DeVries top-sector explanation. A DeVries derivation of  $y_t$ ,  $m_t$ , and top-loop self-energy data remains open.

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